REF Schematic for RK3568 Tablet

Main Functions Introduction

1)Charger/PMIC: 1Cell Battery + RK817-5 Charger
Option: 1Cell Battery + Quick Charger + RK809-5
Option: 2Cell Battery + Quick Charger + RK809-5
2)RAM: LPDDR4/4x 1X32bit(200ball)
Option:DDR3 4x16bit
Option:DDR4 2x16bit
Option:DDR4 4x16bit
Option:LPDDR3 1x32bit(178ball)
3)ROM: eMMC
Option:Nand Flash
4)Support:1 x Micro SD Card3.0
5)Support:1 x USB TypeC(with USB3.0 OTG)
Option:1 x MircoUSB2.0
6)Support:1 x USB3.0 HOST1 + 1 x USB2.0 HOST2
7)Support:4G module Via MiniPCIe2.0 Slot With PCIE2.0 and USB2.0 HOST3 function
8)Support:PCIE3.0 M.2 SSD KEY B/M Connector
9)Support:1 x HDMI2.0 TX
10)Support:1 x LCM MIPI DSI TX0
Option:1 x LCM MIPI DSI TX1
Option:1 x LCM LVDS TX
Option:1 x LCM Dual MIPI DSI TX
Option:1 x LCM eDP TX
11)Support:1 x 4Lanes Camera MIPI CSI RX + Parallel CIF Camera
Option:2 x 2Lanes Camera MIPI CSI RX
Option:1 x 4Lanes or 1 x 2Lanes Camera MIPI CSI RX
12)Support:a/b/g/n/ac 2X2 SDIO WIFI5+BT5.0+PCM
Option:a/b/g/n/ac 1X1 SDIO WIFI+BT+PCM
Option:a/b/g/n/ac/ax 2X2 PCIe WIFI6+BT5.0+PCM
Option:PCIe2.0 M.2 KEY E Connector
13)Support:1 x Headphone output
14)Support:1 x ECM MIC + 1 x Speaker out
Option:4 x MEMS MIC + 1 x Speaker out + Loopback or 2 x MEMS MIC + 1 x Speaker out + Loopback
Option:4 x MEMS MIC + 2 x Speaker out + Loopback
15)Support:Gyroscope + Accelerometer + Motion + Hall + Ambient & Proximity
16)Support:Array Key(MENU,VOL+,VOL-,ESC),Reset,Power on/off Key
17)Support:Debug UART

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Project:	RK3568_Tablet_REF_SCH				
File:	00.Cover Page				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	1 of 61

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Description

Note

Option

Generate Bill of Materials

Header:

Item\tPart\tDescription\tPCB Footprint\tReference\tQuantity\tOption

Combined property string:

{Item}\t{Value}\t{Description}\t{PCB Footprint}\t{Reference}\t{Quantity}\t{Option}

Notes

NOTE 1:
Component parameter description
1. DNP stands for component not mounted temporarily
2. If Value or option is DNP, which means the area is reserved without being mounted

NOTE 2:
Please use our recommended components to avoid too many changes.
For more informations about the second source,please refer to our AVL.

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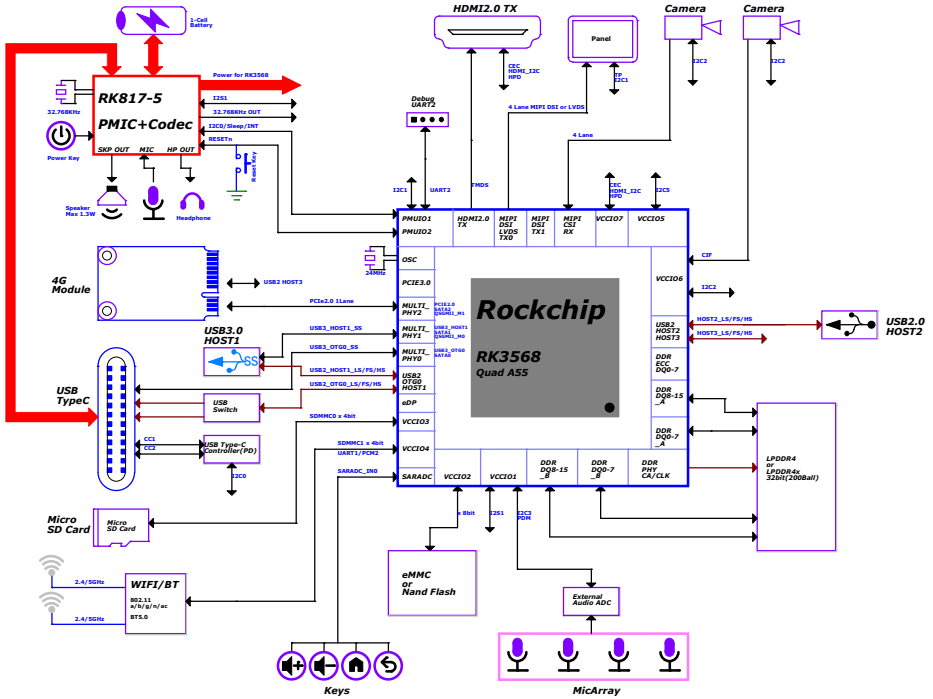
Rockchip 瑞芯微电子 Rockchip Electronics Co., Ltd					
Project:	RK3568_Tablet_REF_SCH				
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Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	2 of 61

Revision History

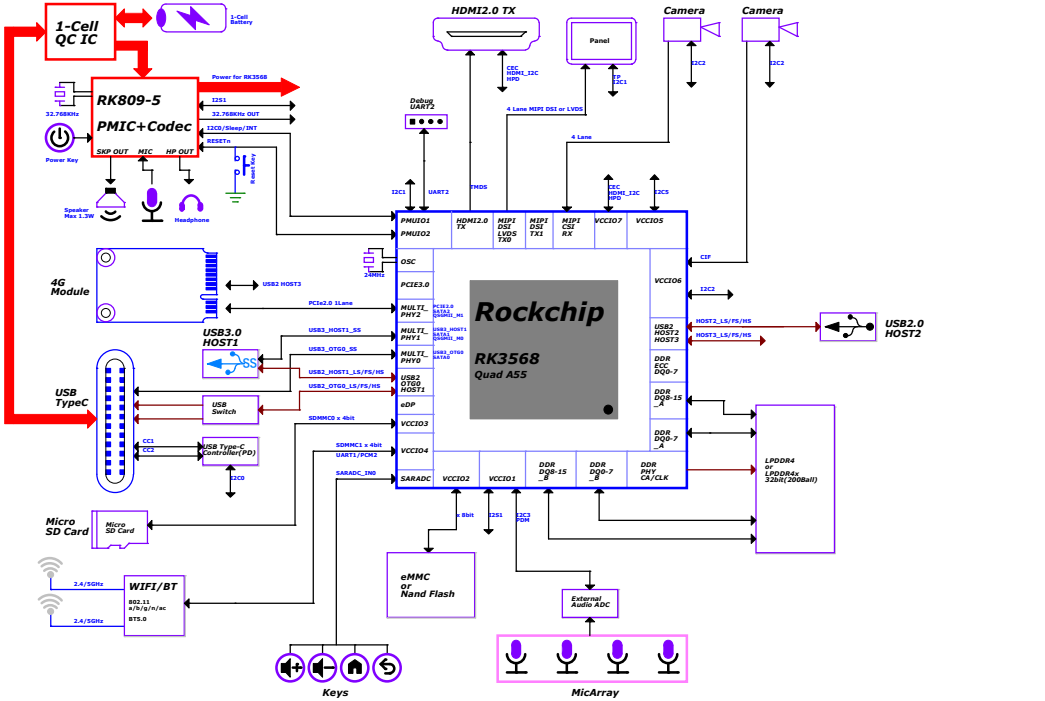
Version	Date	By	Change Dsecription	Approved
V1.0	2021-07-20	Zhangdz	1:Revision preliminary version	

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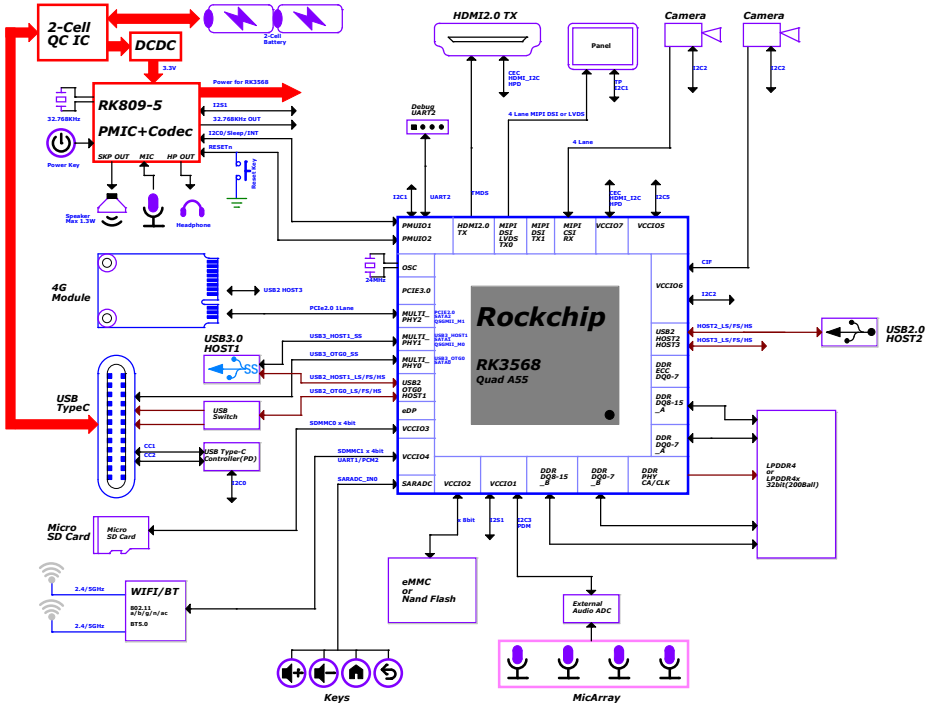
RK3568 Tablet Ref Block Diagram for 1-Cell Charger use RK817-5



RK3568 Tablet Ref Block Diagram for 1-Cell Quick Charger use BQ25890/SY6970

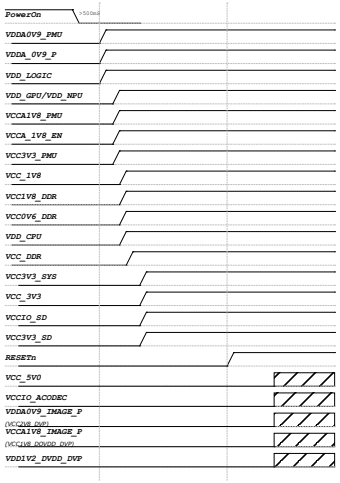


RK3568 Tablet Ref Block Diagram for 2-Cell Quick Charger use BQ25703/SC8886



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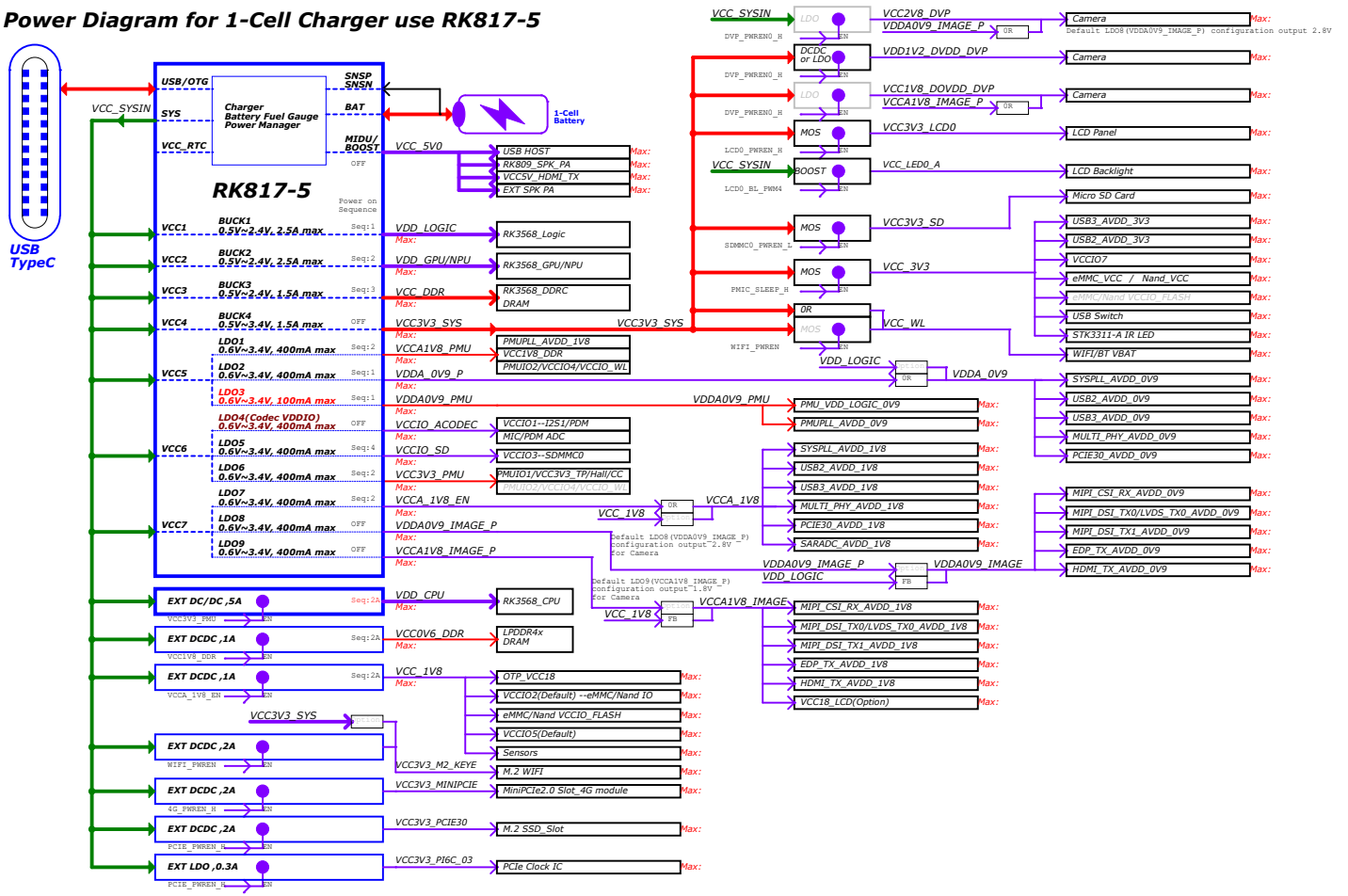
Power Sequence(Use RK817-5)



Power description(Use RK817-5)

Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Work Voltage	Peak Current	Sleep Current
VCC_SYSIN	RK817_BUCK1	2.5A	VDD_LOGIC	Slot:1	0.9V	ON	0.9V	TBD	TBD
VCC_SYSIN	RK817_BUCK2	2.5A	VDD_GPU/VDD_NPU	Slot:2	0.9V	ON	DVFS	TBD	TBD
VCC_SYSIN	RK817_BUCK3	1.5A	VCC_DDR	Slot:3	ADJ (VR=0.8V	ON	1.1V (LPDDR4/4x)	TBD	TBD
VCC_SYSIN	RK817_BUCK4	1.5A	VCC3V3_SYS	Slot:4	3.3V	ON	3.3V	TBD	TBD
VCC_SYSIN	RK817_LDO1	0.4A	VCCA1V8_PMU	Slot:2	1.8V	ON	1.8V	TBD	TBD
VCC_SYSIN	RK817_LDO2	0.4A	VDDA_0V9_P	Slot:1	0.9V	ON	0.9V	TBD	TBD
VCC_SYSIN	RK817_LDO3	0.1A	VDDA0V9_PMU	Slot:1	0.9V	ON	0.9V	TBD	TBD
VCC_SYSIN	RK817_LDO4	0.4A	VCCIO_ACODEC	N/A	0V	OFF	3.3V	TBD	TBD
VCC_SYSIN	RK817_LDO5	0.4A	VCCIO_SD	Slot:4	3.3V	ON	3.3V or 1.8V (SD Card)	TBD	TBD
VCC_SYSIN	RK817_LDO6	0.4A	VCC3V3_PMU	Slot:2	3.3V	ON	3.3V	TBD	TBD
VCC_SYSIN	RK817_LDO7	0.4A	VCCA_1V8_EN	Slot:2	1.8V	ON	1.8V	TBD	TBD
VCC_SYSIN	RK817_LDO8	0.4A	VDDA0V9_IMAGE_P (VCCA1V8_DVP)	N/A	0V	OFF	2.8V (Camera)	TBD	TBD
VCC_SYSIN	RK817_LDO9	0.4A	VCCA1V8_IMAGE_P (VCCA1V8_DOVDD_DVP)	N/A	0V	OFF	1.8V (Camera)	TBD	TBD
VCC_SYSIN	BOOST	1.5A	VCC_5V0	N/A	0V	OFF	5.0V	TBD	TBD
	RK817_RESETn			Slot:4+5					
VCC_SYSIN	EXT BUCK	5.0A	VDD_CPU	Slot:2A	1.025V	ON	DVFS	TBD	TBD
VCC_SYSIN	EXT BUCK	1.0A	VCC_1V8	Slot:2A	1.8V	ON	1.8V	TBD	TBD
VCC_SYSIN	EXT LDO	0.3A	VCC1V8_DDR	Slot:2A	1.8V	ON	1.8V	TBD	TBD
VCC_SYSIN	EXT BUCK	2.0A	VCC0V6_DDR	Slot:2A	0.6V	ON	0.6V	TBD	TBD
VCC3V3_SYS	EXT MOS		VCC_3V3	Slot:4	3.3V	ON	3.3V	TBD	TBD
VCC3V3_SYS	EXT MOS		VCC3V3_SD	Slot:4	3.3V	ON	3.3V	TBD	TBD
VCC_SYSIN	EXT LDO or BUCK	0.5A /1.0A	VDD1V2_DVDD_DVP	N/A	0V	OFF	1.2V (Camera)	TBD	TBD

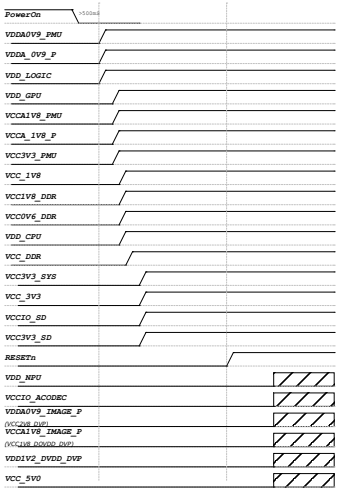
Power Diagram for 1-Cell Charger use RK817-5



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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd
Project:	RK3566_Tablet_REF_SCH
File:	04.Power Diagram-1Cell RK817
Date:	Tuesday, July 20, 2021
Designed by:	Zhangjk
Reviewed by:	Default
Rev:	V1.0
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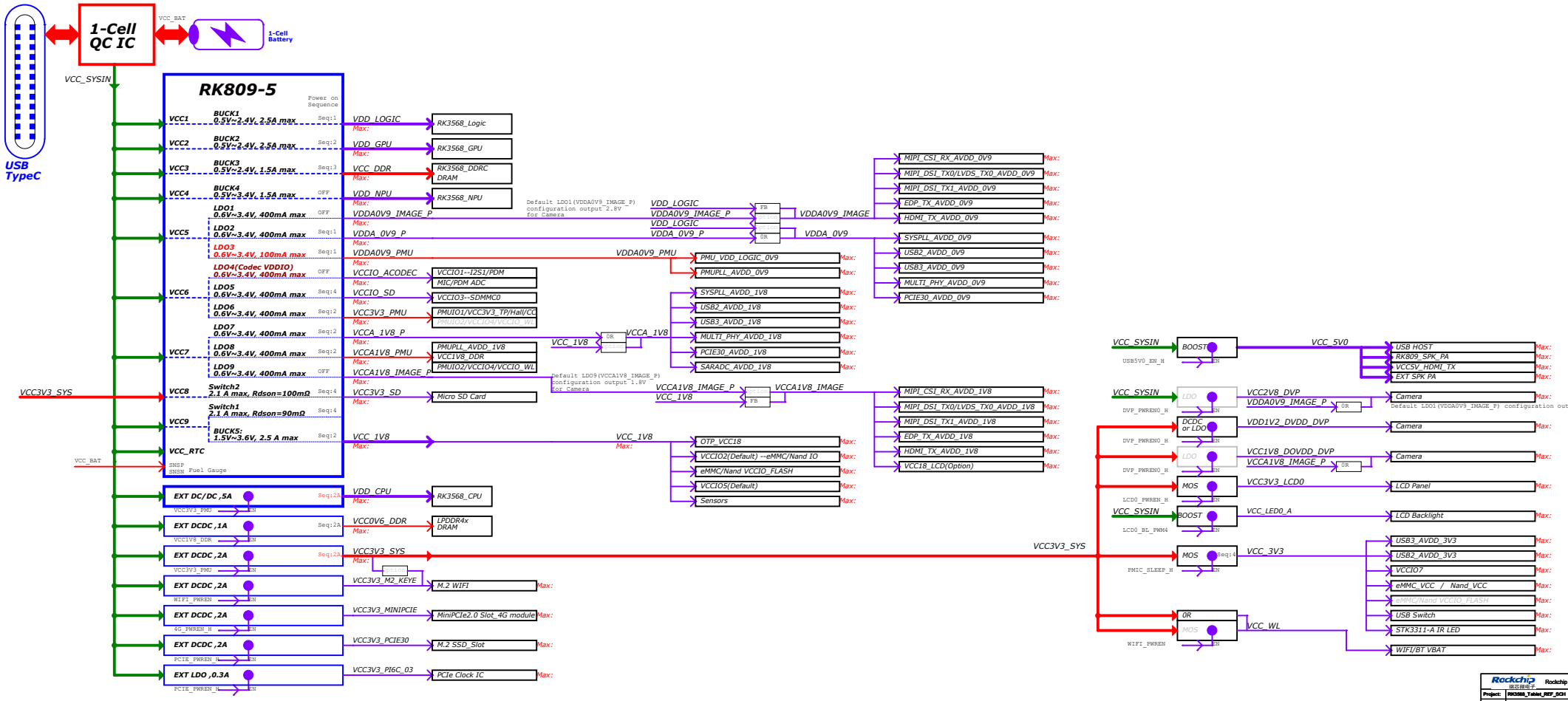
Power Sequence(Use RK809-5)



Power description(Use RK809-5)

Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Work Voltage	Peak Current	Sleep Current
VCC_SYSIN	RK809_BUCK1	2.5A	VDD_LOGIC	Slot:1	0.9V	ON	0.9V	TBD	TBD
VCC_SYSIN	RK809_BUCK2	2.5A	VDD_GPU	Slot:2	0.9V	ON	DVFS	TBD	TBD
VCC_SYSIN	RK809_BUCK3	1.5A	VCC_DDR	Slot:3	ADJ FB=0.8V	ON	1.2V (DDR4)	TBD	TBD
VCC_SYSIN	RK809_BUCK4	1.5A	VDD_NPU	N/A	0V	OFF	DVFS	TBD	TBD
VCC_SYSIN	RK809_LDO1	0.4A	VDDA0V9_IMAGE_P (VCC2V8_DVP)	N/A	0V	OFF	2.8V (Camera)	TBD	TBD
	RK809_LDO2	0.4A	VDDA_0V9	Slot:1	0.9V	ON	0.9V	TBD	TBD
	RK809_LDO3	0.1A	VDDA0V9_PMU	Slot:1	0.9V	ON	0.9V	TBD	TBD
VCC_SYSIN	RK809_LDO4	0.4A	VCCIO_ACODEC	N/A	0V	OFF	3.3V	TBD	TBD
	RK809_LDO5	0.4A	VCCIO_SD	Slot:4	3.3V	ON	3.3V or 1.8V	TBD	TBD
	RK809_LDO6	0.4A	VCC3V3_PMU	Slot:2	3.3V	ON	3.3V	TBD	TBD
VCC_SYSIN	RK809_LDO7	0.4A	VCCA_1V8	Slot:2	1.8V	ON	1.8V	TBD	TBD
VCC_SYSIN	RK809_LDO8	0.4A	VCCA1V8_PMU	Slot:2	1.8V	ON	1.8V	TBD	TBD
	RK809_LDO9	0.4A	VCCA1V8_IMAGE_P (VCC1V8_DOVDD_DVP)	N/A	0V	OFF	1.8V (Camera)	TBD	TBD
VCC3V3_SYS	RK809_SW2	2.1A	VCC3V3_SD	Slot:4	3.3V	ON	3.3V	TBD	TBD
VCC_SYSIN	RK809_SW1 RK809_BUCK5	2.1A 2.5A	N/A VCC_1V8	Slot:2	1.8V	ON	1.8V	TBD	TBD
	RK809_RESETn			Slot:4+5					
VCC_SYSIN	EXT BUCK	5.0A	VDD_CPU	Slot:2A	1.025V	ON	DVFS	TBD	TBD
VCC_SYSIN	EXT LDO	0.3A	VCC1V8_DDR	Slot:2A	1.8V	ON	1.8V	TBD	TBD
VCC_SYSIN	EXT BUCK	2.0A	VCC0V6_DDR	Slot:2A	0.6V	ON	0.6V	TBD	TBD
VCC3V3_SYS	EXT MOS		VCC_3V3	Slot:4	3.3V	ON	3.3V	TBD	TBD
VCC_SYSIN	EXT LDO or BUCK	0.5A /1.0A	VDD1V2_DVDD_DVP	N/A	0V	OFF	1.2V (Camera)	TBD	TBD
VCC_SYSIN	EX BOOST	1.5A	VCC_5V0	N/A	0V	OFF	5.0V	TBD	TBD

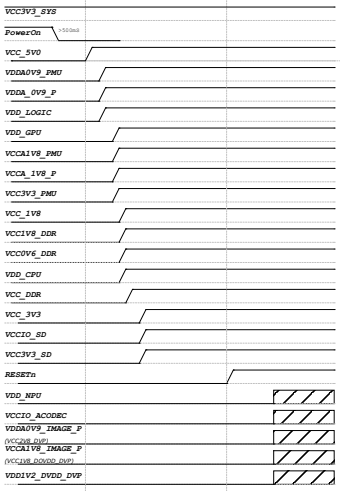
Power Diagram for 1-Cell Quick Charger use BQ25890/SY6970



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Rockchip	Rockchip Electronics Co., Ltd 瑞芯微电子
Project:	RK3566_Tablet_REF_SCH
File:	05.Power Diagram-1Cell QC-RK809
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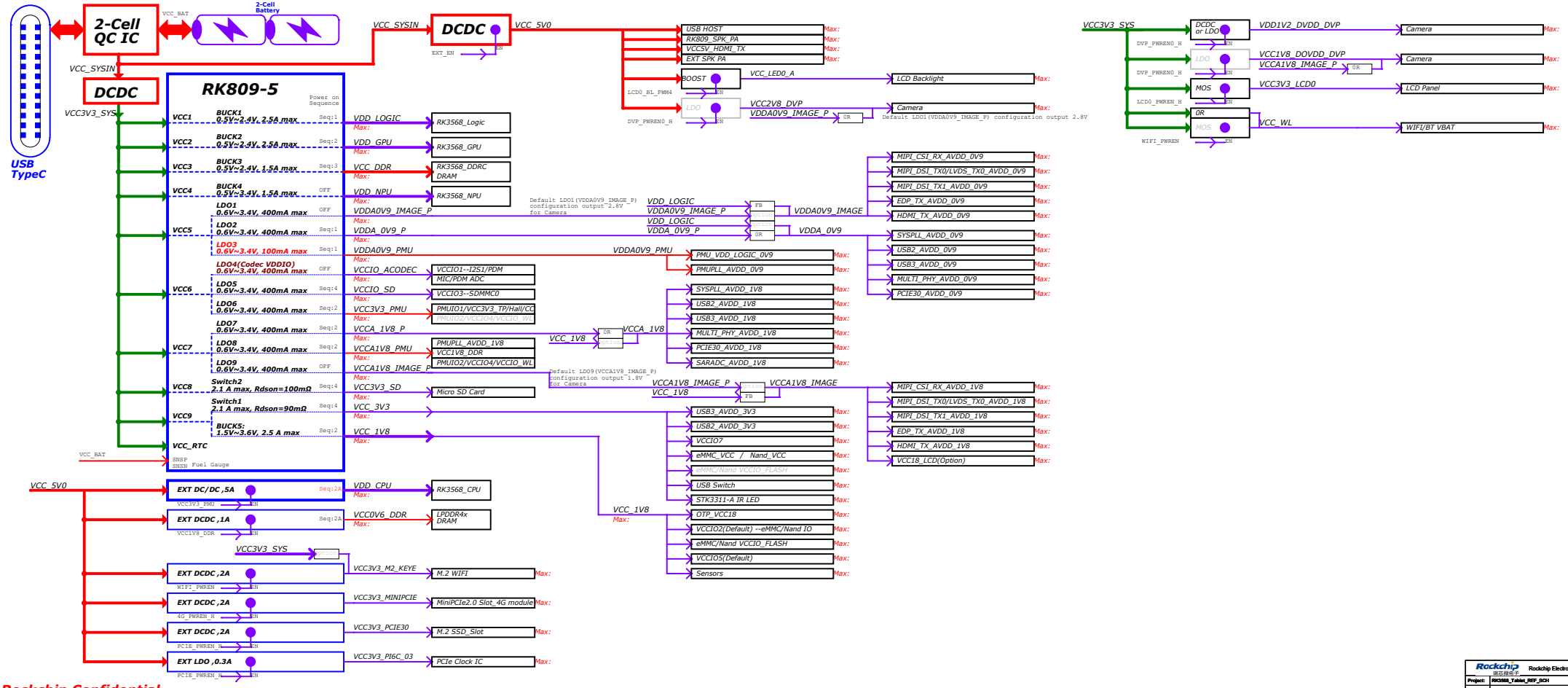
Power Sequence(Use RK809-5)



Power description(Use RK809-5)

Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Work Voltage	Peak Current	Sleep Current
VCC3V3_SYS	RK809_BUCK1	2.5A	VDD_LOGIC	Slot:1	0.9V	ON	0.9V	TBD	TBD
VCC3V3_SYS	RK809_BUCK2	2.5A	VDD_GPU	Slot:2	0.9V	ON	DVFS	TBD	TBD
VCC3V3_SYS	RK809_BUCK3	1.5A	VCC_DDR	Slot:3	ADJ (0.6~0.8V	ON	1.2V (DDR4)	TBD	TBD
VCC3V3_SYS	RK809_BUCK4	1.5A	VDD_NPU	N/A	0V	OFF	DVFS	TBD	TBD
VCC3V3_SYS	RK809_LDO1	0.4A	VDDA0V9_IMAGE_P (VCC2V8_DVP)	N/A	0V	OFF	2.8V (Camera)	TBD	TBD
VCC3V3_SYS	RK809_LDO2	0.4A	VDDA_0V9	Slot:1	0.9V	ON	0.9V	TBD	TBD
VCC3V3_SYS	RK809_LDO3	0.1A	VDDA0V9_PMU	Slot:1	0.9V	ON	0.9V	TBD	TBD
VCC3V3_SYS	RK809_LDO4	0.4A	VCCIO_ACODEC	N/A	0V	OFF	3.3V	TBD	TBD
VCC3V3_SYS	RK809_LDO5	0.4A	VCCIO_SD	Slot:4	3.3V	ON	3.3V or 1.8V	TBD	TBD
VCC3V3_SYS	RK809_LDO6	0.4A	VCC3V3_PMU	Slot:2	3.3V	ON	3.3V	TBD	TBD
VCC3V3_SYS	RK809_LDO7	0.4A	VCCA_1V8	Slot:2	1.8V	ON	1.8V	TBD	TBD
VCC3V3_SYS	RK809_LDO8	0.4A	VCCA1V8_PMU	Slot:2	1.8V	ON	1.8V	TBD	TBD
VCC3V3_SYS	RK809_LDO9	0.4A	VCCA1V8_IMAGE_P (VCC1V8_DOVDD_DVP)	N/A	0V	OFF	1.8V (Camera)	TBD	TBD
VCC3V3_SYS	RK809_SW2 (Default)	2.1A	VCC3V3_SD	Slot:4	3.3V	ON	3.3V	TBD	TBD
VCC3V3_SYS	RK809_SW1 (Default)	2.1A	VCC_3V3	Slot:4	3.3V	ON	3.3V	TBD	TBD
	RK809_BUCK5	2.5A	VCC_1V8	Slot:2	1.8V	ON	1.8V	TBD	TBD
	RK809_RESETn			Slot:4+5					
VCC_SYSIN	EXT BUCK	3A	VCC3V3_SYS	Slot:0	3.3V	ON	3.3V	TBD	TBD
VCC_SYSIN	EXT BUCK	3A	VCC_5V0	Slot:0	5.0V	ON	5.0V	TBD	TBD
VCC_5V0	EXT BUCK	5.0A	VDD_CPU	Slot:2A	1.025V	ON	DVFS	TBD	TBD
VCC_5V0	EXT LDO	0.3A	VCC1V8_DDR	Slot:2A	1.8V	ON	1.8V	TBD	TBD
VCC_5V0	EXT BUCK	2.0A	VCC0V6_DDR	Slot:2A	0.6V	ON	0.6V	TBD	TBD
VCC3V3_SYS	EXT LDO or BUCK	0.5A (1.0A	VDD1V2_DVDD_DVP	N/A	0V	OFF	1.2V (Camera)	TBD	TBD

Power Diagram for 2-Cell Quick Charger use BQ25703/SC8886



Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd
Project:	RK3566_Tablet_REF_SCH
File:	08.Power Diagram-2Cell QC+RK809
Date: Tuesday, July 20, 2021	Rev: V1.0
Designed by: Zhangjh	Reviewed by: Default
	Sheet: 7 of 61

IO Power Domain Map

If IO domain power voltage is adjusted, the software DTS configuration must be updated synchronously, otherwise the IO may be damaged!

IO Domain	Pin Num	Support IO Voltage		Notes	Default IO Domain Voltage		
		3.3V	1.8V		Supply Power Net Name	Power Source	Voltage
PMUIO0 (PMUPLL_AVDD_1V8)	Pin Y21	✗	✓	PMUIO0 are fixed 1.8V level mode, which cannot be configured.	VCCA1V8_PMU	VCCA1V8_PMU	1.8V
PMUIO1	Pin Y20	✓	✗	PMUIO1 are fixed 3.3V level mode, which cannot be configured.	VCC3V3_PMU	VCC3V3_PMU	3.3V
PMUIO2	Pin W19	✓	✓	PMUIO2 supports 1.8V or 3.3V level mode Support configurable but require that their hardware power supply voltages must be consistent with the software configuration correspondingly.[2]	PMUIO2	VCCA1V8_PMU	1.8V
VCCIO1	Pin H17	✓	✓	VCCIO1 supports 1.8V or 3.3V level mode Support configurable but require that their hardware power supply voltages must be consistent with the software configuration correspondingly.[2]	VCCIO_ACODEC	VCCIO_ACODEC	3.3V
VCCIO2	Pin H18	✓	✓	VCCIO2 supports 1.8V or 3.3V level mode Default is configured by hardware,namely PIN "FLASH_VOL_SEL" state determines which mode to work in.[1][2]	VCCIO_FLASH	VCC_1V8	1.8V
VCCIO3	Pin L22	✓	✓	VCCIO3 supports 1.8V or 3.3V level mode Support configurable but require that their hardware power supply voltages must be consistent with the software configuration correspondingly.[2][3]	VCCIO_SD	VCCIO_SD	SD2.0 3.3V SD3.0 1.8V
VCCIO4	Pin J21	✓	✓	VCCIO4 supports 1.8V or 3.3V level mode Support configurable but require that their hardware power supply voltages must be consistent with the software configuration correspondingly.[2]	VCCIO_WL	VCCA1V8_PMU	1.8V
VCCIO5	Pin V10 Pin V11	✓	✓	VCCIO5 supports 1.8V or 3.3V level mode Support configurable but require that their hardware power supply voltages must be consistent with the software configuration correspondingly.[2]	VCC_1V8	VCC_1V8	1.8V
VCCIO6	Pin R9 Pin U9	✓	✓	VCCIO6 supports 1.8V or 3.3V level mode Support configurable but require that their hardware power supply voltages must be consistent with the software configuration correspondingly.[2]	VCC1V8_DOVDD_DVP	VCC1V8_DOVDD_DVP	1.8V
VCCIO7	Pin V12	✓	✓	VCCIO7 supports 1.8V or 3.3V level mode Support configurable but require that their hardware power supply voltages must be consistent with the software configuration correspondingly.[2]	VCC_3V3	VCC_3V3	3.3V

Notes

According to the WiFi module selected:
Option VCC3V3_PMU

If a board needs to be compatible with two voltage choices, recommended to enable BOM ID

For example, the VCCIO4 hardware has been modified to 3.3V power supply, and the corresponding DTS must be modified to 3.3V configuration, otherwise the IO of VCCIO4 will be damaged.

If a board needs to be compatible with two voltage choices, recommended to enable BOM ID

Notes

[1]:When VCCIO2 voltage is connected to 1.8V, FLASH_VOL_SEL must be high
When VCCIO2 voltage is connected to 3.3V, FLASH_VOL_SEL must be low
If VCCIO2 power supply voltage and FLASH_VOL_SEL fails to meet the above relationship, its function will be abnormally(for example, it cannot be started normally) or IO will be damaged.

[2]:When the IO domain power supply voltage is 1.8V, the IO domain voltage configuration in DTS must be set to 1.8V mode.
If it is misconfigured to 3.3V mode, the IO function of this power domain will be abnormally;
When the IO domain power supply voltage is 3.3V, the IO domain voltage configuration in DTS must be set to 3.3V mode.
If it is misconfigured to 1.8V mode, the IO in this power domain will be in overvoltage state, and the IO will be damaged after long-term operation.

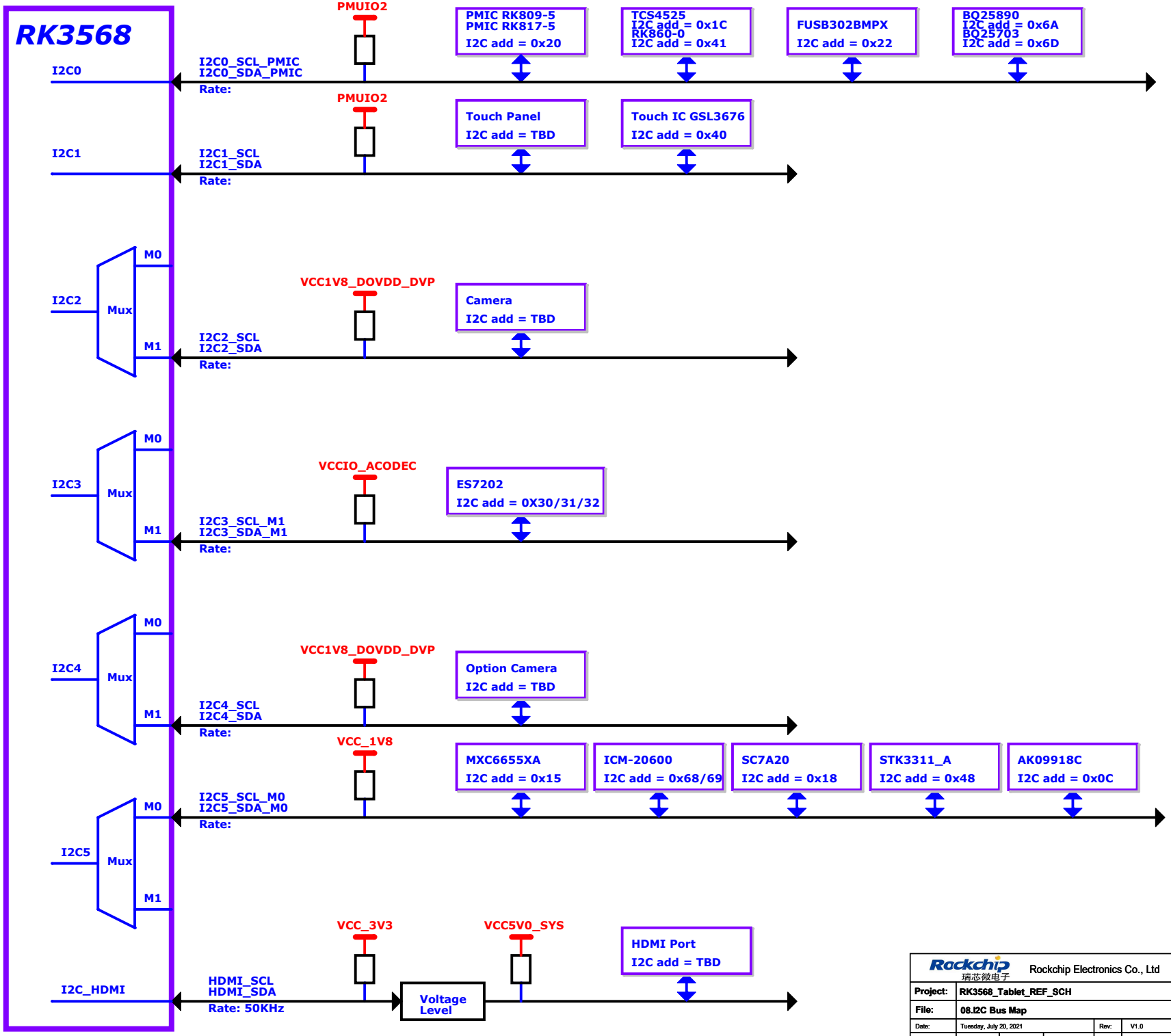
[3]:When VCCIO3 IO domain is assigned as SD card function,:
If SD3.0 mode is to be supported, VCCIO3 power supply voltage must be support configurable, 3.3V in SD2.0 mode and 1.8V in SD3.0 mode.
If only SD2.0 mode is supported (SD3.0 card only works in SD2.0 mode), VCCIO3 only needs fixed power supply of 3.3V.
When VCCIO3 IO domain is assigned as other function,:
Such as uart5 and uart6, then note [2] should be followed

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Project:	RK3568_Tablet_REF_SCH				
File:	07.IO Power Domain Map				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	8 of 61

Default I2C Map

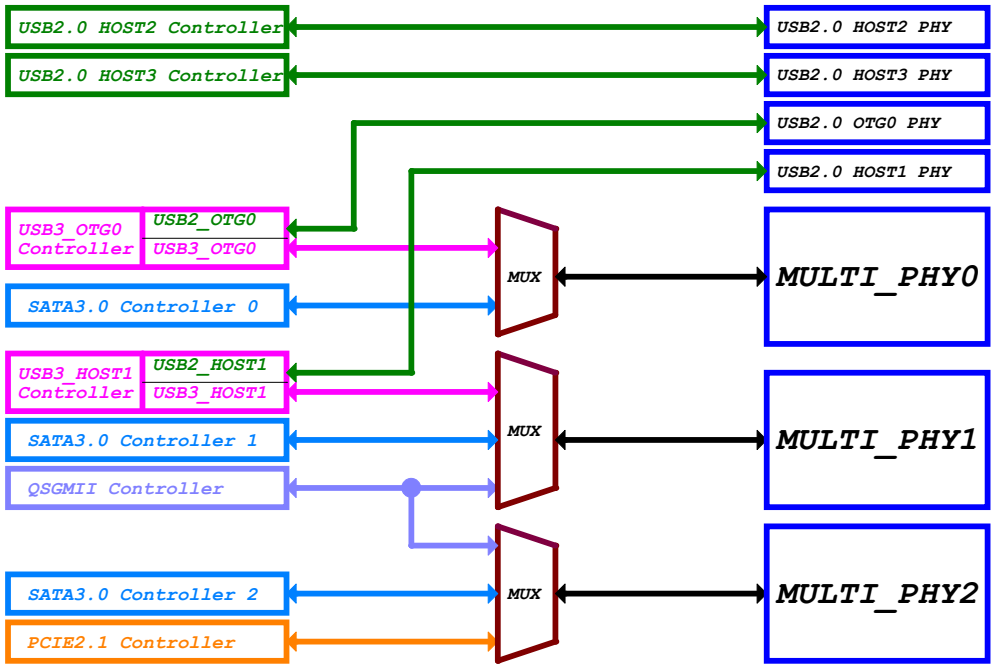
Note:
M0 M1 M2 indicates that the same function is multiplexed to different IO. When selecting, only M0 or M1 or M2 can be selected
eg:
Not supported I2C1_SCL_M0 and I2C1_SDA_M1 combination



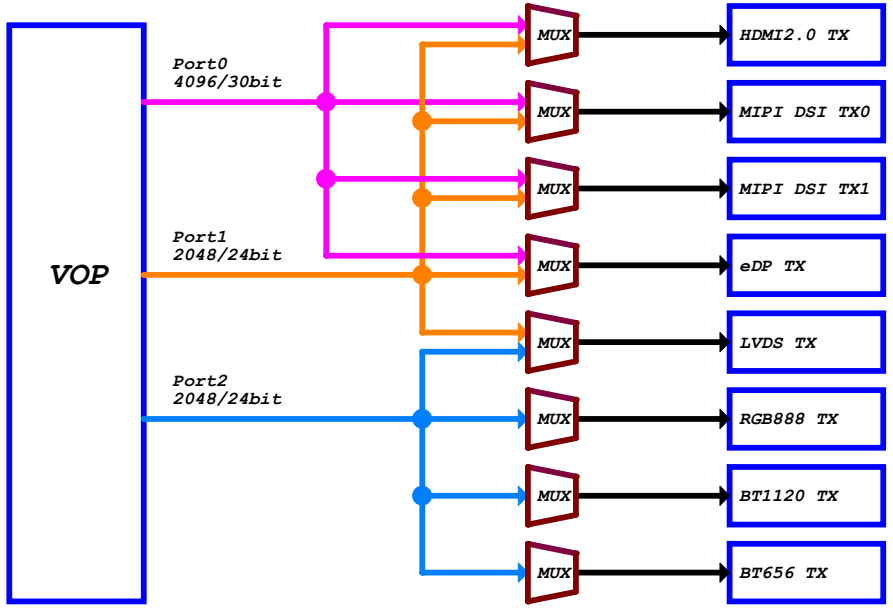
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Rockchip 瑞芯微电子 Rockchip Electronics Co., Ltd					
Project:	RK3568_Tablet_REF_SCH				
File:	08.I2C Bus Map				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	9 of 61

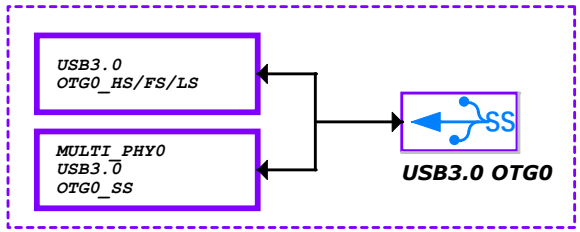
MULTI_PHY0/1/2 Path Map



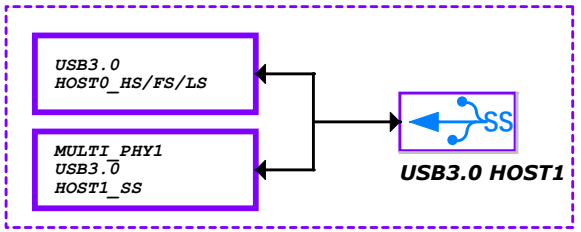
VOP Path Map



USB3.0 OTG0



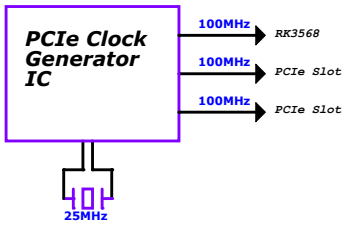
USB3.0 HOST1



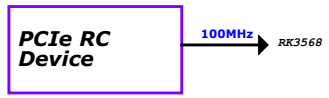
PCIe3.0 PHY

Option1	PCIe3.0 x2Lane	PCIE30_REFCLK (RC/EP:input)	PCIE30_TX0 PCIE30_RX0 PCIE30_TX1 PCIE30_RX1	PCIE30X2_CLKREQn PCIE30X2_WAKEn PCIE30X2_PERSTn PCIE30X2_BUTTONRSTn	RC or EP
Option2	PCIe3.0 x1Lane + PCIe3.0 x1Lane	PCIE30_REFCLK (RC:input)	PCIE30_TX0 PCIE30_RX0	PCIE30X2_CLKREQn PCIE30X2_WAKEn PCIE30X2_PERSTn PCIE30X2_BUTTONRSTn	Only RC
			PCIE30_TX1 PCIE30_RX1	PCIE30X1_CLKREQn PCIE30X1_WAKEn PCIE30X1_PERSTn PCIE30X1_BUTTONRSTn	Only RC

PCIe3.0 REFCLK-RC Mode



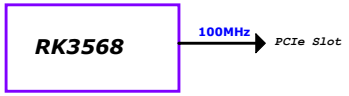
PCIe3.0 REFCLK-EP Mode



PCIe2.1 PHY

MULTI_PHY2	PCIe2.1 x1Lane	PCIE20_REFCLK (RC:output)	PCIE20_TX PCIE20_RX	PCIE20_CLKREQn PCIE20_WAKEn PCIE20_PERSTn PCIE20_BUTTONRSTn	Only RC

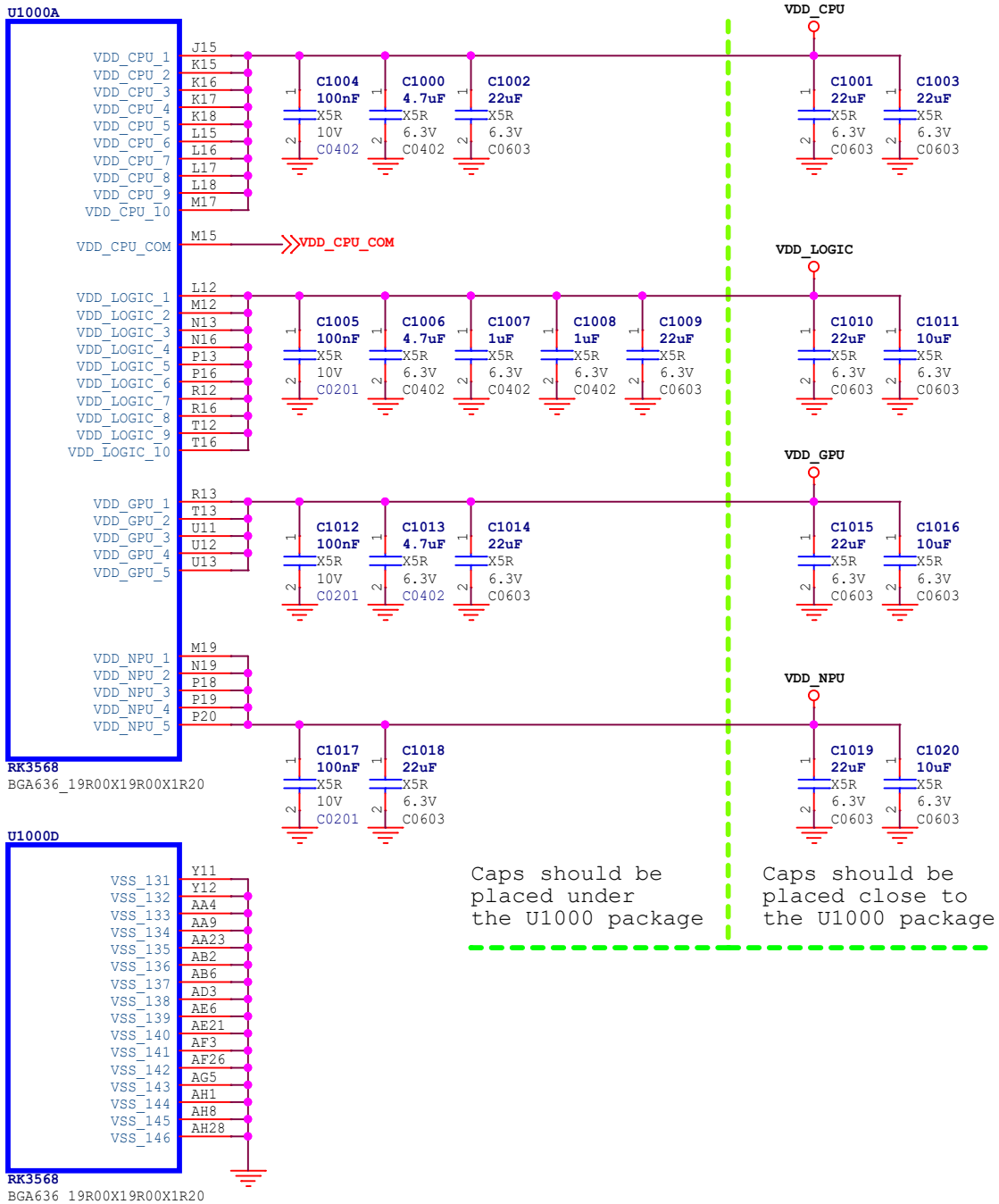
PCIe2.1 REFCLK-RC Mode

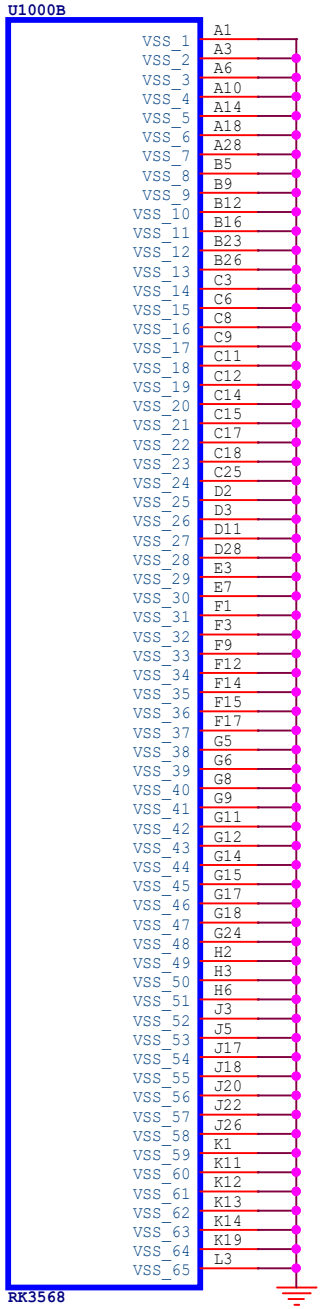


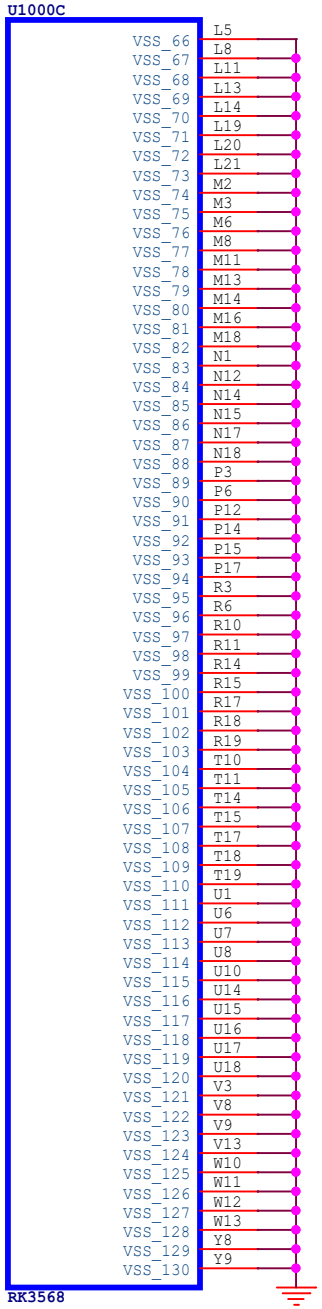
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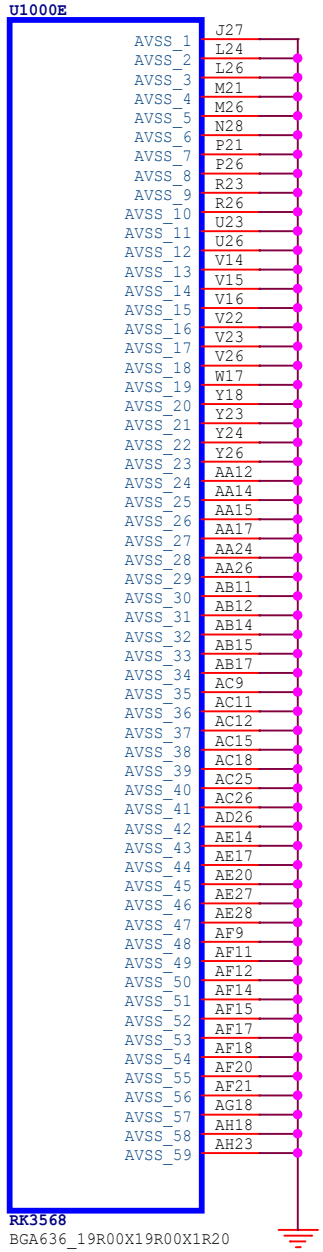
Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd
Project:	RK3568_Tablet_REF_SCH
File:	09.PCIE30/MULTI_PHY/VOP Fun Map
Date:	Tuesday, July 20, 2021
Rev:	V1.0
Designed by:	Zhangtz
Reviewed by:	Default
Sheet:	10 of 61

RK3568_ABCDE (Power&Gnd)





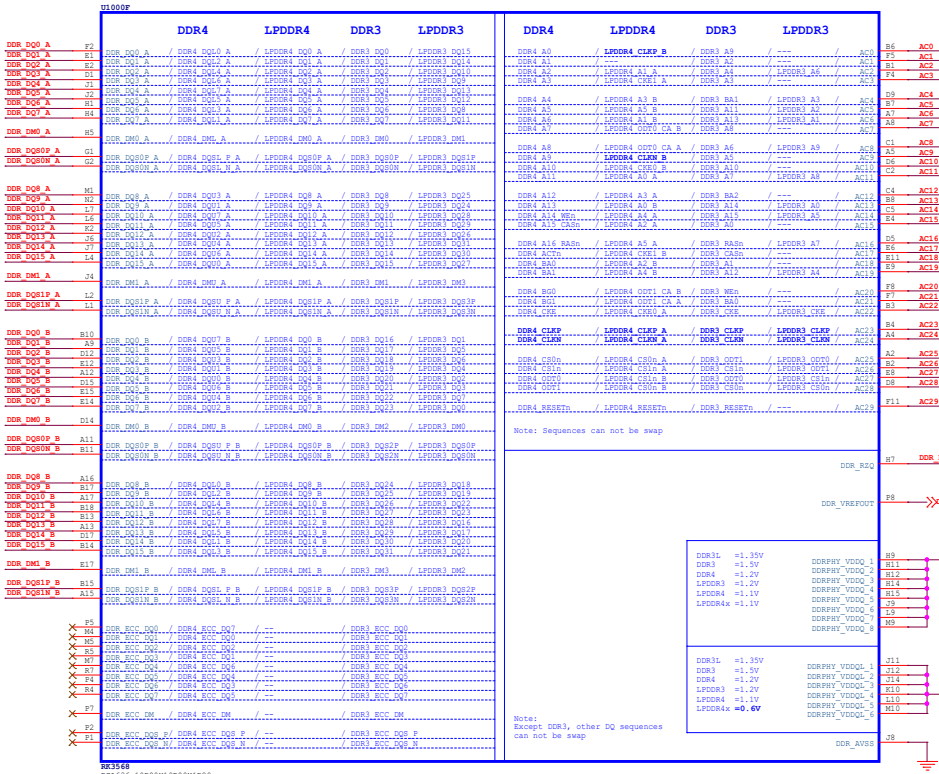




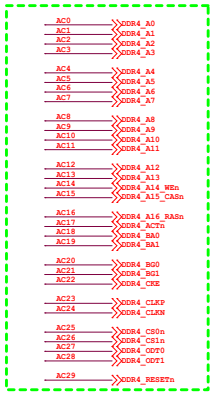
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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	10.RK3568_Power/GND				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	11 of 61

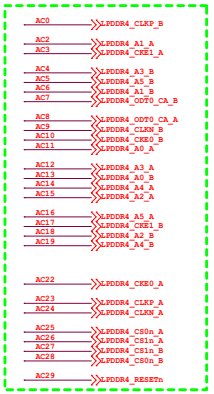
RK3568_F (DDR PHY)



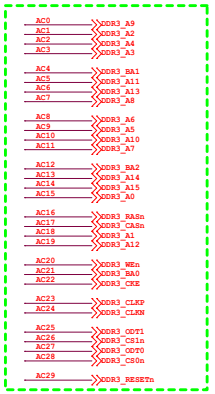
Default DDR4 AC



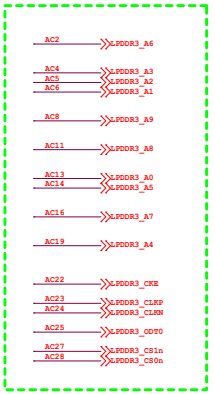
LPDDR4/4x AC

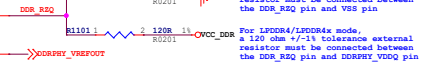


DDR3 AC

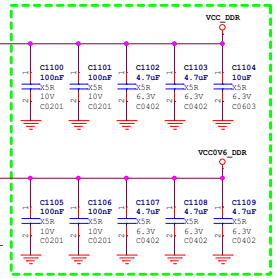


LPDDR3 AC





Default LPDDR4x

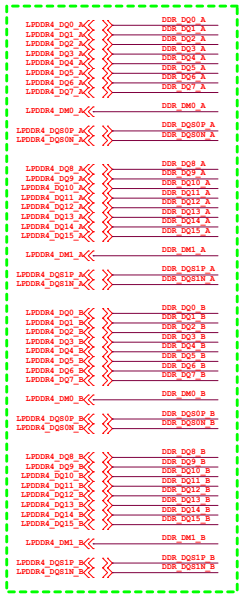


Caps should be placed under the U1000 package

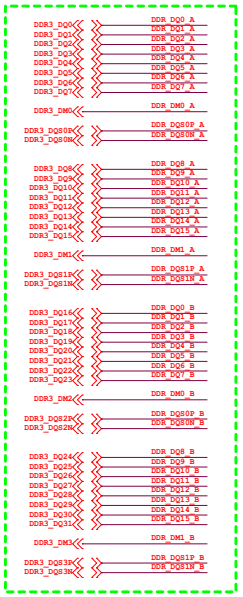
Default DDR4 DQ



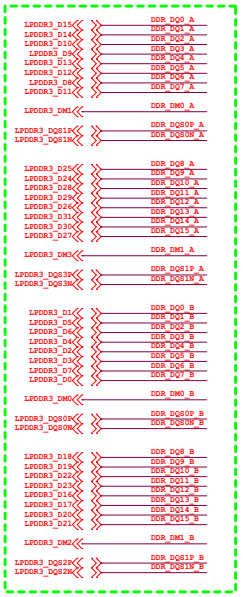
LPDDR4/4x DQ



DDR3 DQ



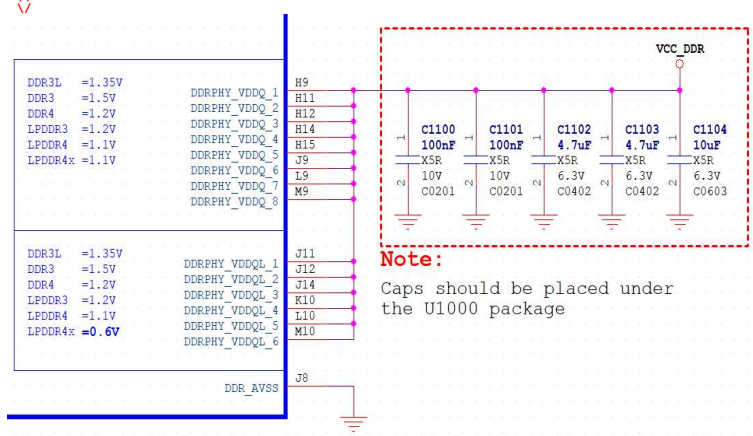
LPDDR3 DQ



Note:

DDR3/DDR3L/DDR4/LPDDR4(only) mode:
DDRPHY_VDDQ and DDRPHY_VQQL Power supply can be combined

For example



Note:

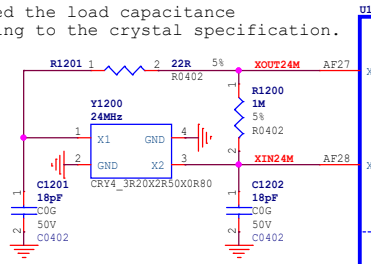
Caps should be placed under the U1000 package

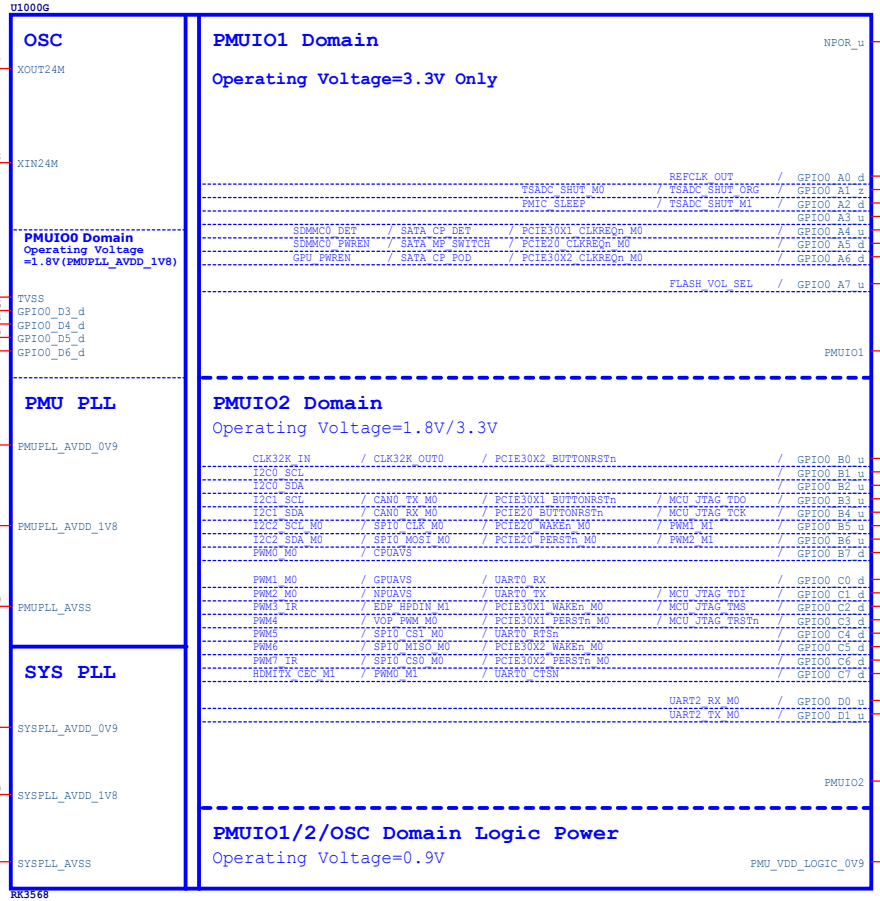
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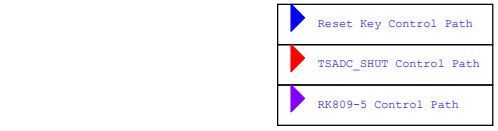
Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd
Project:	RK3568_Tablet_REF_SCH
File:	11.RK3568_DDR PHY
Date:	Tuesday, July 20, 2021
Rev:	V1.0
Designed by:	Zhangtz
Reviewed by:	Default
Sheet:	12 of 61

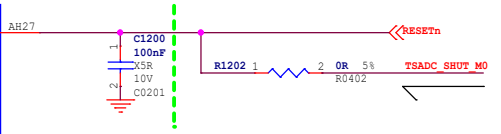
RK3568_G(OSC/PLL/PMUIO1/2)

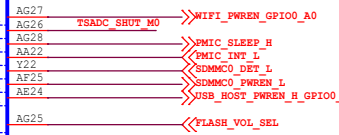
Note:
Adjusted the load capacitance according to the crystal specification.

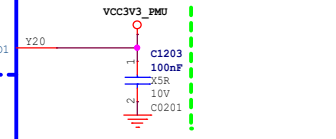


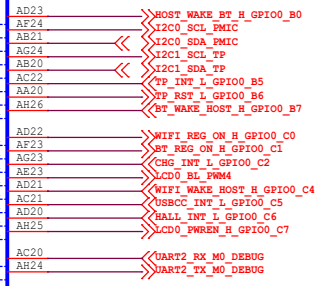


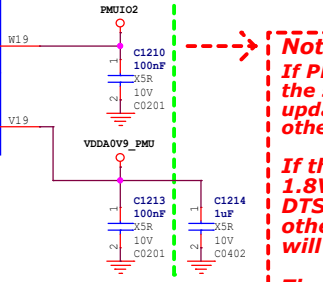


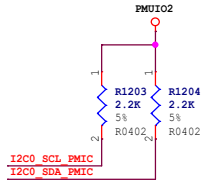


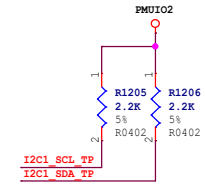


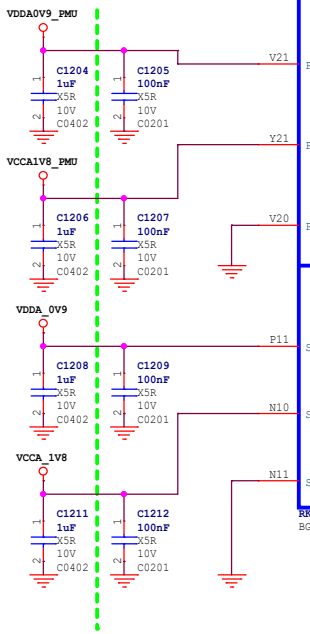












Note:
If PMUIO2 domain power voltage is adjusted, the software DTS configuration must be updated synchronously, otherwise the IO may be damaged!

If the PMUIO2 hardware has been modified to 1.8V power supply, and the corresponding DTS must be modified to 1.8V configuration, otherwise the IO function of PMUIO2 will be abnormally.

The PMUIO2 hardware has been modified to 3.3V power supply, if the software DTS configuration is still 1.8V configuration, the IO of PMUIO2 will be damaged!

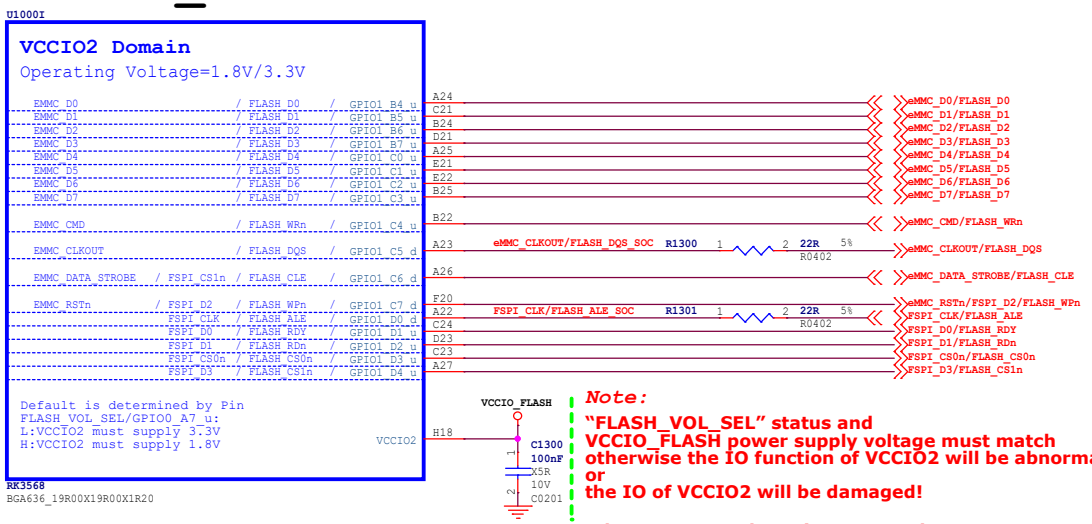
If a board needs to be compatible with two voltage choices, recommended to enable BOM_ID

Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	12.RK3568_OSC/PLL/PMUIO				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	13 of 61

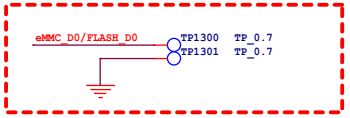
RK3568_I(VCCIO2 Domain)



Note:

"FLASH_VOL_SEL" status and
VCCIO_FLASH power supply voltage must match
otherwise the IO function of VCCIO2 will be abnormally
or
the IO of VCCIO2 will be damaged!

When VCCIO2 voltage is connected to 1.8V, FLASH_VOL_SEL must be high
When VCCIO2 voltage is connected to 3.3V, FLASH_VOL_SEL must be low
If VCCIO2 power supply voltage and FLASH_VOL_SEL fails to meet the above relationship,
its function will be abnormally(for example, it cannot be started normally) or IO will be damaged.

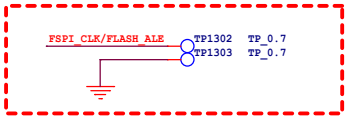


Note:

For eMMC or Nand Flash:
If eMMC_D0/FLASH_D0=0V at after power on and reset,
then system will enter into Maskrom mode.

Layout note:

Test point must be placed on the line, and no branch can be added



Note:

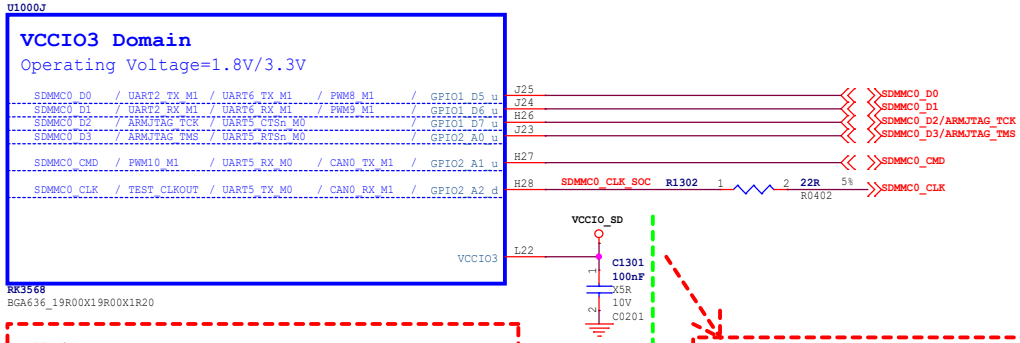
For SPI Flash:
If FSPI_CLK=0V at after power on and reset,
then system will enter into Maskrom mode.

Note:

Reserve TestPoint for put the system into Maskrom mode to update the firmware
When writing mismatched firmware or other conditions result in boot failure,
use this test point

Except in this case, please use Recovery Key
Put the system into loader mode to update the firmware

RK3568_J(VCCIO3 Domain)



Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

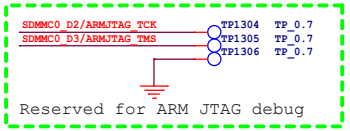
Note:

If VCCIO3 domain power voltage is adjusted, the software DTS configuration must be updated synchronously, otherwise the IO may be damaged!

If the VCCIO3 hardware has been modified to 1.8V power supply, and the corresponding DTS must be modified to 1.8V configuration, otherwise the IO function of VCCIO3 will be abnormally.

The VCCIO3 hardware has been modified to 3.3V power supply, if the software DTS configuration is still 1.8V configuration, the IO of VCCIO3 will be damaged!

If a board needs to be compatible with two voltage choices, recommended to enable BOM_ID

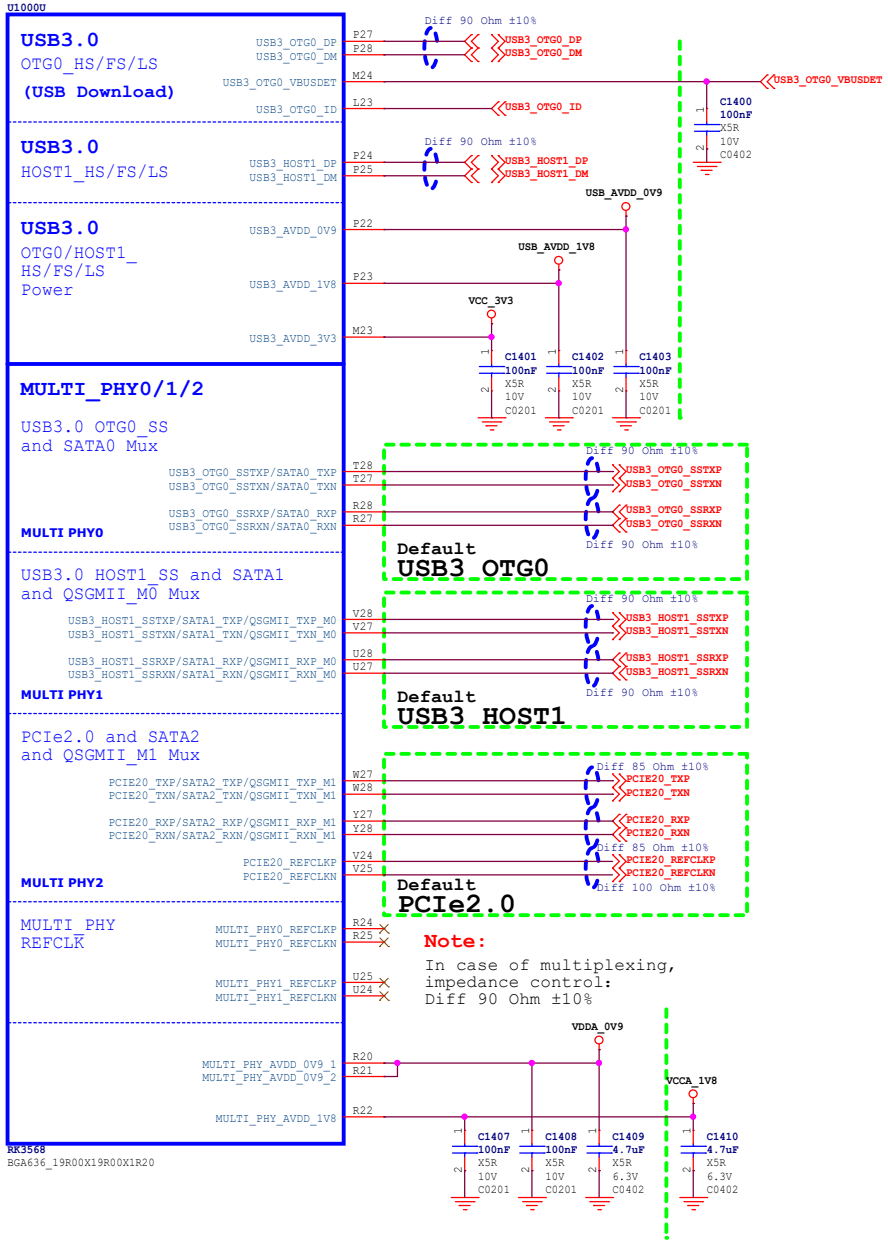


Reserved for ARM JTAG debug

Rockchip Confidential

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	13.RK3568_Flash/SD Controller				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	14 of 61

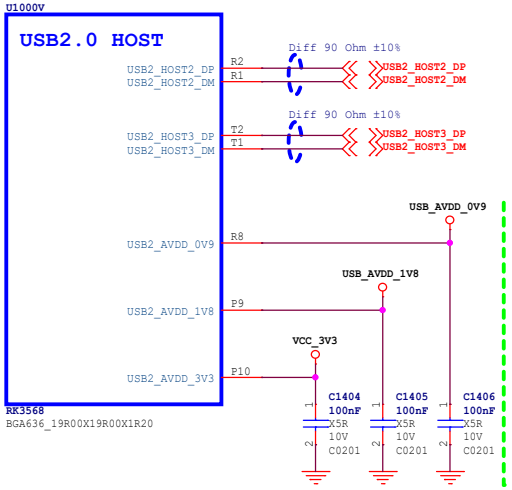
RK3568_U(USB3.0/SATA/QSGMII/PCIe2.0 x1)



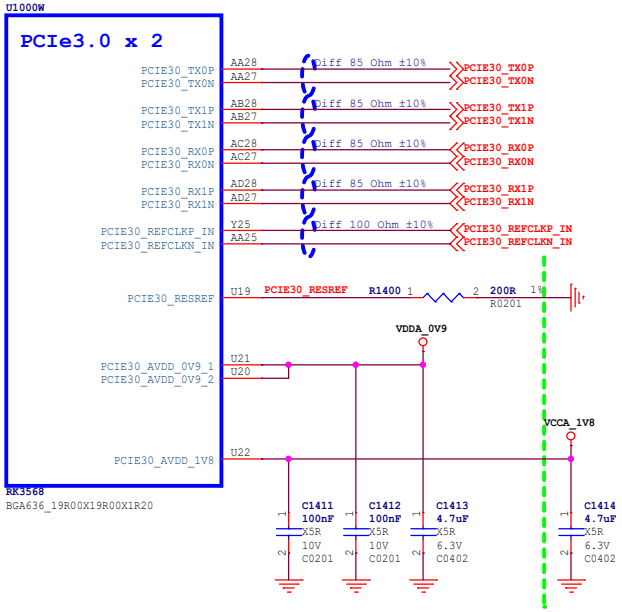
Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

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RK3568_V(USB2.0 HOST)

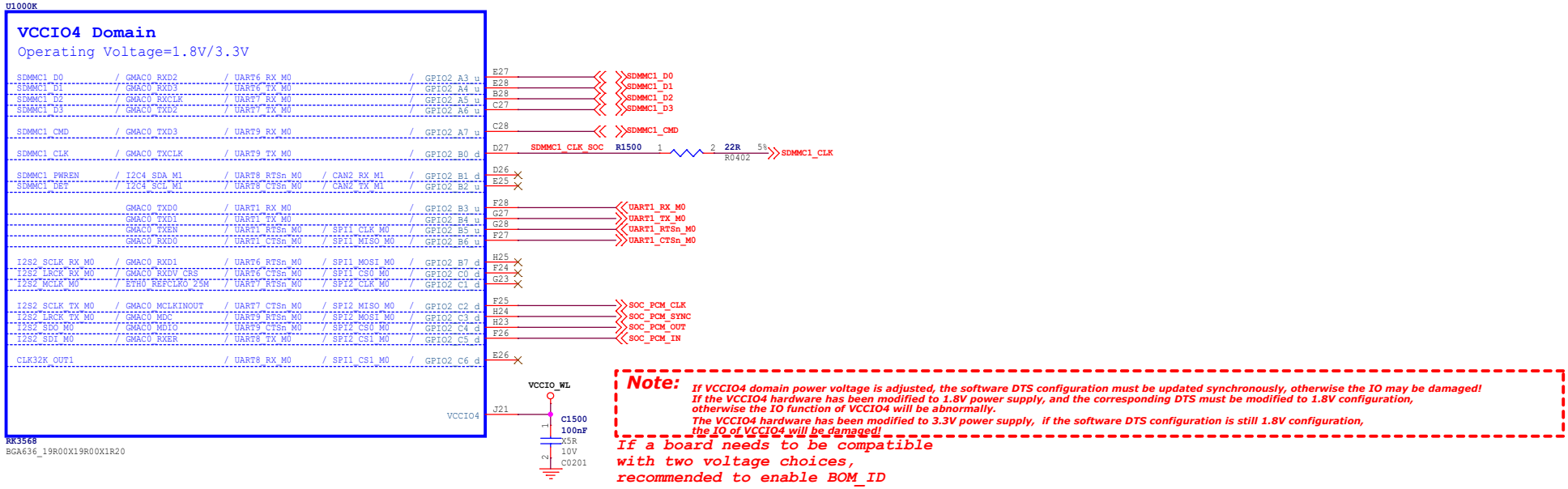


RK3568_W(PCIe3.0 x2)



Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	14.RK3568_USB/PCIe/SATA PHY				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	15 of 61

RK3568_K(VCCIO4 Domain)

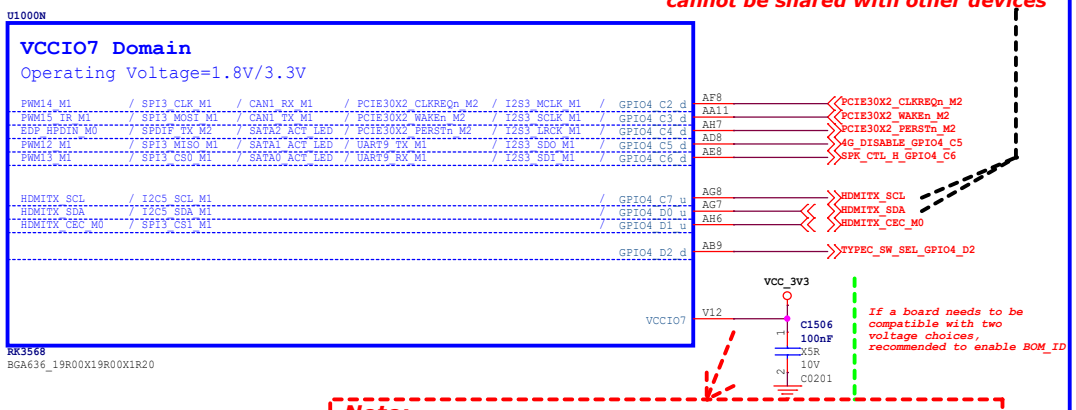


Note: *If VCCIO4 domain power voltage is adjusted, the software DTS configuration must be updated synchronously, otherwise the IO may be damaged!*
If the VCCIO4 hardware has been modified to 1.8V power supply, and the corresponding DTS must be modified to 1.8V configuration, otherwise the IO function of VCCIO4 will be abnormally.
The VCCIO4 hardware has been modified to 3.3V power supply, if the software DTS configuration is still 1.8V configuration, the IO of VCCIO4 will be damaged!

If a board needs to be compatible with two voltage choices, recommended to enable BOM_ID

RK3568_N(VCCIO7 Domain)

Note: *When use HDMI, HDMITX_SCL/SDA cannot be shared with other devices*



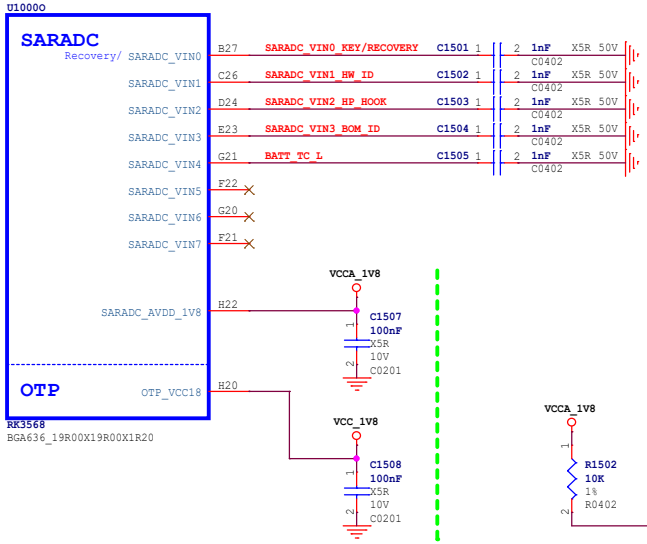
If a board needs to be compatible with two voltage choices, recommended to enable BOM_ID

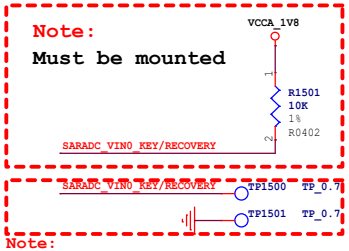
Note:
If VCCIO7 domain power voltage is adjusted, the software DTS configuration must be updated synchronously, otherwise the IO may be damaged!

If the VCCIO7 hardware has been modified to 1.8V power supply, and the corresponding DTS must be modified to 1.8V configuration, otherwise the IO function of VCCIO7 will be abnormally.

The VCCIO7 hardware has been modified to 3.3V power supply, if the software DTS configuration is still 1.8V configuration, the IO of VCCIO7 will be damaged!

RK3568_O(SARADC/OTP)

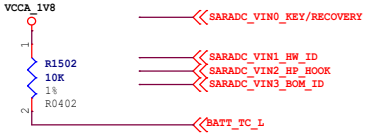




Note:
If there is no Key requirement, two test points must be reserved to facilitate firmware update

It is suggested to reserve a Key to facilitate the development debug

If SARADC_VIN0=0V at after power on and reset, then system will enter into loader mode.

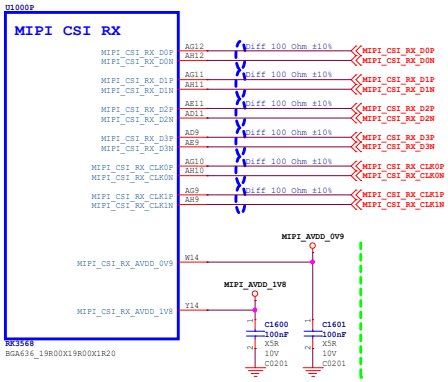


Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	15.RK3568_SARADC/GPIO				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	16 of 61

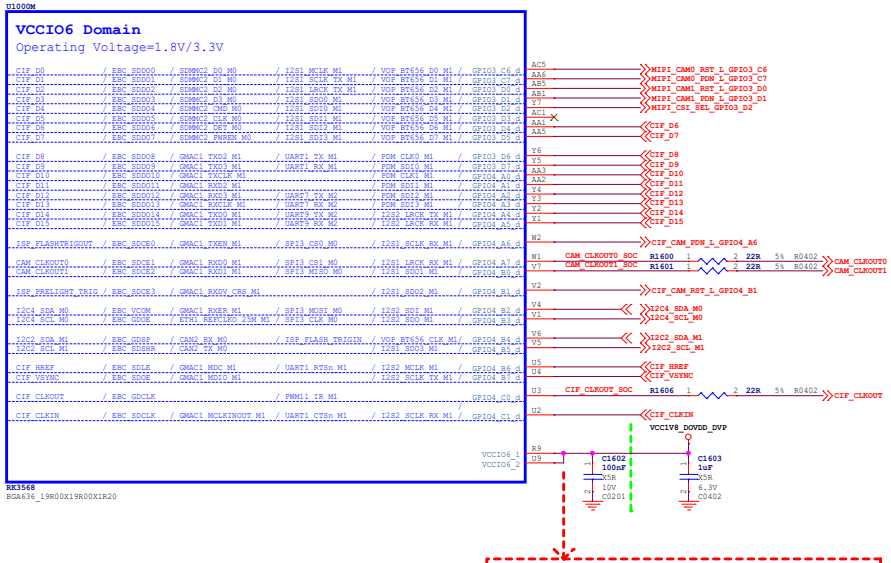
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RK3568_P(MIPI_CSI_RX)



Option1	Sensor1 x4Lane	MIPI_CSI_RX_D0-3 MIPI_CSI_RX_CLK0
Option2	Sensor1 x2Lane + Sensor2 x2Lane	MIPI_CSI_RX_D0-1 MIPI_CSI_RX_CLK0
		MIPI_CSI_RX_D2-3 MIPI_CSI_RX_CLK1

RK3568_M(VCCIO6 Domain)



Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

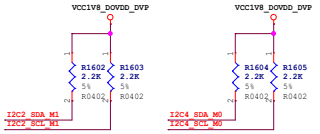
Note:
If VCCIO6 domain power voltage is adjusted,
the software DTS configuration must be
updated synchronously,
otherwise the IO may be damaged!

If the VCCIO6 hardware has been modified to
1.8V power supply, and the corresponding
DTS must be modified to 1.8V configuration,
otherwise the IO function of VCCIO6
will be abnormally.

The VCCIO6 hardware has been modified to
3.3V power supply, if the software DTS
configuration is still 1.8V configuration,
the IO of VCCIO6 will be damaged!

Note:
Camera MCLK can select the following clock:
1:CAM_CLKOUT0
2:CAM_CLKOUT1
3:CIF_CLKOUT
4:REFCLK_OUT(24MHz)

Attention to the voltage matching



Note:
According to the actual choice of mounted
Cannot be mounted at the same time

Default:1.8V
Select the voltage according to the application

If the IO domain is to be used as FEPHY,
since some FEPHY only support 3.3V IO,
it is recommended to reallocate GPIO
to reduce the cost of level conversion

If a board needs to be compatible
with two voltage choices,
recommended to enable BOM_ID

Mode	16bit	12bit	10bit	8bit
CIF_D0	D0	--	--	--
CIF_D1	D1	--	--	--
CIF_D2	D2	--	--	--
CIF_D3	D3	--	--	--
CIF_D4	D4	D0	--	--
CIF_D5	D5	D1	--	--
CIF_D6	D6	D2	D0	--
CIF_D7	D7	D3	D1	--
CIF_D8	D8	D4	D2	D0
CIF_D9	D9	D5	D3	D1
CIF_D10	D10	D6	D4	D2
CIF_D11	D11	D7	D5	D3
CIF_D12	D12	D8	D6	D4
CIF_D13	D13	D9	D7	D5
CIF_D14	D14	D10	D8	D6
CIF_D15	D15	D11	D9	D7

Support BT601 YCbCr 422 8bit input
Support BT656 YCbCr 422 8bit input
Support RAW 8/10/12bit input
Support BT1120 YCbCr 422 8/10/12/16bit input, single/dual-edge sampling
Support 2/4 mixed BT656/BT1120 YCbCr 422 8bit input

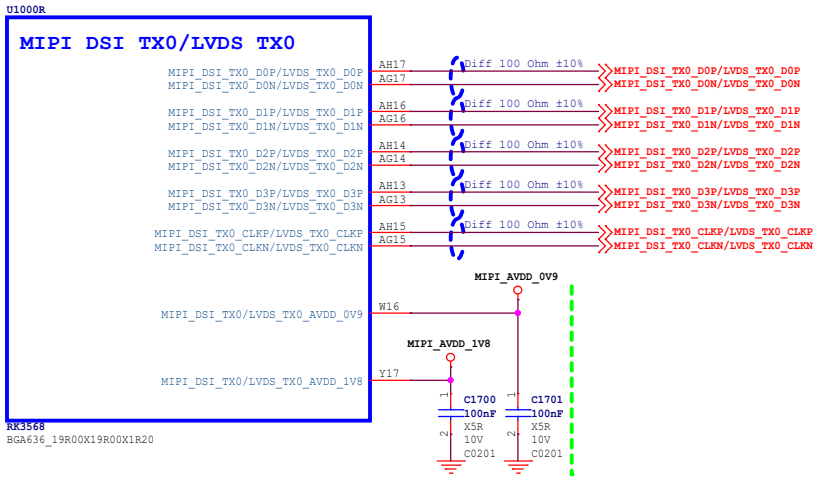
BT1120 16bit Mode:
Default: D0-D7 <--> Y0-Y7 , D8-D15 <--> C0-C7
Swap ON: D0-D7 <--> C0-C7 , D8-D15 <--> Y0-Y7

GMAC	Direction	GEPHY	GMAC	Direction	FEPHY
GMACx_TXD0	------->	PHYx_TXD0	GMACx_TXD0	------->	PHYx_TXD0
GMACx_TXD1	------->	PHYx_TXD1	GMACx_TXD1	------->	PHYx_TXD1
GMACx_TXD2	------->	PHYx_TXD2			
GMACx_TXD3	------->	PHYx_TXD3			
GMACx_TXEN	------->	PHYx_TXEN	GMACx_TXEN	------->	PHYx_TXEN
GMACx_TXCLK	------->	PHYx_TXCLK			
GMACx_RXD0	<-------	PHYx_RXD0	GMACx_RXD0	<-------	PHYx_RXD0
GMACx_RXD1	<-------	PHYx_RXD1	GMACx_RXD1	<-------	PHYx_RXD1
GMACx_RXD2	<-------	PHYx_RXD2			
GMACx_RXD3	<-------	PHYx_RXD3			
GMACx_RXDV	<-------	PHYx_RXDV	GMACx_RXDV	<-------	PHYx_CRS_DV
GMACx_RXCLK	<-------	PHYx_RXCLK			
GMACx_RXER			GMACx_RXER	<-------	PHYx_RXER
GMACx_MDC	------->	PHYx_MDC	GMACx_MDC	------->	PHYx_MDC
GMACx_MDIO	<------->	PHYx_MDIO	GMACx_MDIO	<------->	PHYx_MDIO
ETHx_REFCLKO_25M	------->	PHYx OSC	ETHx_REFCLKO_25M	------->	PHYx OSC
GMACx_MCLKINOUT	<-------	PHYx_CLKOUT125(Option)	GMACx_MCLKINOUT	<------->	PHYx_TXC
GPIO	------->	PHYx_RSTn	GPIO	------->	PHYx_RSTn
GPIO	<-------	PHYx_INT/PMEB	GPIO	<-------	PHYx_INT/PMEB

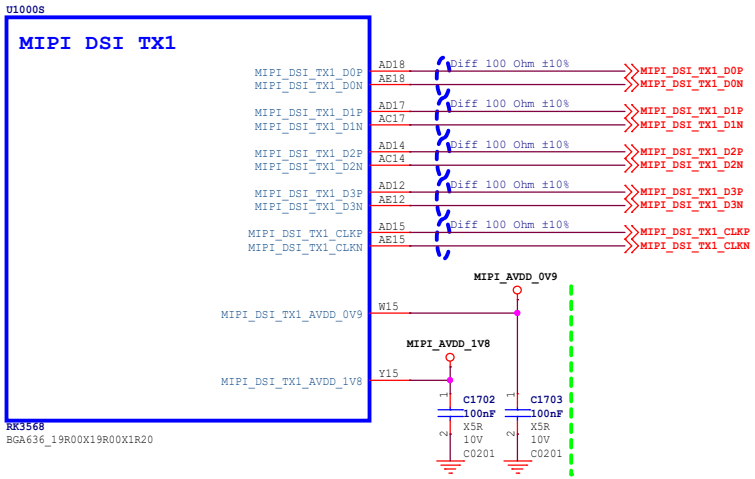
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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd
Project:	RK3568_Tablet_REF_SCH
File:	16.RK3568_VI Interface
Date:	Tuesday, July 20, 2021
Rev:	V1.0
Designed by:	Zhangtz
Reviewed by:	Default
Sheet:	17 of 61

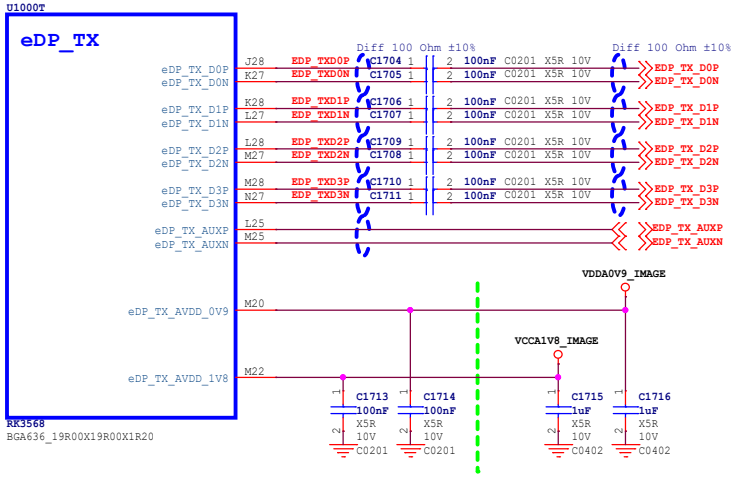
RK3568_R(MIPI_DSI_TX0/LVDS_TX0)



RK3568_S(MIPI_DSI_TX1)



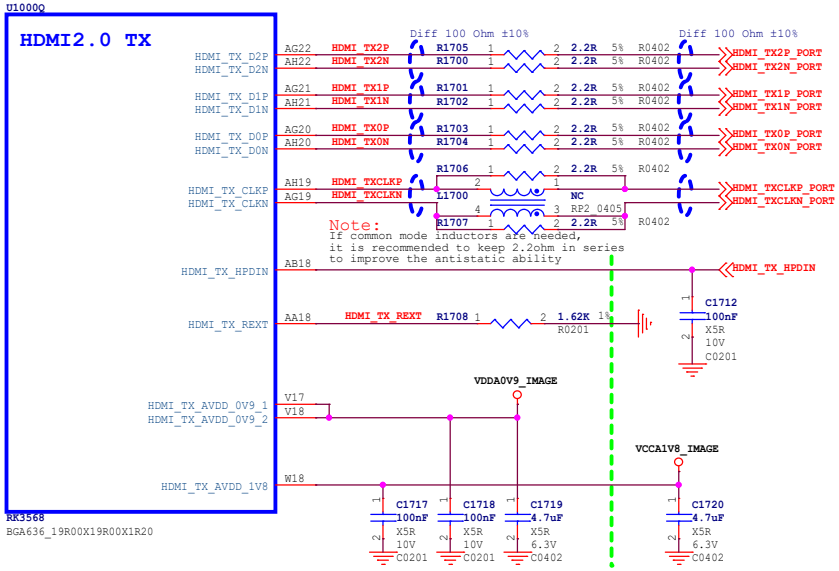
RK3568_T(eDP TX)



Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

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RK3568_Q(HDMI2.0 TX)



Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	17.RK3568_VO Interface_1				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	18 of 61

RK3568_L(VCCIO5 Domain)

U1000L

VCCIO5 Domain

Operating Voltage=1.8V/3.3V

LCDC_D0	/ VOP_BT656_D0_M0	/ SPI0_MISO_M1	/ PCIE20_CLKREQn_M1	/ I2S1_MCLK_M2	/ GPIO2_D0_d
LCDC_D1	/ VOP_BT656_D1_M0	/ SPI0_MOSI_M1	/ PCIE20_WAKEn_M1	/ I2S1_SCLK_TX_M2	/ GPIO2_D1_d
LCDC_D2	/ VOP_BT656_D2_M0	/ SPI0_CS0_M1	/ PCIE30X1_CLKREQn_M1	/ I2S1_LRCK_TX_M2	/ GPIO2_D2_d
LCDC_D3	/ VOP_BT656_D3_M0	/ SPI0_CLK_M1	/ PCIE30X1_WAKEn_M1	/ I2S1_SDI0_M2	/ GPIO2_D3_d
LCDC_D4	/ VOP_BT656_D4_M0	/ SPI2_CS1_M1	/ PCIE30X2_CLKREQn_M1	/ I2S1_SDI1_M2	/ GPIO2_D4_d
LCDC_D5	/ VOP_BT656_D5_M0	/ SPI2_CS0_M1	/ PCIE30X2_WAKEn_M1	/ I2S1_SDI2_M2	/ GPIO2_D5_d
LCDC_D6	/ VOP_BT656_D6_M0	/ SPI2_MOSI_M1	/ PCIE30X2_PERSTn_M1	/ I2S1_SDI3_M2	/ GPIO2_D6_d
LCDC_D7	/ VOP_BT656_D7_M0	/ SPI2_MISO_M1	/ UART8_TX_M1	/ I2S1_SDO0_M2	/ GPIO2_D7_d
LCDC_CLK	/ VOP_BT656_CLK_M0	/ SPI2_CLK_M1	/ UART8_RX_M1	/ I2S1_SDO1_M2	/ GPIO3_A0_d
LCDC_D8	/ VOP_BT1120_D0	/ SPI1_CS0_M1	/ PCIE30X1_PERSTn_M1	/ SDMMC2_D0_M1	/ GPIO3_A1_d
LCDC_D9	/ VOP_BT1120_D1	/ GMAC1_TXD2_M0	/ I2S3_MCLK_M0	/ SDMMC2_D1_M1	/ GPIO3_A2_d
LCDC_D10	/ VOP_BT1120_D2	/ GMAC1_TXD3_M0	/ I2S3_SCLK_M0	/ SDMMC2_D2_M1	/ GPIO3_A3_d
LCDC_D11	/ VOP_BT1120_D3	/ GMAC1_RXD2_M0	/ I2S3_LRCK_M0	/ SDMMC2_D3_M1	/ GPIO3_A4_d
LCDC_D12	/ VOP_BT1120_D4	/ GMAC1_RXD3_M0	/ I2S3_SDO_M0	/ SDMMC2_CMD_M1	/ GPIO3_A5_d
LCDC_D13	/ VOP_BT1120_CLK	/ GMAC1_TXCLK_M0	/ I2S3_SDI_M0	/ SDMMC2_CLK_M1	/ GPIO3_A6_d
LCDC_D14	/ VOP_BT1120_D5	/ GMAC1_RXCLK_M0		/ SDMMC2_DET_M1	/ GPIO3_A7_d
LCDC_D15	/ VOP_BT1120_D6	/ ETH1_REFCLKO_25M_M0		/ SDMMC2_PWREN_M1	/ GPIO3_B0_d
LCDC_D16	/ VOP_BT1120_D7	/ GMAC1_RXD0_M0	/ UART4_RX_M1	/ PWM8_M0	/ GPIO3_B1_d
LCDC_D17	/ VOP_BT1120_D8	/ GMAC1_RXD1_M0	/ UART4_TX_M1	/ PWM9_M0	/ GPIO3_B2_d
LCDC_D18	/ VOP_BT1120_D9	/ GMAC1_RXDV_CRS_M0	/ I2C5_SCL_M0	/ PDM_SDI0_M2	/ GPIO3_B3_d
LCDC_D19	/ VOP_BT1120_D10	/ GMAC1_RXER_M0	/ I2C5_SDA_M0	/ PDM_SDI1_M2	/ GPIO3_B4_d
LCDC_D20	/ VOP_BT1120_D11	/ GMAC1_TXD0_M0	/ I2C3_SCL_M1	/ PWM10_M0	/ GPIO3_B5_d
LCDC_D21	/ VOP_BT1120_D12	/ GMAC1_TXD1_M0	/ I2C3_SDA_M1	/ PWM11_IR_M0	/ GPIO3_B6_d
LCDC_D22	/ PWM12_M0	/ GMAC1_TXEN_M0	/ UART3_TX_M1	/ PDM_SDI2_M2	/ GPIO3_B7_d
LCDC_D23	/ PWM13_M0	/ GMAC1_MCLKINOUT_M0	/ UART3_RX_M1	/ PDM_SDI3_M2	/ GPIO3_C0_d
LCDC_HSYNC	/ VOP_BT1120_D13	/ SPI1_MOSI_M1	/ PCIE20_PERSTn_M1	/ I2S1_SDO2_M2	/ GPIO3_C1_d
LCDC_VSYNC	/ VOP_BT1120_D14	/ SPI1_MISO_M1	/ UART5_TX_M1	/ I2S1_SDO3_M2	/ GPIO3_C2_d
LCDC_DEN	/ VOP_BT1120_D15	/ SPI1_CLK_M1	/ UART5_RX_M1	/ I2S1_SCLK_RX_M2	/ GPIO3_C3_d
PWM14_M0	/ VOP_PWM_M1	/ GMAC1_MDC_M0	/ UART7_TX_M1	/ PDM_CLK1_M2	/ GPIO3_C4_d
PWM15_IR_M0	/ SPDIF_TX_M1	/ GMAC1_MDIO_M0	/ UART7_RX_M1	/ I2S1_LRCK_RX_M2	/ GPIO3_C5_d

AG6 PCIE20_CLKREQn_M1
AD7 PCIE20_WAKEn_M1
AC8 GSENSOR_INT_L_GPIO2_D2
AC7 ALPS_INT_L_GPIO2_D3
AF5 GYRO_INT_L_GPIO2_D4
AF6 LCD_EN_H_GPIO2_D5
AD6 LCD_RST_L_GPIO2_D6
AH5
AH4
AB8
AE5
AG4
AF4
AH3
AG3
AH2
AG2
AG1
AF2
AF1 I2C5_SCL_M0
AE1 I2C5_SDA_M0
AE2
AE3
AD4
AD2
AD1 PCIE20_PERSTn_M1
AA7 HP_DET_L_GPIO3_C2
AC4 BQ_OTG_EN_H_GPIO3_C3
AC3
AC2

V10 VCCIO5_1
V11 VCCIO5_2

VCC_1V8
C1800 100nF X5R 10V C0201

VCC_1V8
R1800 2.2K 5% R0402
R1801 2.2K 5% R0402
I2C5_SCL_M0
I2C5_SDA_M0

RK3568
BGA636_19R00X19R00X1R20

If a board needs to be compatible with two voltage choices, recommended to enable BOM_ID

Note:

If VCCIO5 domain power voltage is adjusted, the software DTS configuration must be updated synchronously, otherwise the IO may be damaged!

If the VCCIO5 hardware has been modified to 1.8V power supply, and the corresponding DTS must be modified to 1.8V configuration, otherwise the IO function of VCCIO5 will be abnormally.

The VCCIO5 hardware has been modified to 3.3V power supply, if the software DTS configuration is still 1.8V configuration, the IO of VCCIO5 will be damaged!

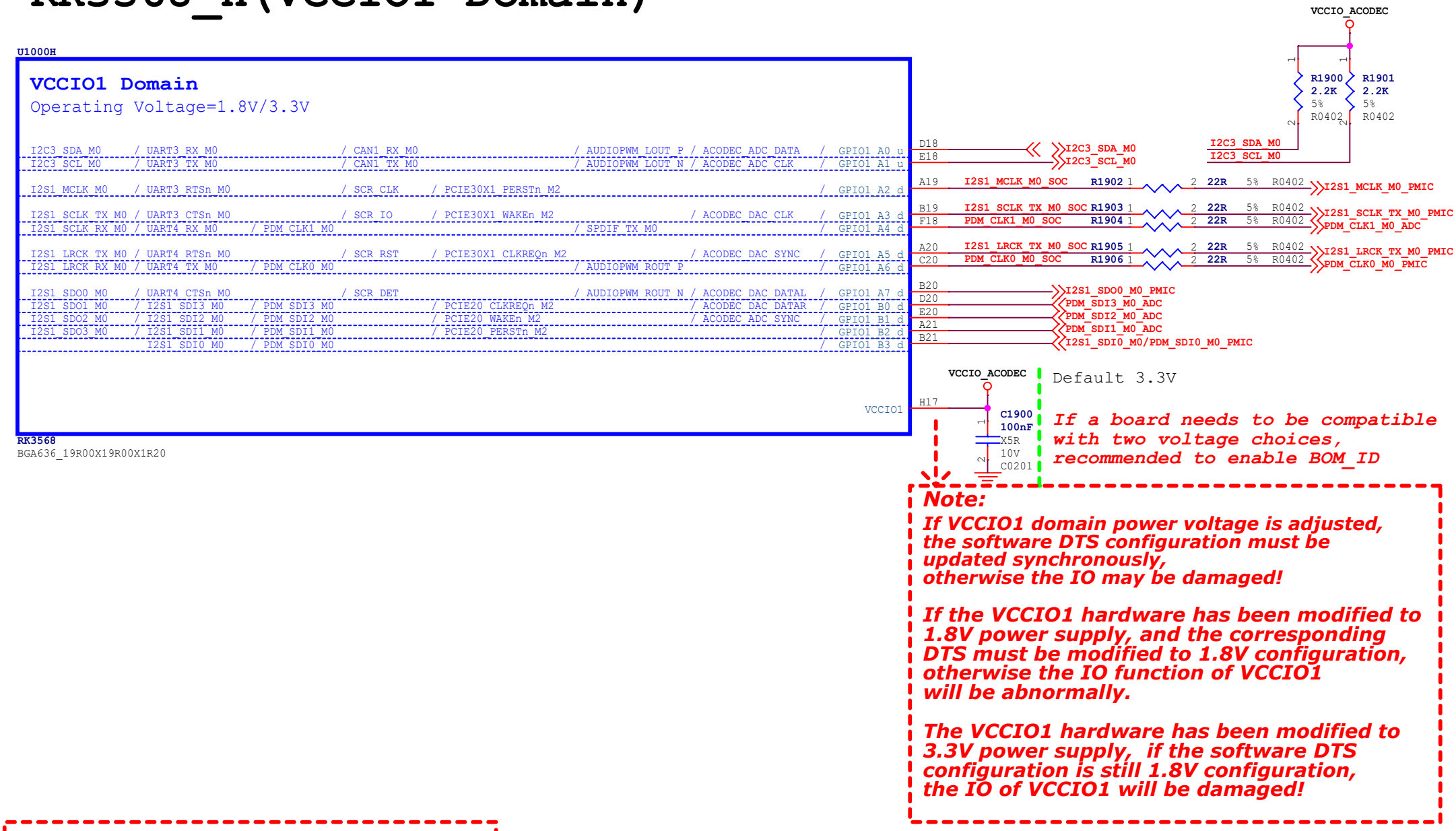
Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	18.RK3568_VO Interface_2				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	19 of 61

RK3568_H(VCCIO1 Domain)



Note:
If VCCIO1 domain power voltage is adjusted, the software DTS configuration must be updated synchronously, otherwise the IO may be damaged!

If the VCCIO1 hardware has been modified to 1.8V power supply, and the corresponding DTS must be modified to 1.8V configuration, otherwise the IO function of VCCIO1 will be abnormally.

The VCCIO1 hardware has been modified to 3.3V power supply, if the software DTS configuration is still 1.8V configuration, the IO of VCCIO1 will be damaged!

Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

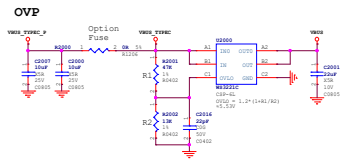
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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	19.RK3568_Audio Interface				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	20 of 61

Default:
USB Type-C Port



OVP

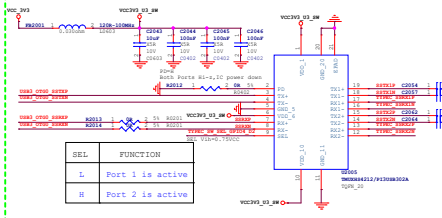


ESD



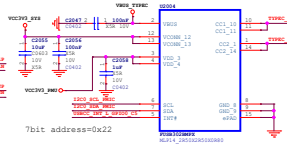
Note:
If common mode inductors are needed,
it is recommended to keep 2.2ohm in series
to improve the antistatic ability

Differential Channel Crossbar Switch



SEL	FUNCTION
L	Port 1 is active
H	Port 2 is active

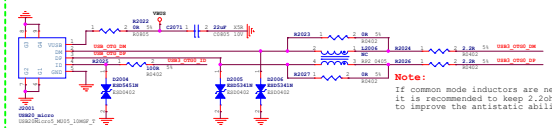
USB Type-C Controller(PD)



7bit address=0x22

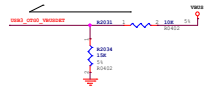
VBUS of FUSB302BMPX: 4 to 21 TYP = 5V
VDD of FUSB302BMPX: 3 to 5.5 TYP = 3.3V
VCONN of FUSB302BMPX: 3 to 5.5

Option:
MicroUSB Port



Note:
If common mode inductors are needed,
it is recommended to keep 2.2ohm in series
to improve the antistatic ability

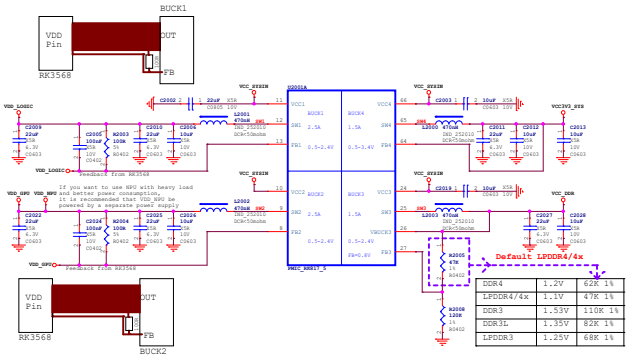
USB3_OTG0_VBUSDET



Peripheral power input source

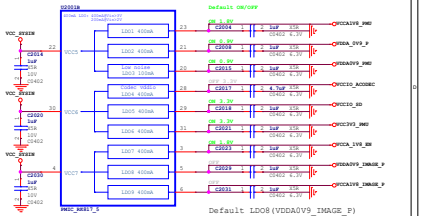


PMIC RK817 DCDC



DDR4	1.2V	62K 1%
LPDDR4/4x	1.1V	47K 1%
DDR3	1.53V	110K 1%
DDR3L	1.35V	82K 1%
LPDDR3	1.25V	68K 1%

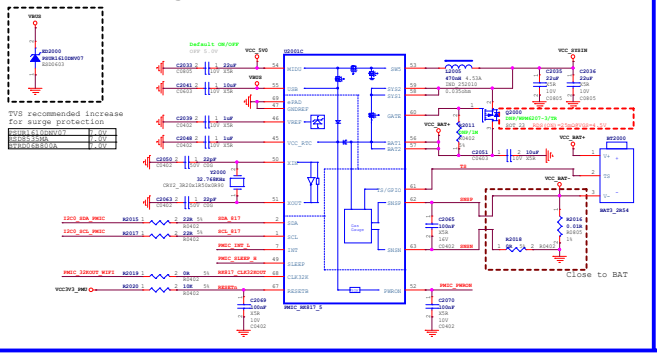
PMIC RK817 LDO



Default LDO8(VDDA0V9_IMAGE_P) configuration output 2.8V for Camera

Default LDO8(VCCA1V8_IMAGE_P) configuration output 1.8V for Camera

PMIC RK817 Managerment

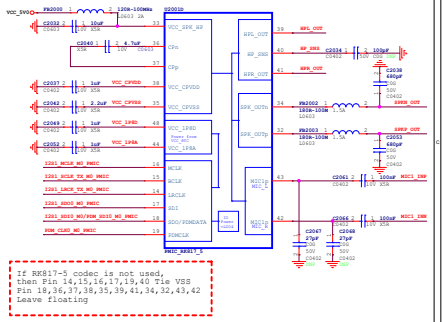


TVS recommended increase for surge protection

ESDA14V2SW6T	14.2V
SRV05-4-HNG	6.0V
TPD4E05U06	6.0V

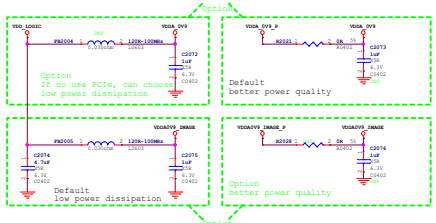
Close to BAT

PMIC RK817 CODEC



If RK817-5 codec is not used,
then Pin 14,15,16,17,19,40 Tie VSS
Pin 18,36,37,38,35,39,41,34,32,43,42
Leave floating

VDDA_0V9, VDDA0V9_IMAGE ------select



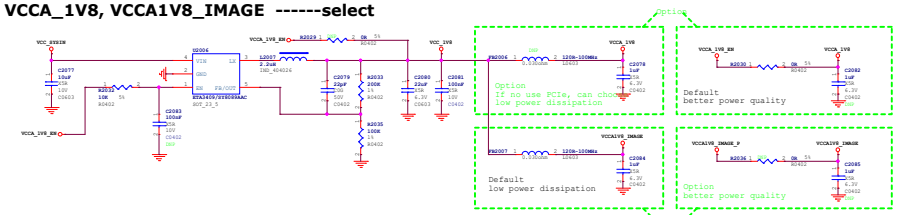
Option
If no use PCIe, can choose low power dissipation

Default
better power quality

Default
low power dissipation

Option
better power quality

VCCA_1V8, VCCA1V8_IMAGE ------select



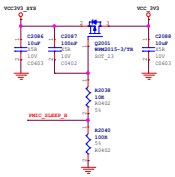
Option
If no use PCIe, can choose low power dissipation

Default
better power quality

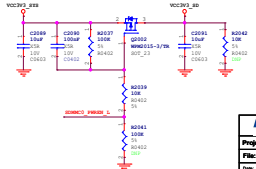
Default
low power dissipation

Option
better power quality

VCC_3V3 (Power OFF under SLEEP)



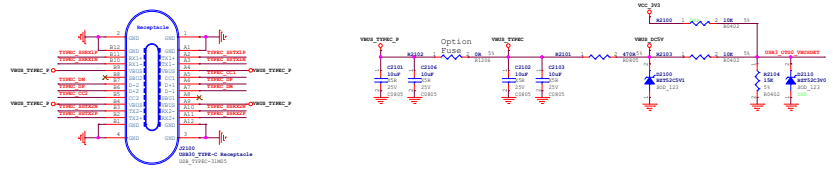
VCC3V3_SD



Rockchip	Rockchip Electronics Co., Ltd
Project:	RK3566_Tablet_REF_SCH
File:	38.Power_5Cell_CHG_RK817-5
Date:	Tuesday, July 20, 2021
Rev:	V1.1
Designed by:	Zhangjh
Reviewed by:	Default
Sheet:	31 of 41

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USB Type-C Port

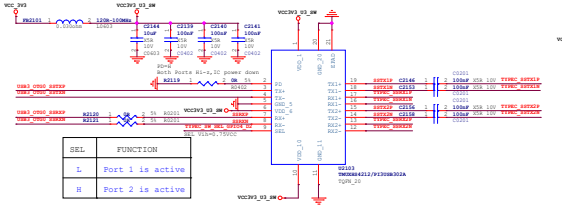


ESD



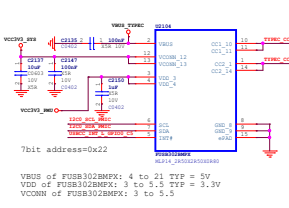
Note:
If common mode inductors are needed,
it is recommended to keep 2.2ohm in series
to improve the antistatic ability

Differential Channel Crossbar Switch



SEL	FUNCTION
L	Port 1 is active
H	Port 2 is active

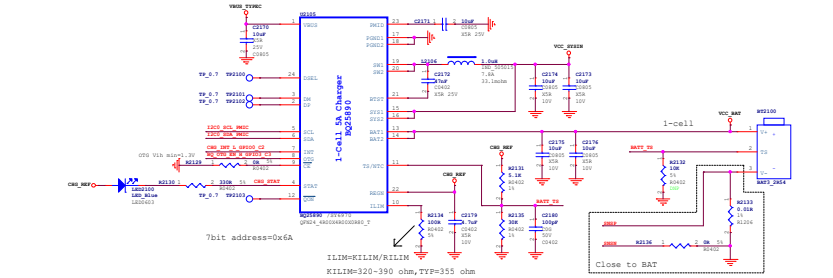
USB Type-C Controller(PD)



7bit address=0x22

VBUS of FUSB302BMPX: 4 to 21 TYP = 5V
VDD of FUSB302BMPX: 3 to 5.5 TYP = 3.3V
VCONN of FUSB302BMPX: 3 to 5.5

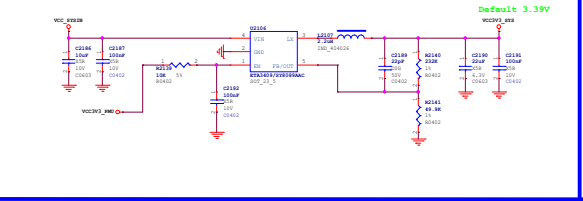
Charger IC



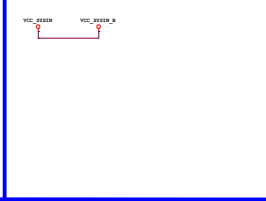
7bit address=0x6A

ILIM=KILIM/RILIM
KILIM=320-390 ohm,TYP=355 ohm

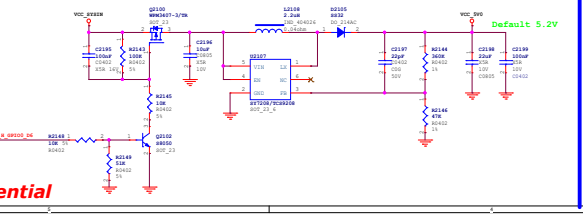
VCC3V3_SYS



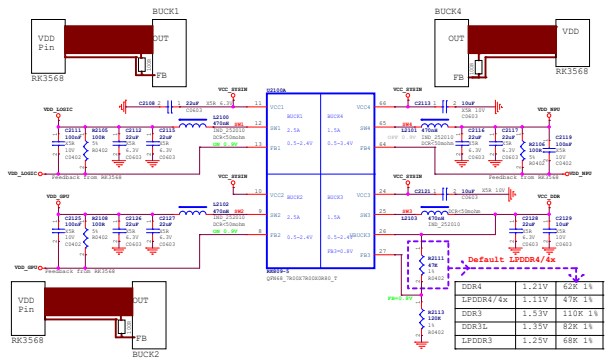
Peripheral power input source



VCC_5V0

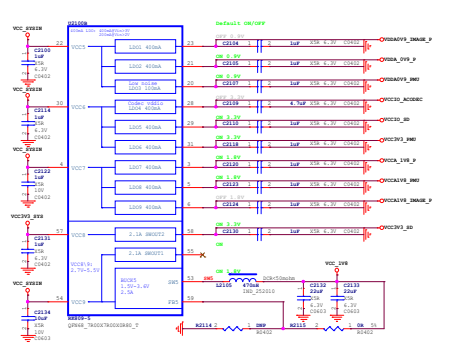


PMIC RK809 DCDC

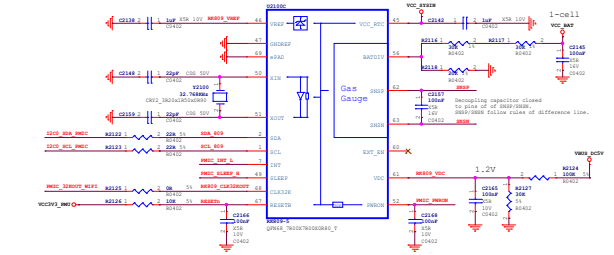


DDR		
LPDDR4/4x	1.21V	62K 1%
LPDDR4/4x	1.11V	47K 1%
DDR3	1.53V	110K 1%
DDR3L	1.35V	82K 1%
LPDDR3	1.25V	68K 1%

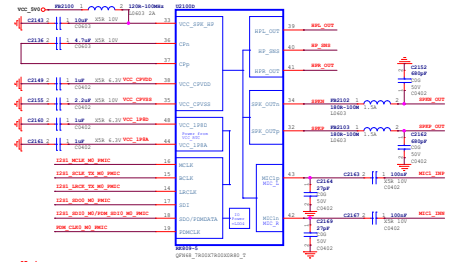
PMIC RK809 LDO



PMIC RK809 Managerment

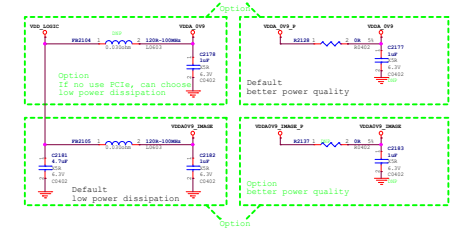


PMIC RK809 CODEC

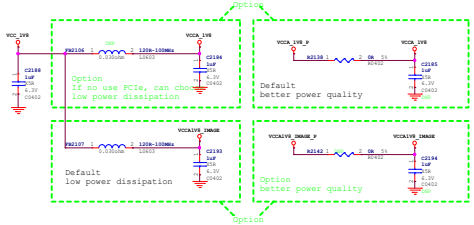


Note:
If RK809-5 codec is not used,
then Pin 14,15,16,17,19,40 Tie VSS
Pin 18,36,37,38,35,39,41,34,32,43,42
Leave floating

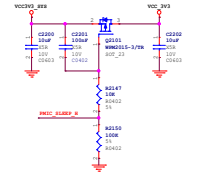
VDDA_0V9, VDDA0V9_IMAGE ------select



VCCA_1V8, VCCA1V8_IMAGE ------select



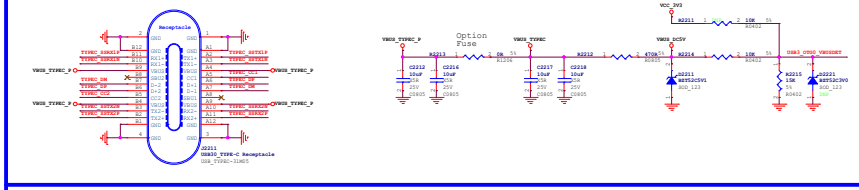
VCC_3V3 (Power OFF under SLEEP)



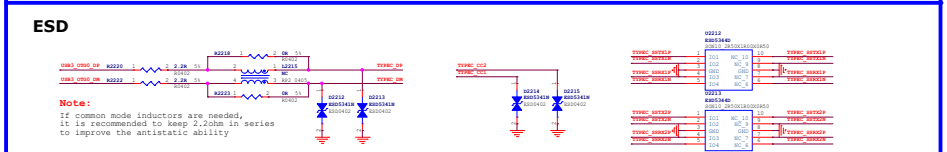
Rockchip Confidential

Rockchip Electronics Co., Ltd	
Project:	RK3566_Tablet_REF_SCH
File:	21.Power_1Cell_QC_RK809-5(Opt)
Date:	Tuesday, July 20, 2021
Rev:	V1.1
Designed by:	Zhangjh
Reviewed by:	Default
Sheet:	20 of 41

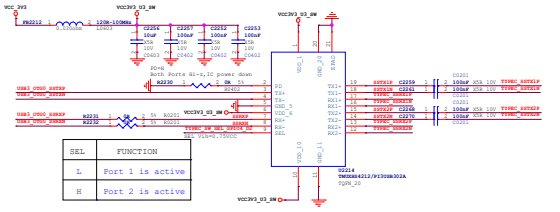
USB Type-C Port



ESD

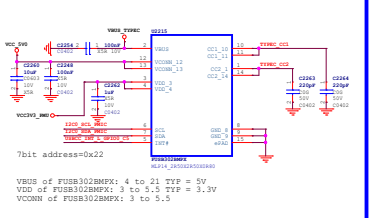


Differential Channel Crossbar Switch

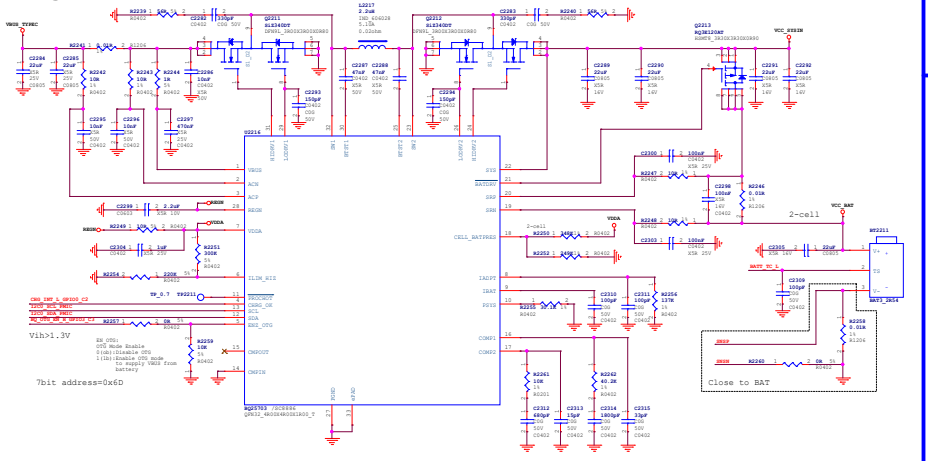


SEL	FUNCTION
L	Port 1 is active
H	Port 2 is active

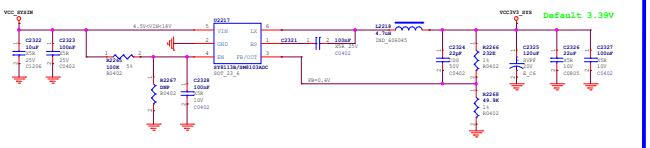
USB Type-C Controller(PD)



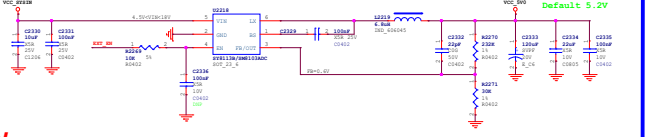
Charger IC



VCC3V3_SYS



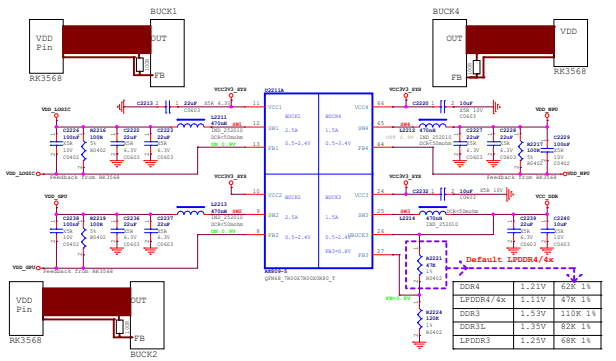
VCC_5V0



Peripheral power input source

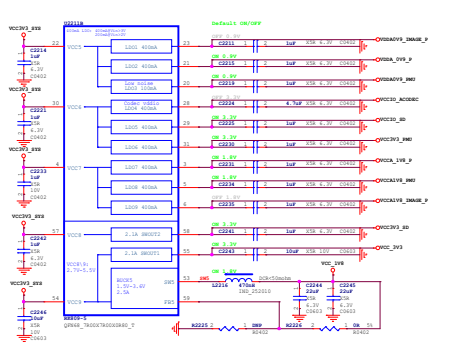


PMIC RK809 DCDC

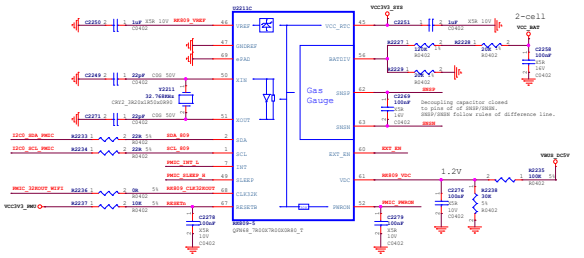


DDR4	1.21V	62K 1%
LPDDR4/4x	1.11V	47K 1%
DDR3	1.53V	110K 1%
DDR3L	1.35V	82K 1%
LPDDR3	1.25V	68K 1%

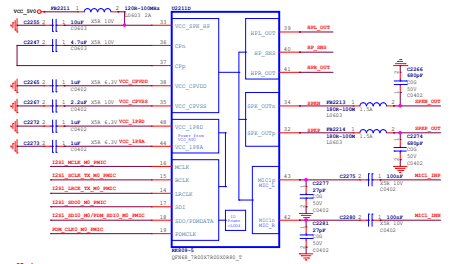
PMIC RK809 LDO



PMIC RK809 Managerment

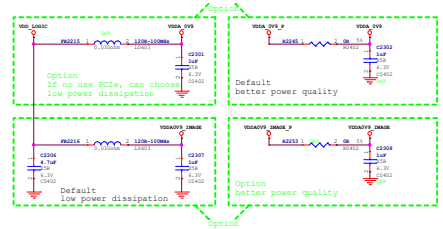


PMIC RK809 CODEC

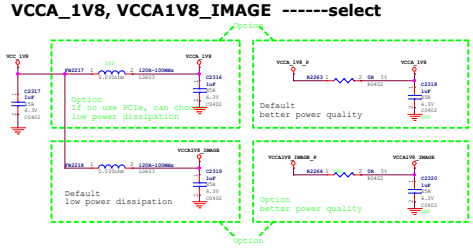


Note:
If RK809-5 codec is not used,
then Pin 14,15,16,17,19,40 Tie VSS
Pin 18,36,37,38,35,39,41,34,32,43,42
Leave floating

VDDA_0V9, VDDA0V9_IMAGE ------select



VCCA_1V8, VCCA1V8_IMAGE ------select

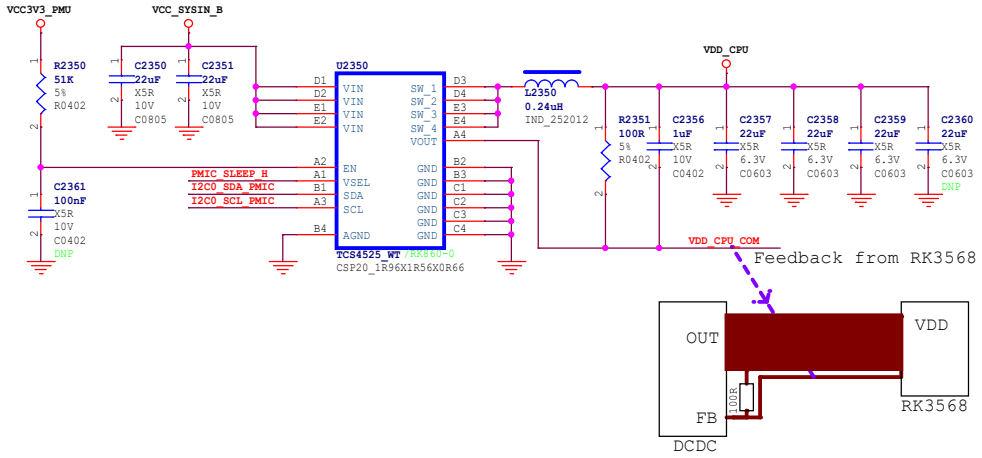


Rockchip Confidential

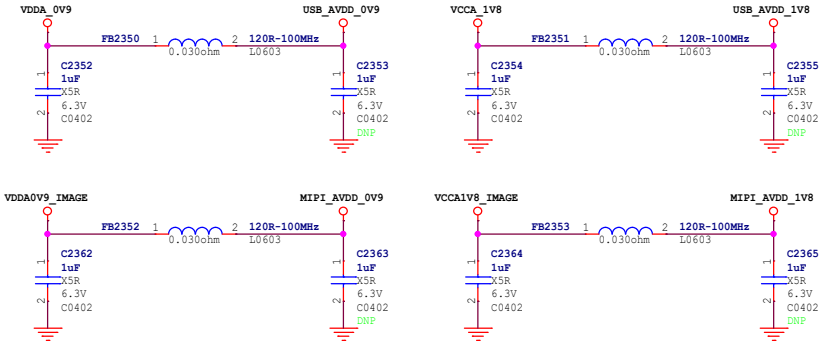
Rockchip Electronics Co., Ltd	
Project:	RK3566_Tablet_REF_SCH
File:	22.Power_2Cell_QC_RK809-5(Opt)
Date:	Tuesday, July 20, 2021
Rev:	V1.1
Designed by:	Zhangjh
Reviewed by:	Default
Sheet:	20 of 41

I2C0_SCL_PMIC
I2C0_SDA_PMIC
PMIC_SLEEP_H
VDD_CPU_COM
LCD0_PWREN_H_GPIO0_C7

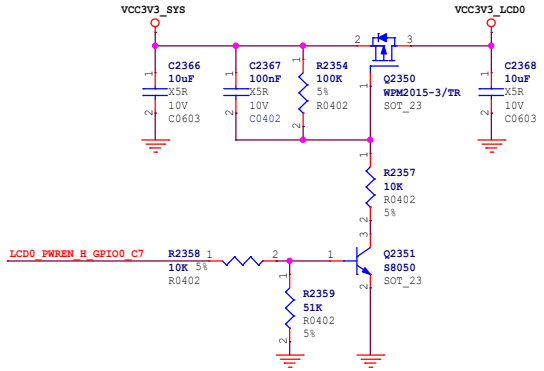
VDD_CPU



USB/MIPI PHY Power



LCD Power



WIFI/BT IO and CTL IO Power

PMUIO2,VCCIO4 IO Domain Power Supply select



Note:

Please choose IO voltage values according to the real mounted WIFI module and modify the corresponding software configuration.

If a board needs to be compatible with two voltage choices, recommended to enable BOM_ID

For RTL8723DS Module, PMUIO2 = VCC3V3_PMU and VCCIO_WL connect to VCC_WL to ensure the power on sequence.

Rockchip Confidential

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	23.Power_Ext Discrete				
Date:	Tuesday, July 20, 2021			Rev:	V1.1
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	24 of 61

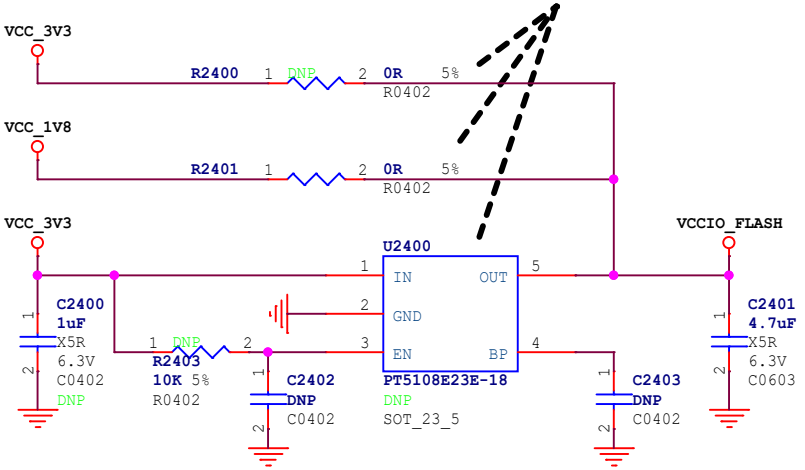
FLASH_VOL_SEL

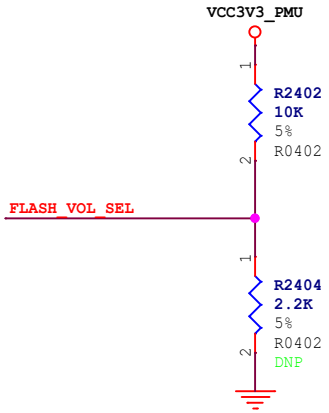
Flash Power Manage

	VCCIO2 domain voltage: Recommend voltage value (VCCIO_FLASH)	FLASH_VOL_SEL state decided to VCCIO2 domain IO driven by default
eMMC	1.8V	FLASH_VOL_SEL --> Logic=H
Nand flash	Default 3.3V, Optional 1.8V	FLASH_VOL_SEL --> Logic=L(Default)
SPI flash	Default 1.8V, Optional 3.3V	FLASH_VOL_SEL --> Logic=H(Default)

Note:

According to the actual choice of mounted
Cannot be mounted at the same time



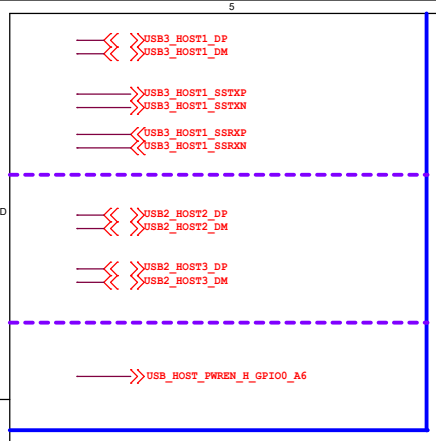


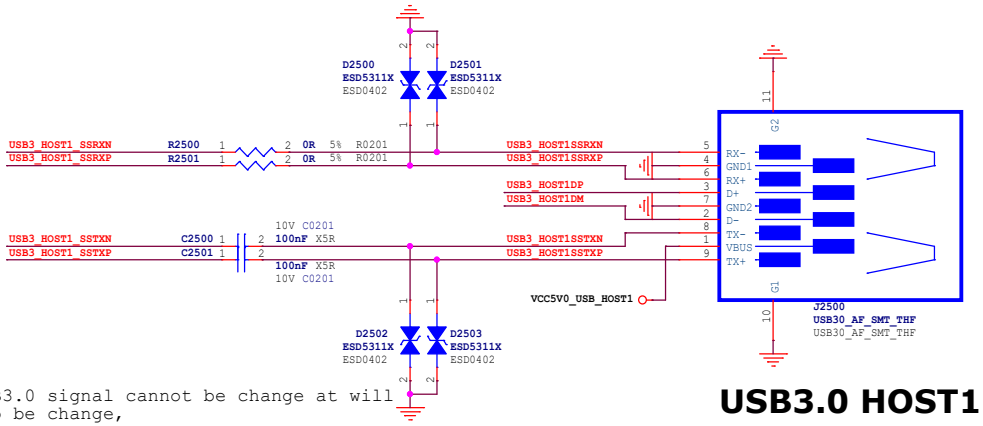
Note:
FLASH_VOL_SEL state decided
to VCCIO2 domain IO driven by default
Logic=L: 3.3V IO driven
Logic=H: 1.8V IO driven

When VCCIO2 voltage is connected to 1.8V, FLASH_VOL_SEL must be high
When VCCIO2 voltage is connected to 3.3V, FLASH_VOL_SEL must be low
If VCCIO2 power supply voltage and FLASH_VOL_SEL fails to meet the above relationship,
its function will be abnormally(for example, it cannot be started normally) or IO will be damaged.

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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	24.Power_Flash Power Manage				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	25 of 61

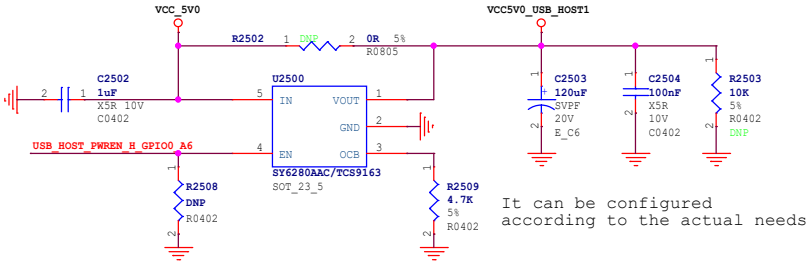




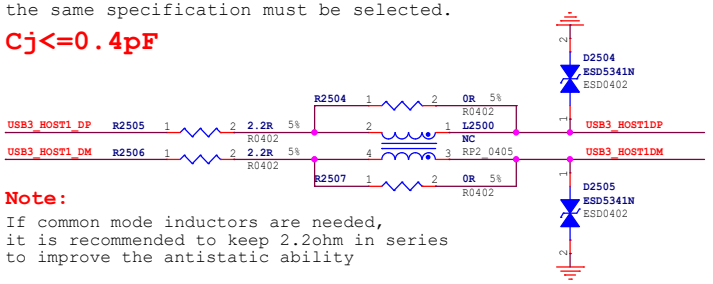
USB3.0 HOST1

Note:
The ESD of USB3.0 signal cannot be change at will
If it needs to be change,
the same specification must be selected.

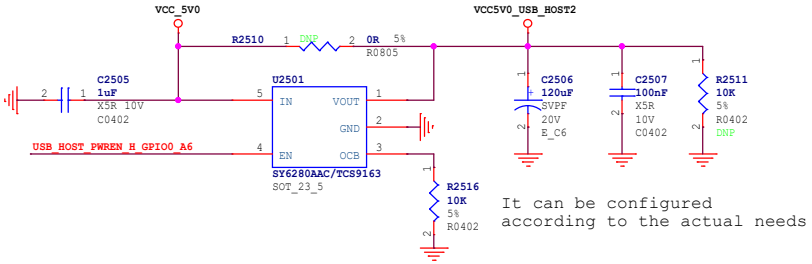
Cj<=0.4pF



It can be configured
according to the actual needs

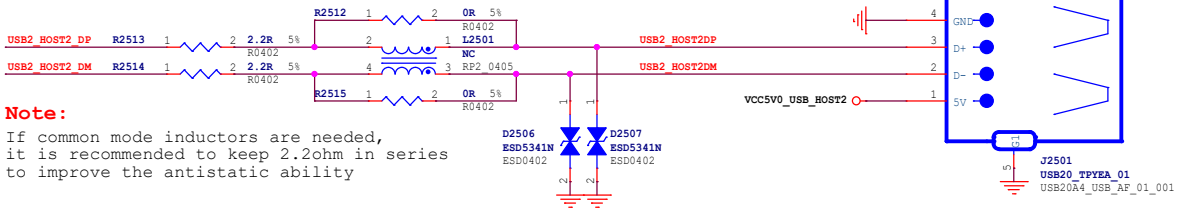


Note:
If common mode inductors are needed,
it is recommended to keep 2.2ohm in series
to improve the antistatic ability

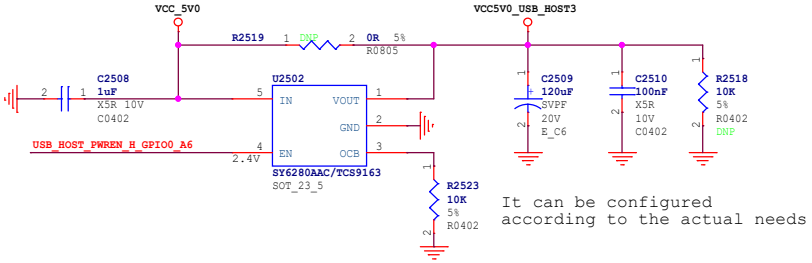


It can be configured
according to the actual needs

USB2.0 HOST2
And M.2 module Option

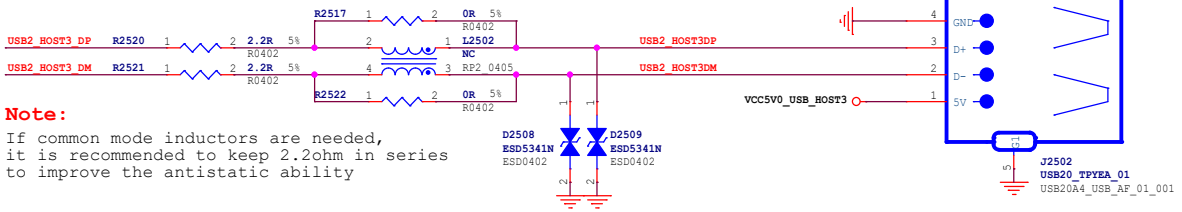


Note:
If common mode inductors are needed,
it is recommended to keep 2.2ohm in series
to improve the antistatic ability



It can be configured
according to the actual needs

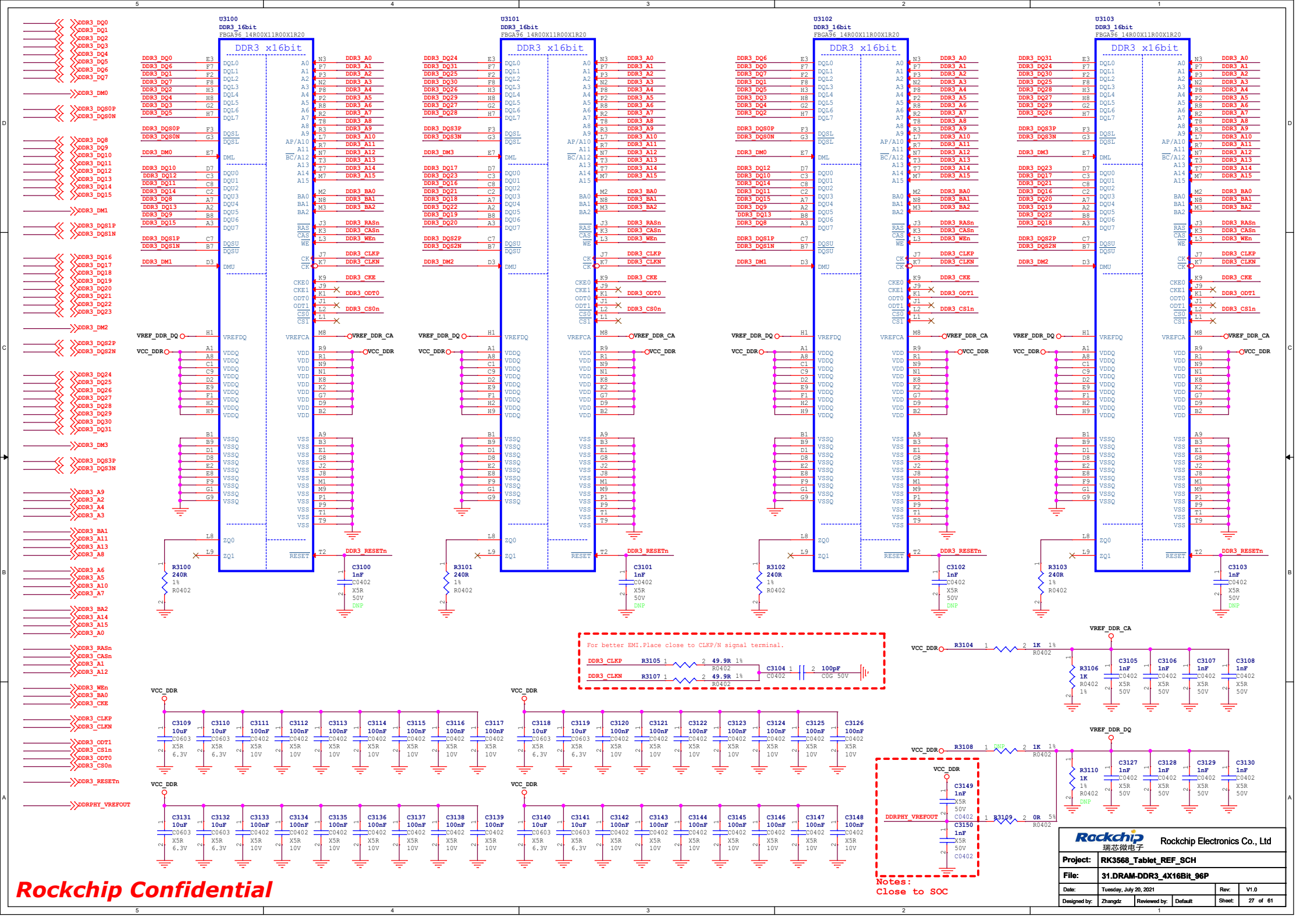
USB2.0 HOST3
And 4G module Option



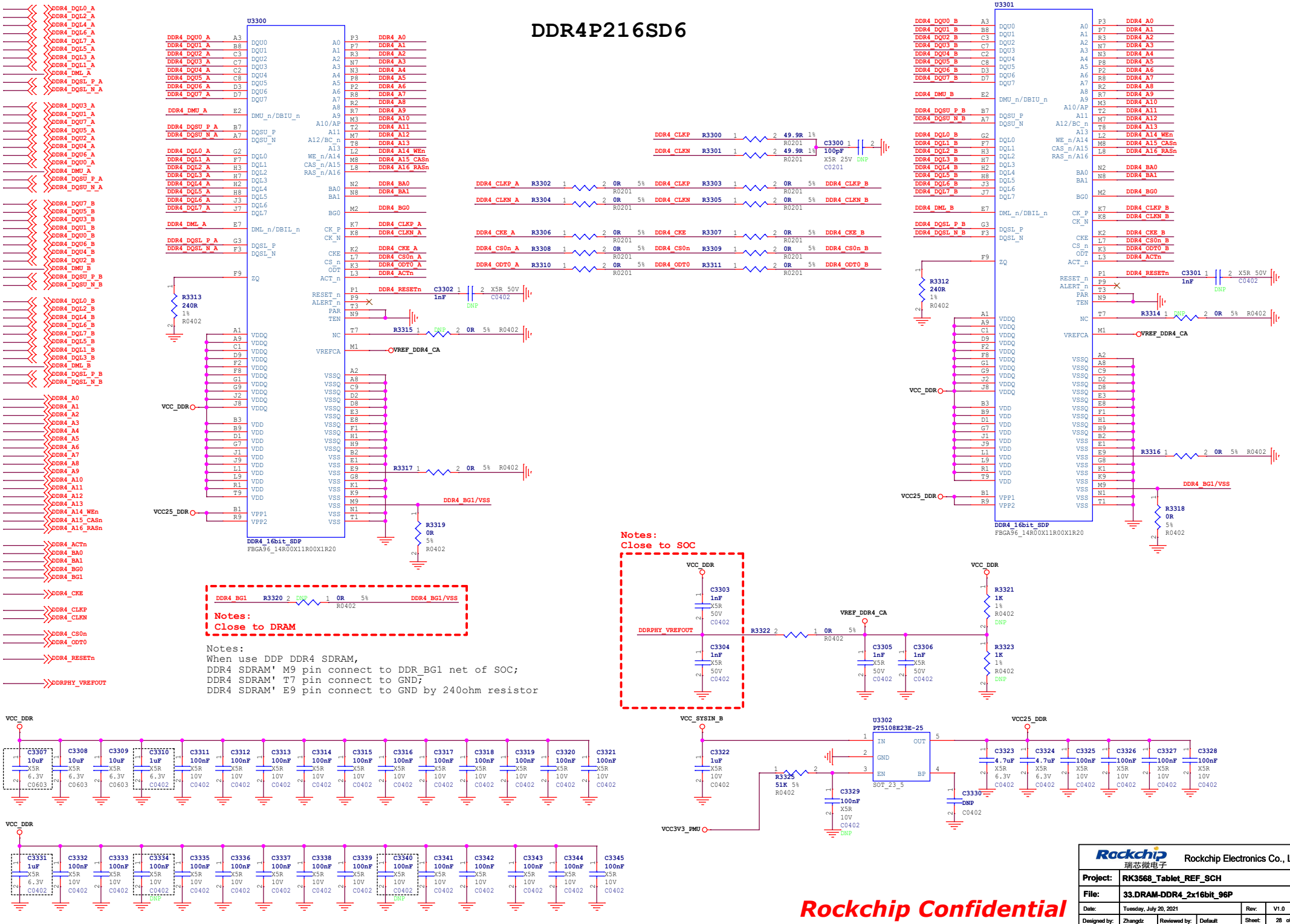
Note:
If common mode inductors are needed,
it is recommended to keep 2.2ohm in series
to improve the antistatic ability

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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	25.USB2/USB3 HOST Port				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	26 of 61

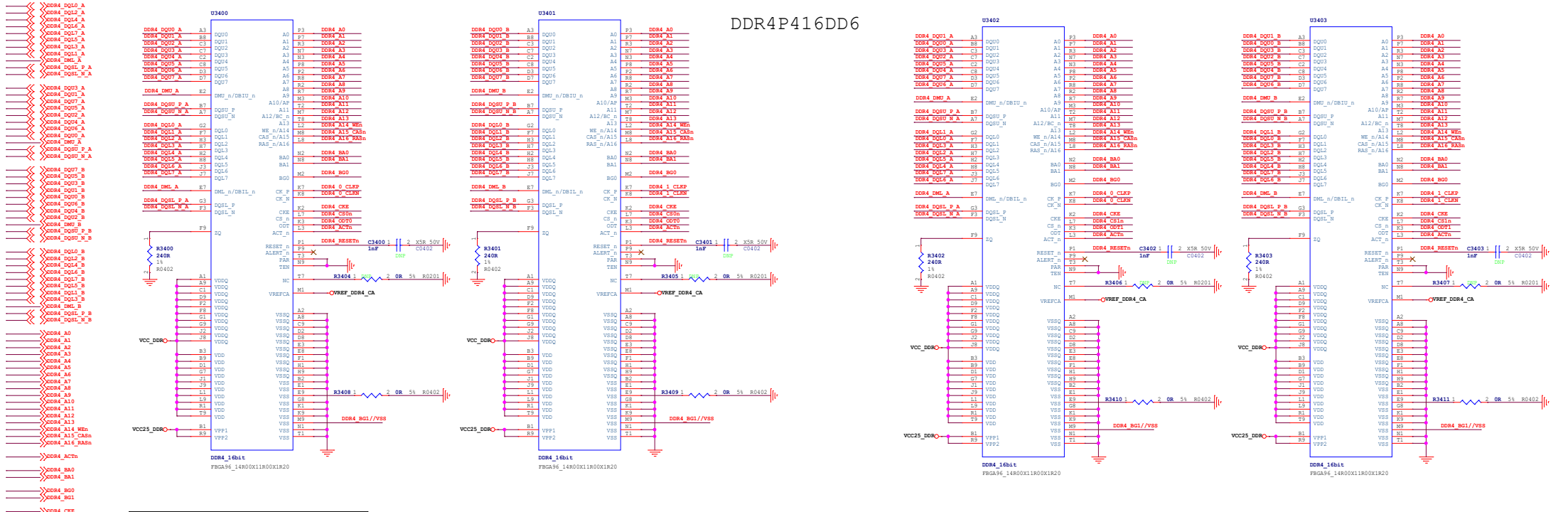

U3100
DDR3_16bit
FBGA96_14R00X11R00X1R20
DDR3 x16bit
U3101
DDR3_16bit
FBGA96_14R00X11R00X1R20
DDR3 x16bit
U3102
DDR3_16bit
FBGA96_14R00X11R00X1R20
DDR3 x16bit
U3103
DDR3_16bit
FBGA96_14R00X11R00X1R20
DDR3 x16bit
For better EMI.Place close to CLKP/N signal terminal.
Notes:
Close to SOC
Rockchip Confidential
Rockchip Electronics Co., Ltd
Project: RK3568_Tablet_REF_SCH
File: 31.DRAM-DDR3_4X16Bit_96P
Date: Tuesday, July 20, 2021
Rev: V1.0
Designed by: Zhangdz
Reviewed by: Default
Sheet: 27 of 61

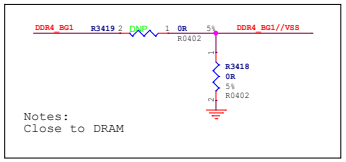
DDR4P216SD6



Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	33.DRAM-DDR4_2x16bit_96P				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	28 of 61

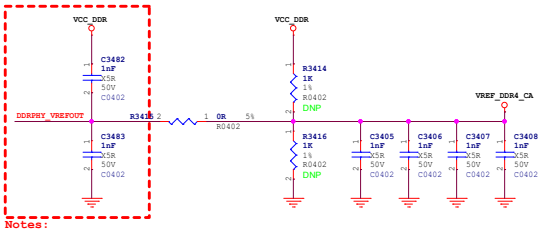
DDR4P416DD6

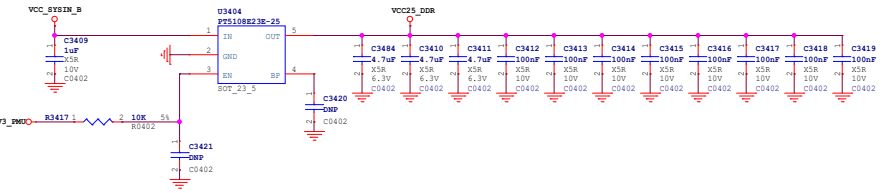


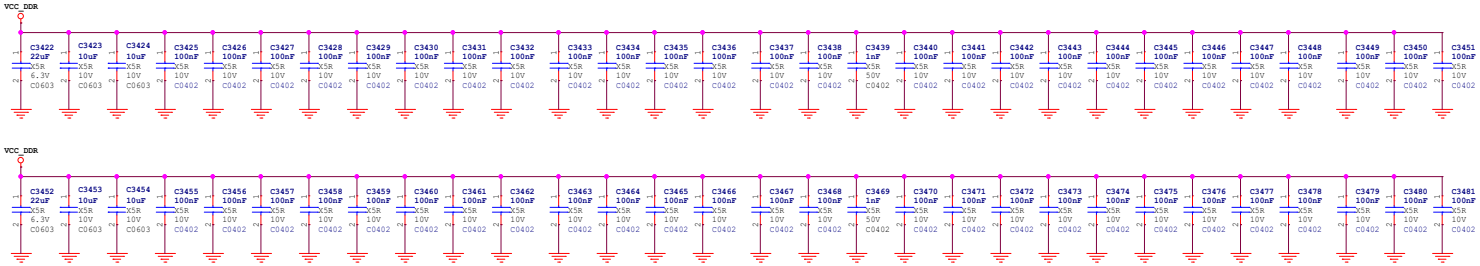


Notes:
When use 16Gbit/32Gbit DDR4 SDRAM,
DDR4 SDRAM' M9 pin connect to DDR_BG1 net of SOC;
DDR4 SDRAM' T7 pin connect to GND;
DDR4 SDRAM' E9 pin connect to GND by 240ohm resistor



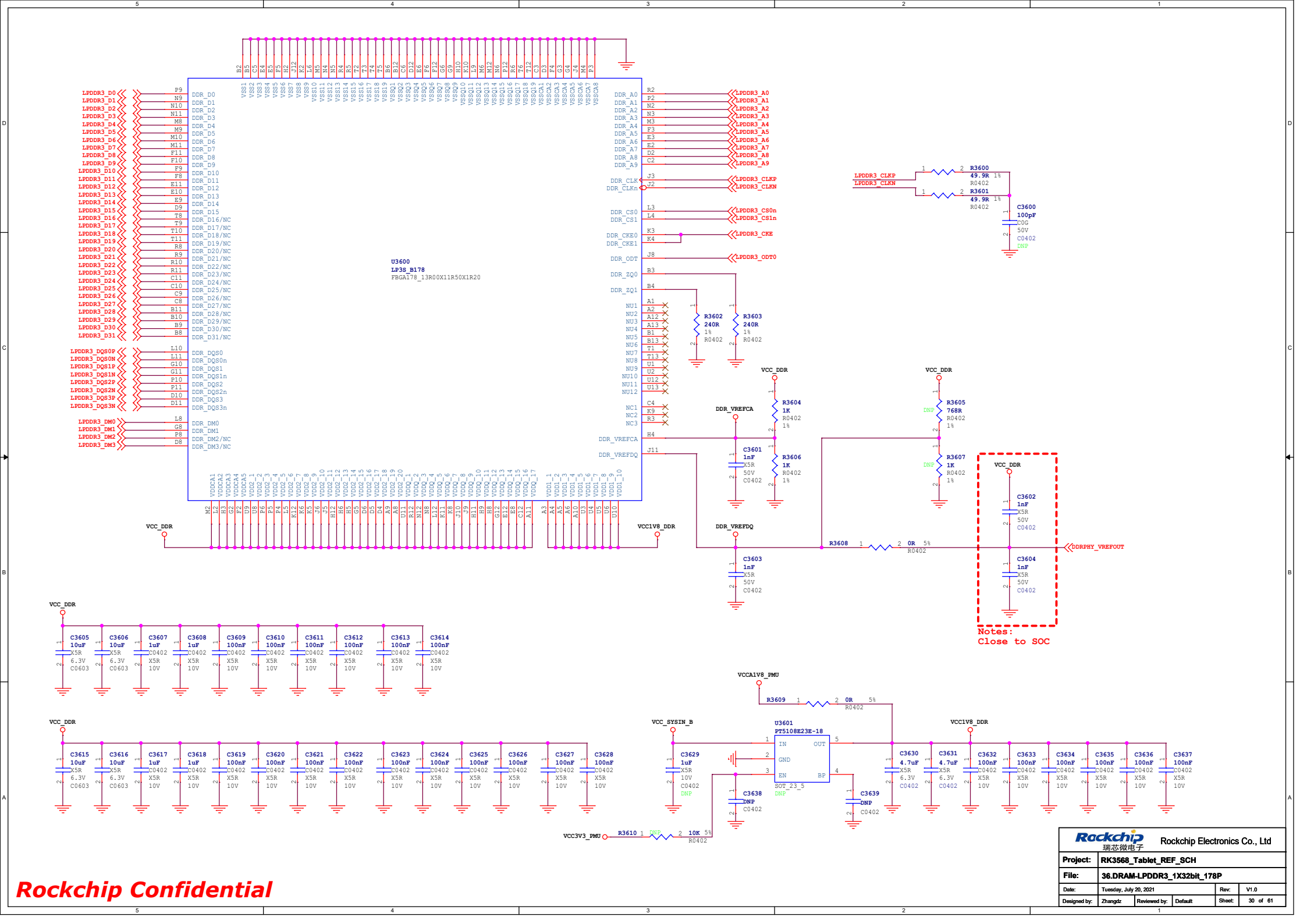




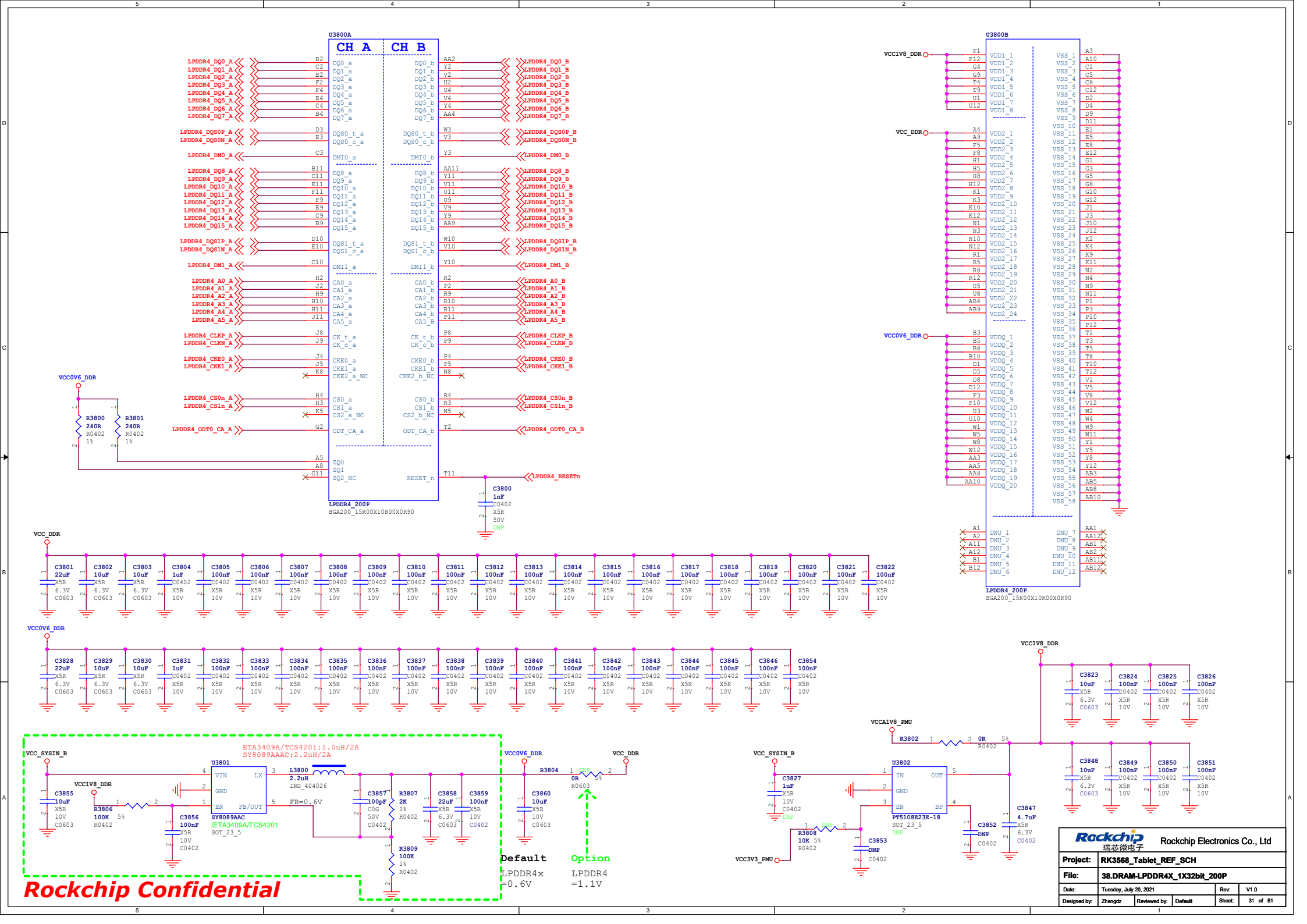


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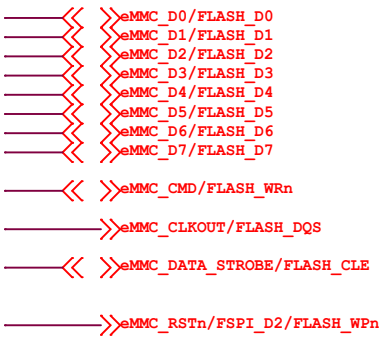
Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd
Project:	RK3568_Tablet_REF_SCH
File:	34.DRAM-DDR4_4x16Bit_96P
Date:	Tuesday, July 20, 2021
Rev:	V1.0
Designed by:	Zhangtz
Reviewed by:	Default
Sheet:	29 of 61

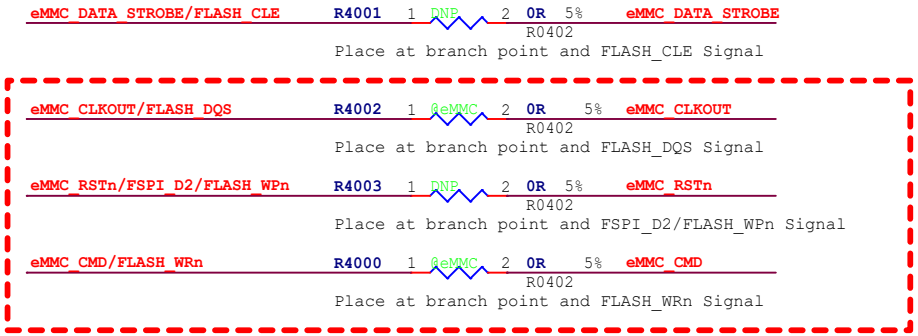


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U3800A
CH A
CH B
LPDDR4_200P
BGA200_15R00X10R00X0R90
U3800B
LPDDR4_200P
BGA200_15R00X10R00X0R90
VCC_DDR
VCC0V6_DDR
VCC1V8_DDR
R3800 240R R0402 1%
R3801 240R R0402 1%
C3800 1nF C0402 X5R 50V DNP
C3801 22uF X5R 6.3V C0603
C3802 10uF X5R 6.3V C0603
C3803 10uF X5R 6.3V C0603
C3804 1uF C0402 X5R 10V
C3805 100nF C0402 X5R 10V
C3806 100nF C0402 X5R 10V
C3807 100nF C0402 X5R 10V
C3808 100nF C0402 X5R 10V
C3809 100nF C0402 X5R 10V
C3810 100nF C0402 X5R 10V
C3811 100nF C0402 X5R 10V
C3812 100nF C0402 X5R 10V
C3813 100nF C0402 X5R 10V
C3814 100nF C0402 X5R 10V
C3815 100nF C0402 X5R 10V
C3816 100nF C0402 X5R 10V
C3817 100nF C0402 X5R 10V
C3818 100nF C0402 X5R 10V
C3819 100nF C0402 X5R 10V
C3820 100nF C0402 X5R 10V
C3821 100nF C0402 X5R 10V
C3822 100nF C0402 X5R 10V
C3828 22uF X5R 6.3V C0603
C3829 10uF X5R 6.3V C0603
C3830 10uF X5R 6.3V C0603
C3831 1uF C0402 X5R 10V
C3832 100nF C0402 X5R 10V
C3833 100nF C0402 X5R 10V
C3834 100nF C0402 X5R 10V
C3835 100nF C0402 X5R 10V
C3836 100nF C0402 X5R 10V
C3837 100nF C0402 X5R 10V
C3838 100nF C0402 X5R 10V
C3839 100nF C0402 X5R 10V
C3840 100nF C0402 X5R 10V
C3841 100nF C0402 X5R 10V
C3842 100nF C0402 X5R 10V
C3843 100nF C0402 X5R 10V
C3844 100nF C0402 X5R 10V
C3845 100nF C0402 X5R 10V
C3846 100nF C0402 X5R 10V
C3854 100nF C0402 X5R 10V
ETA3409A/TCS4201:1.0uH/2A
SY8089AAAC:2.2uH/2A
VCC_SYSIN_B
VCC1V8_DDR
U3801
SY8089AAC
/ETA3409A/TCS4201
SOT_23_5
L3800 2.2uH IND_404026
R3806 100K 5% R0402
C3855 10uF X5R 10V C0603
C3856 100nF X5R 10V C0402
C3857 100pF C0G 50V C0402
R3807 2K 1% R0402
R3809 100K 1% R0402
C3858 22uF X5R 6.3V C0603
C3859 100nF X5R 10V C0402
FB=0.6V
VCC0V6_DDR
VCC_DDR
R3804 0R R0603 DNP
C3860 10uF X5R 10V C0603
Default
LPDDR4x
=0.6V
Option
LPDDR4
=1.1V
Rockchip Confidential
VCCA1V8_PMU
VCC_SYSIN_B
U3802
PT5108E23E-18
SOT_23_5
R3802 0R R0402
C3827 1uF X5R 10V C0402 DNP
R3808 10K 5% R0402 DNP
C3853 DNP C0402
C3852 DNP C0402
C3847 4.7uF X5R 6.3V C0402
VCC3V3_PMU
C3823 10uF X5R 6.3V C0603
C3824 100nF C0402 X5R 10V
C3825 100nF C0402 X5R 10V
C3826 100nF C0402 X5R 10V
C3848 10uF X5R 6.3V C0603
C3849 100nF C0402 X5R 10V
C3850 100nF C0402 X5R 10V
C3851 100nF C0402 X5R 10V
Rockchip 瑞芯微电子
Rockchip Electronics Co., Ltd
Project: RK3568_Tablet_REF_SCH
File: 38.DRAM-LPDDR4X_1X32bit_200P
Date: Tuesday, July 20, 2021
Rev: V1.0
Designed by: Zhangdz
Reviewed by: Default
Sheet: 31 of 61

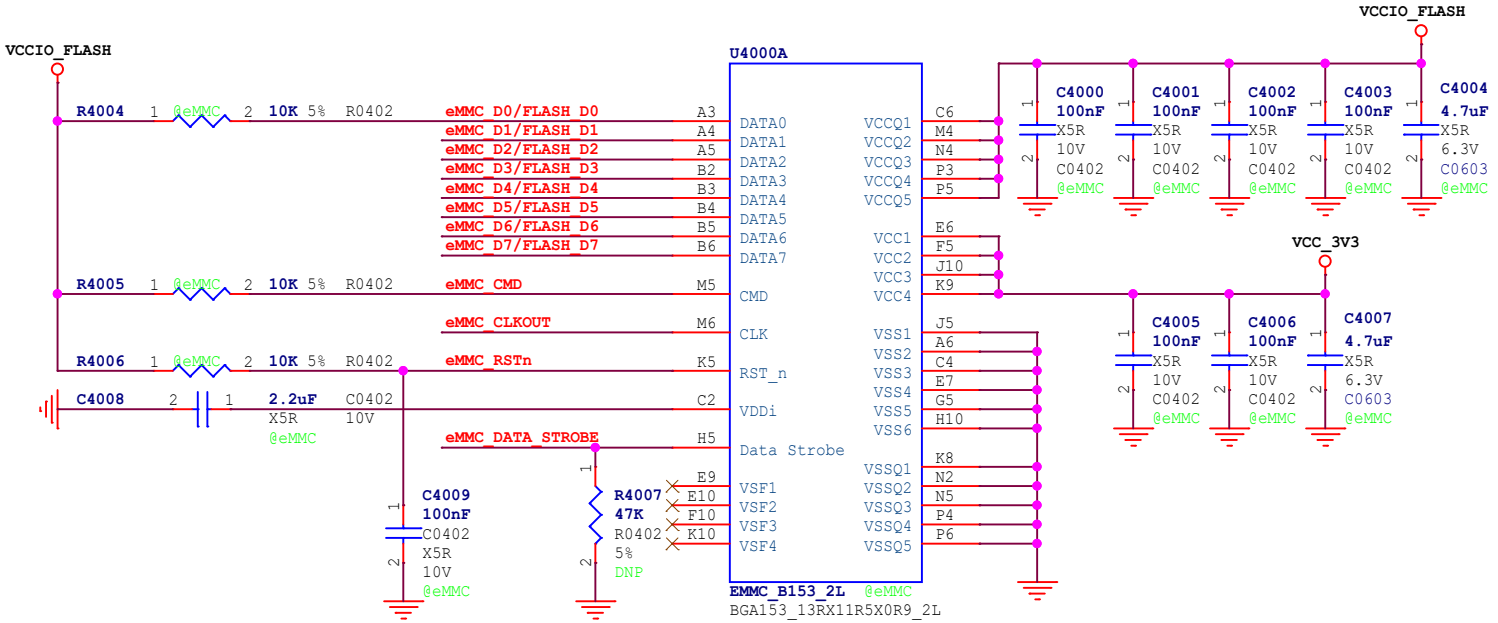
eMMC Flash

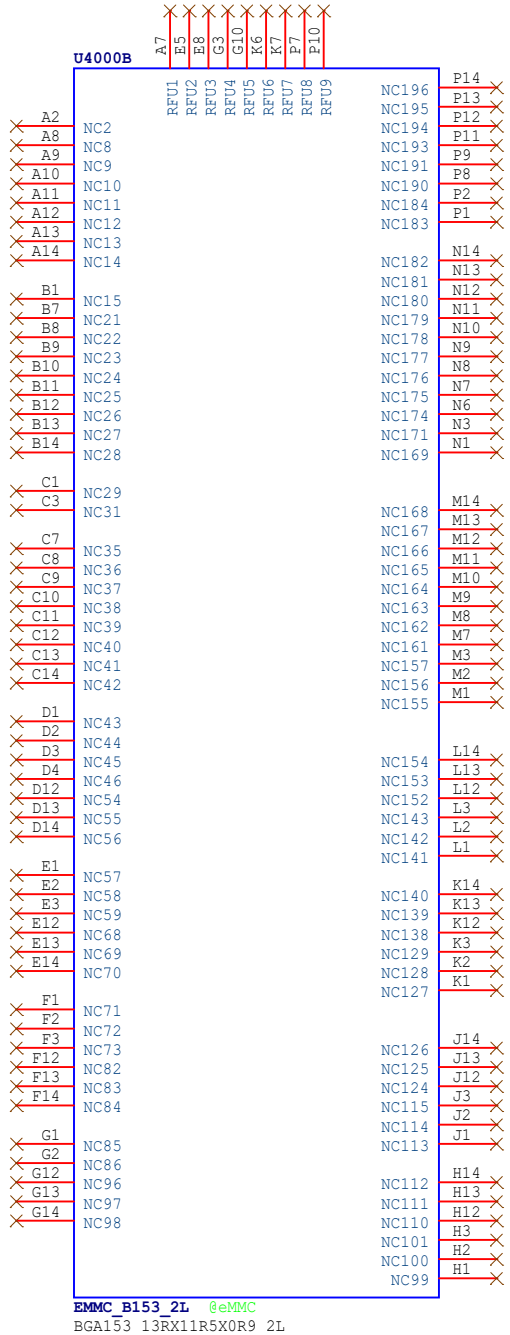




Note:

No need to double layout with Nand Flash, 0R resistor can be omitted





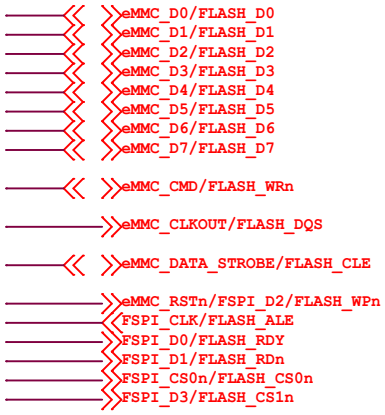
Note:

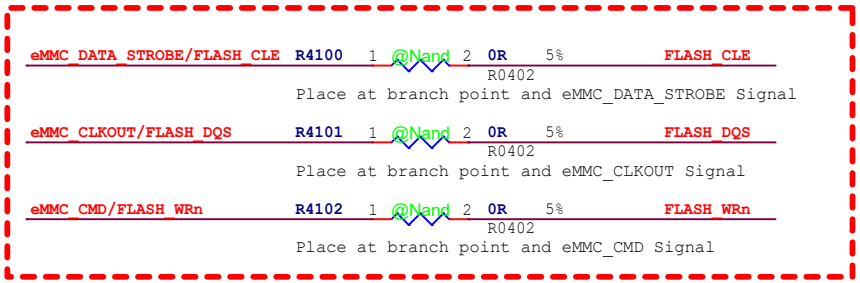
If Flash is compatible, please notice
when eMMC is used, the option is that @eMMC is mounted, @Nand is not mounted,@SPI Flash is not mounted
when Nand is used, the option is that @Nand is mounted, @eMMC is not mounted,@SPI Flash is not mounted
when SPI Flash is used, the option is that SPI Flash is mounted, @eMMC is not mounted,@Nand is not mounted

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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	40.Flash-eMMC Flash				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	32 of 61

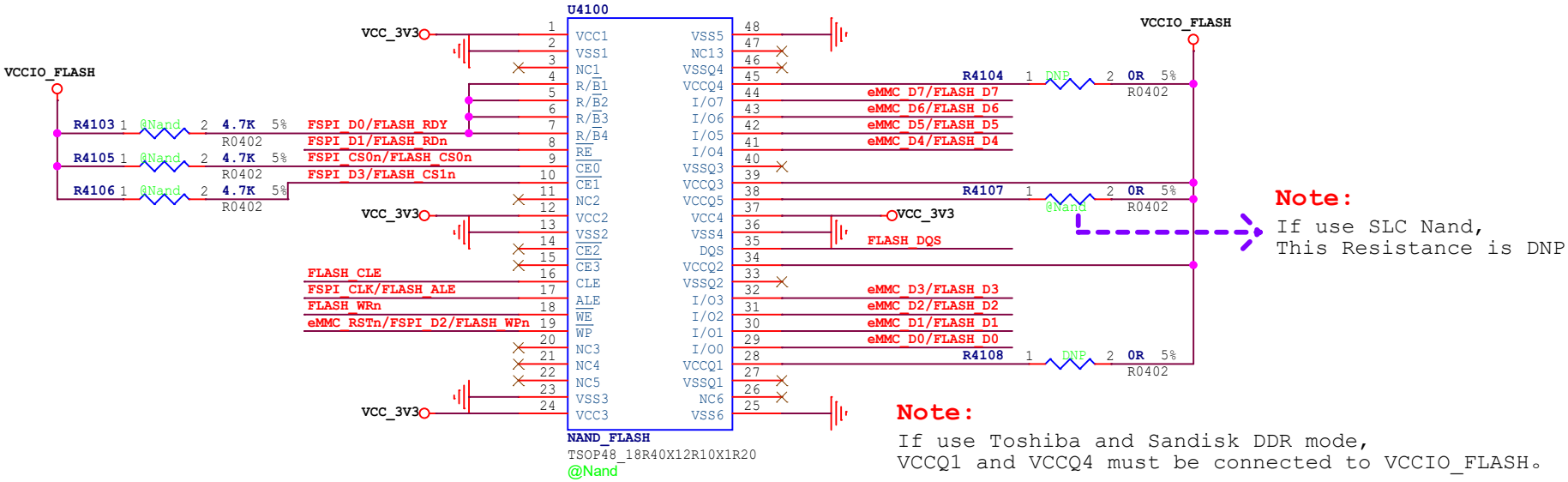
Nand Flash





Note:

No need to double layout with eMMC, 0R resistor can be omitted

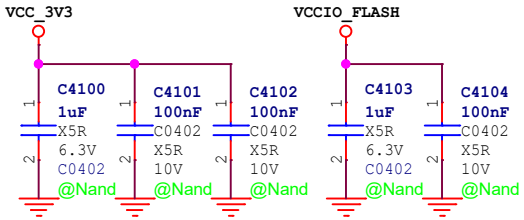


Note:

If use SLC Nand,
This Resistance is DNP

Note:

If use Toshiba and Sandisk DDR mode,
VCCQ1 and VCCQ4 must be connected to VCCIO_FLASH。



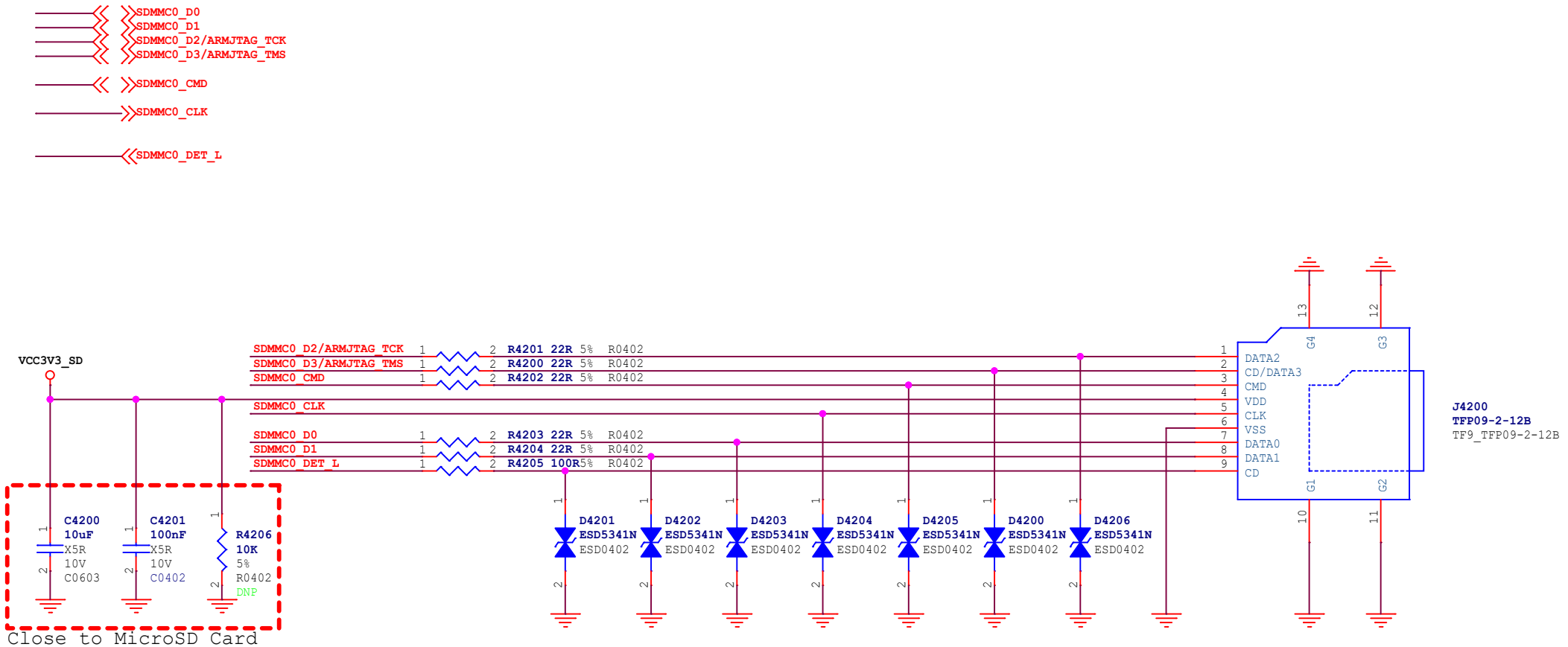
Note:

If Flash is compatible, please notice
when eMMC is used, the option is that @eMMC is mounted, @Nand is not mounted,@SPI Flash is not mounted
when Nand is used, the option is that @Nand is mounted, @eMMC is not mounted,@SPI Flash is not mounted
when SPI Flash is used, the option is that SPI Flash is mounted, @eMMC is not mounted,@Nand is not mounted

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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	41.Flash-Nand Flash(Option)				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	33 of 61

MicroSD Card



MicroSD Card

Rockchip Confidential

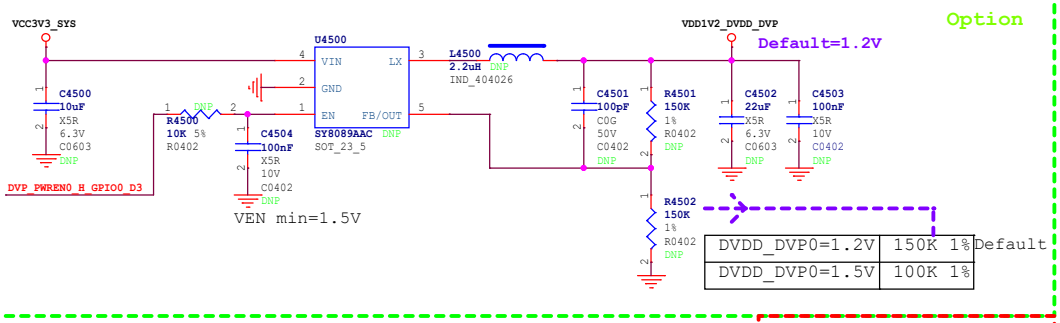
Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	42.Flash-MicroSD Card				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	34 of 61

DVP_PWREN0_H_GPIO0_D3

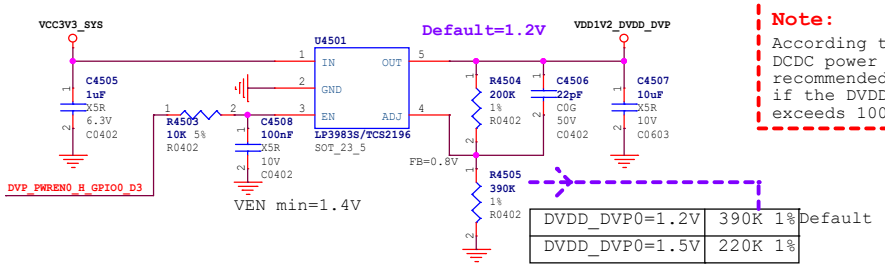
Camera Power supply

If there are multiple cameras,
Please evaluate whether the supply current is sufficient.
Please confirm whether the voltage and power on sequence of different cameras are the same

Option



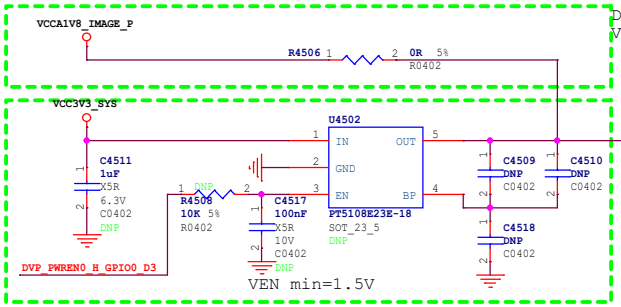
DVDD_DVP0=1.2V	150K 1%	Default
DVDD_DVP0=1.5V	100K 1%	



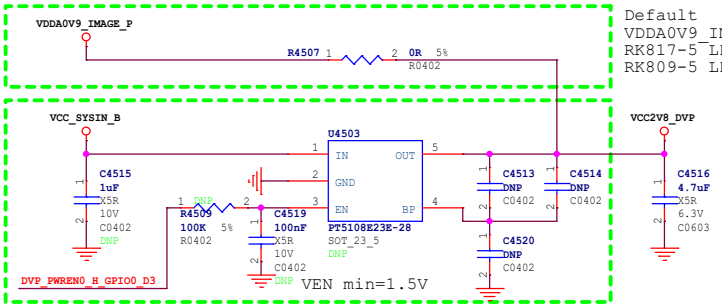
DVDD_DVP0=1.2V	390K 1%	Default
DVDD_DVP0=1.5V	220K 1%	

Note:
The power supply of DVDD refer to camera module specs.

Note:
According to camera model, DCDC power supply is recommended if the DVDD current exceeds 100mA



Default:
VCCA1V8_IMAGE When select VCC_1V8



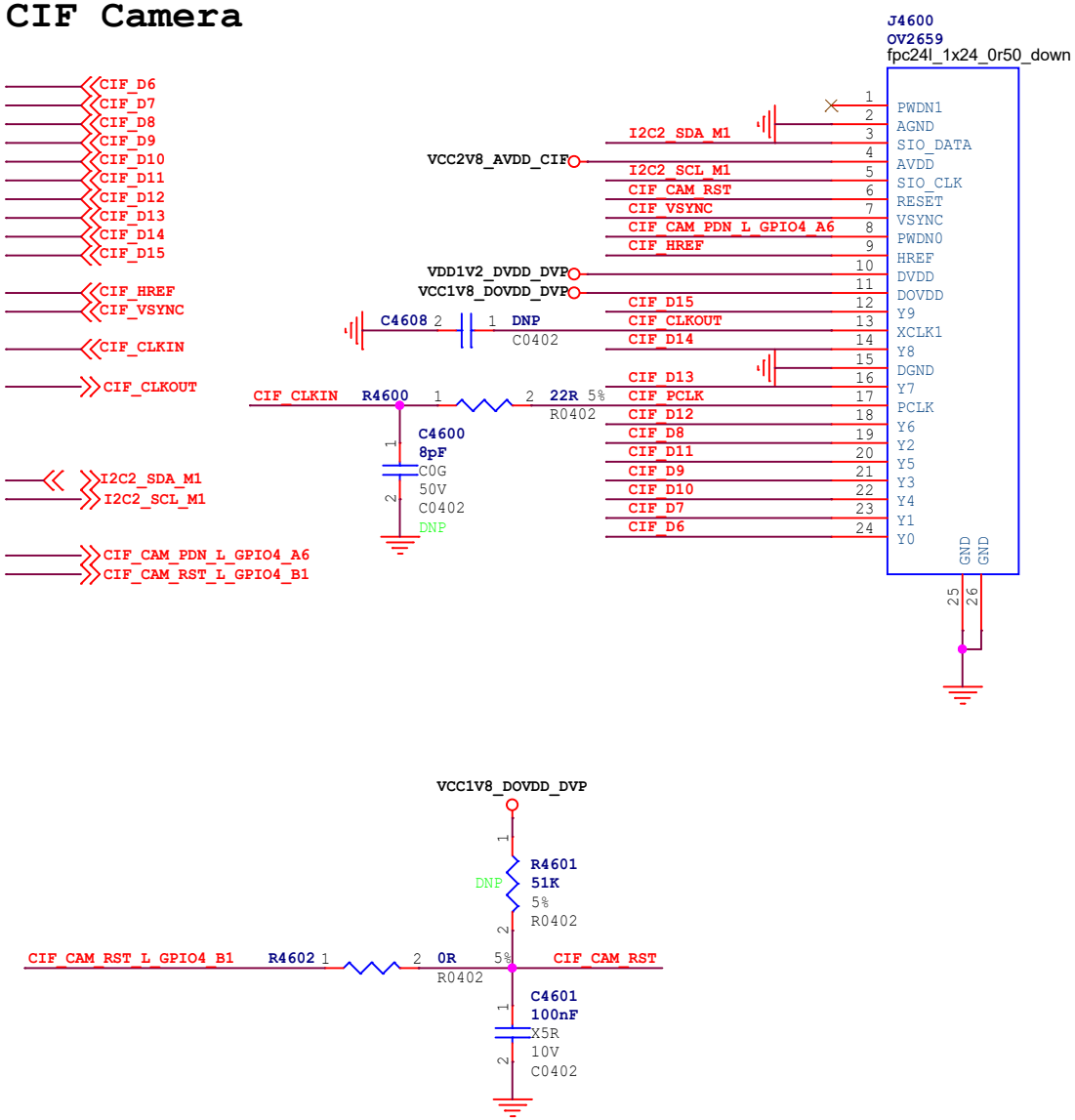
Default
VDDA0V9_IMAGE When select VDD_LOGIC
RK817-5 LDO8 Configure 2.8V output
RK809-5 LDO1 Configure 2.8V output

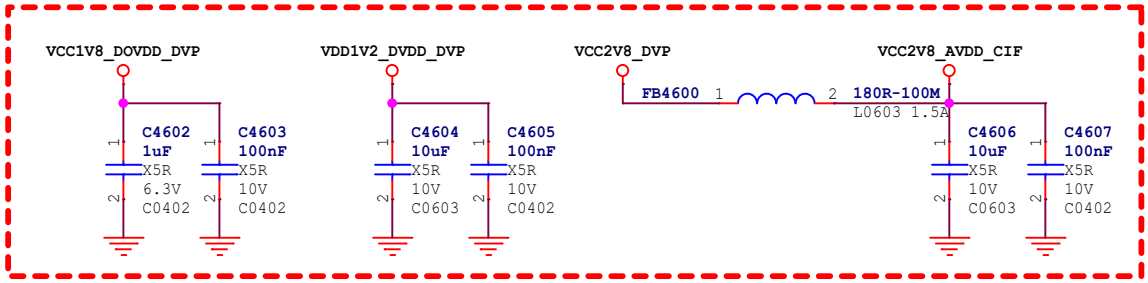
Note:
Adjust the power on sequence according to the camera model
eg:GC8034
Power on Sequence
1.8V-->1.2V-->2.8V--->MCLK-->PWDN--->RST

Rockchip Confidential

Rockchip 瑞芯微电子 Rockchip Electronics Co., Ltd					
Project:	RK3568_Tablet_REF_SCH				
File:	45.VI-Camera_Power				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	35 of 61

CIF Camera





Close to FPC Connector

Mode	16bit	12bit	10bit	8bit
CIF_D0	D0	--	--	--
CIF_D1	D1	--	--	--
CIF_D2	D2	--	--	--
CIF_D3	D3	--	--	--
CIF_D4	D4	D0	--	--
CIF_D5	D5	D1	--	--
CIF_D6	D6	D2	D0	--
CIF_D7	D7	D3	D1	--
CIF_D8	D8	D4	D2	D0
CIF_D9	D9	D5	D3	D1
CIF_D10	D10	D6	D4	D2
CIF_D11	D11	D7	D5	D3
CIF_D12	D12	D8	D6	D4
CIF_D13	D13	D9	D7	D5
CIF_D14	D14	D10	D8	D6
CIF_D15	D15	D11	D9	D7

Support BT601 YCbCr 422 8bit input
Support BT656 YCbCr 422 8bit input
Support RAW 8/10/12bit input
Support BT1120 YCbCr 422 8/10/12/16bit input, single/dual-edge samplin
Support 2/4 mixed BT656/BT1120 YCbCr 422 8bit input

BT1120 16bit Mode:
Default: D0-D7 <--> Y0-Y7 , D8-D15 <--> C0-C7
Swap ON: D0-D7 <--> C0-C7 , D8-D15 <--> Y0-Y7

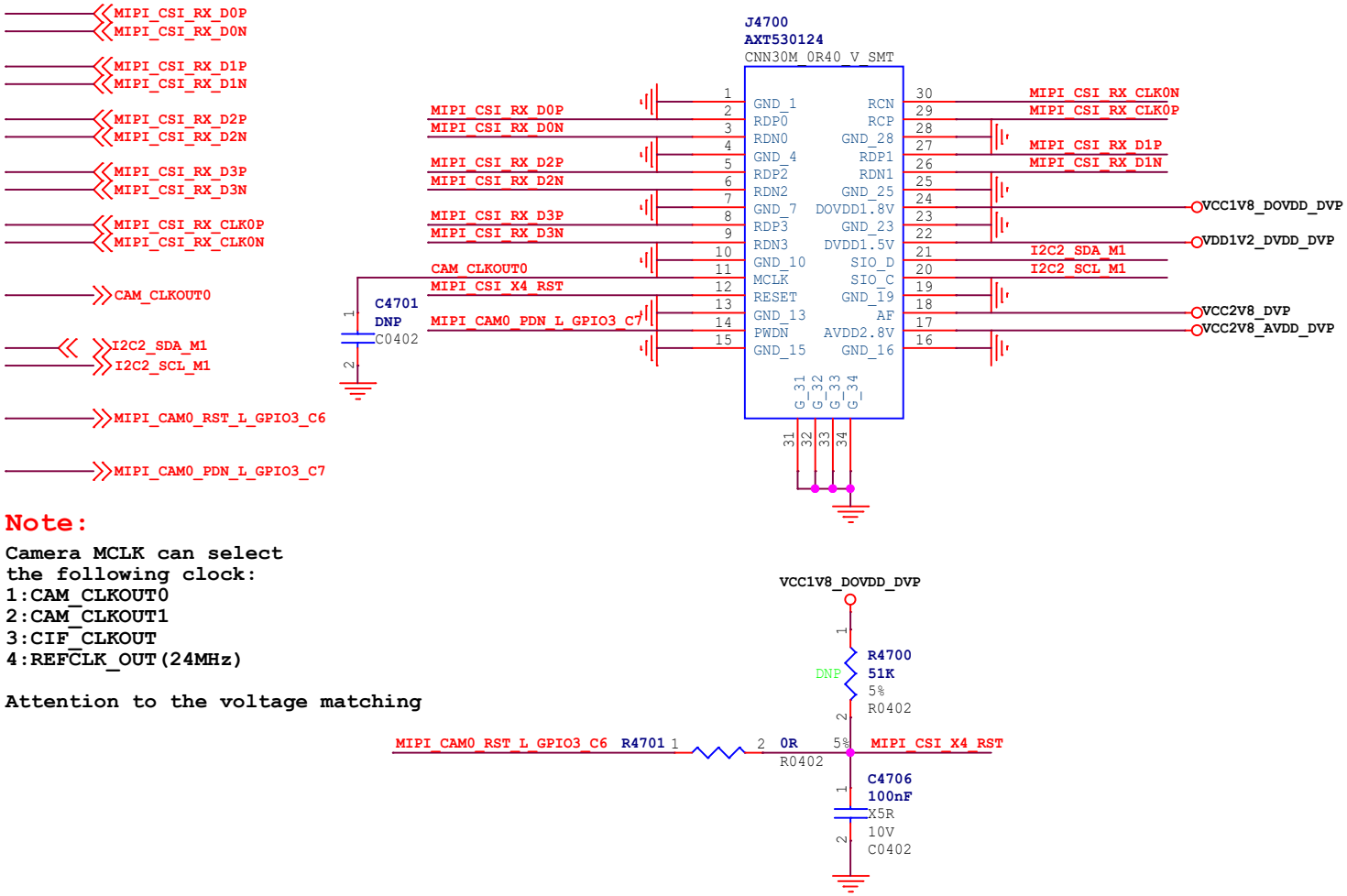
Note:

Camera MCLK can select
the following clock:
1:CAM_CLKOUT0
2:CAM_CLKOUT1
3:CIF_CLKOUT
4:REFCLK_OUT(24MHz)

Attention to the voltage matching

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	46.VI-Camera_CIF				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	36 of 61

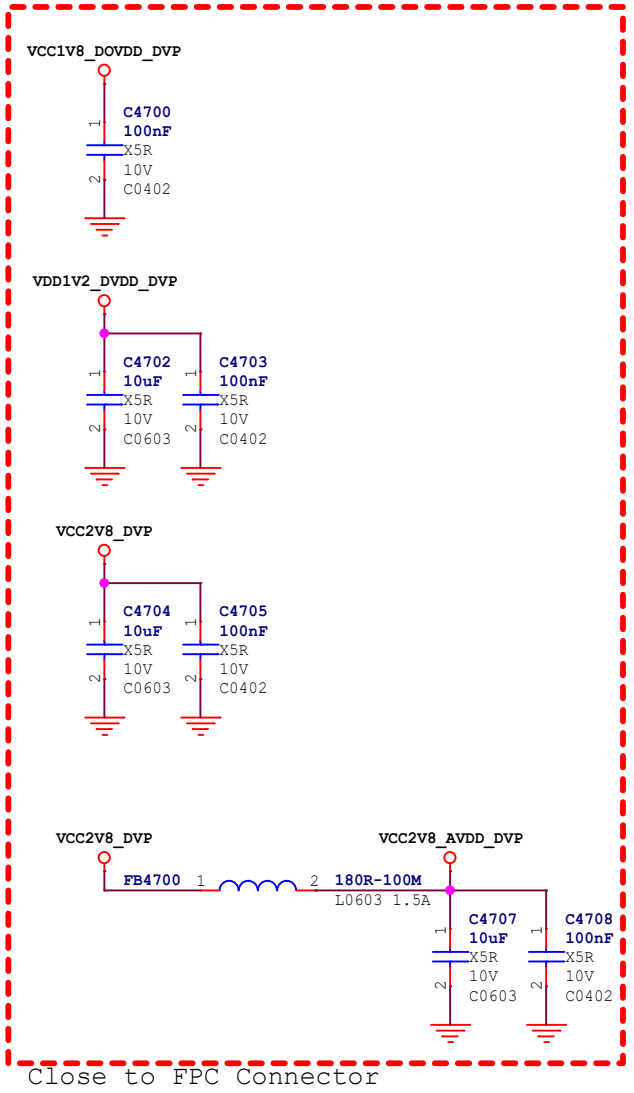
MIPI Camera:MIPI_CSI_RX 1 x 4Lanes



Note:

Camera MCLK can select
the following clock:
1:CAM_CLKOUT0
2:CAM_CLKOUT1
3:CIF_CLKOUT
4:REFCLK_OUT(24MHz)

Attention to the voltage matching

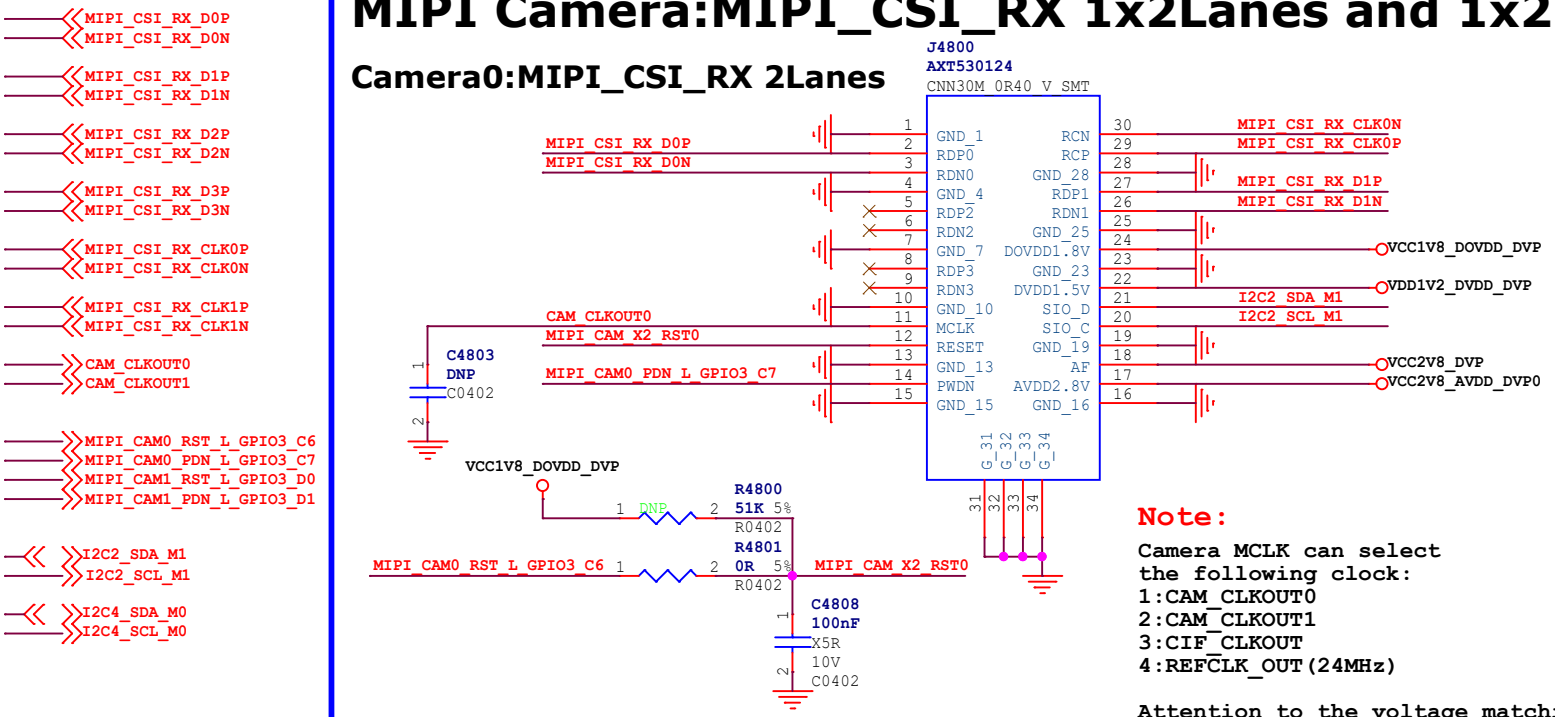


Rockchip Confidential

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	47.VI-Camera_MIPI_CSI_1x 4Lanes				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	37 of 61

MIPI Camera:MIPI_CSI_RX 1x2Lanes and 1x2Lanes

Camera0:MIPI_CSI_RX 2Lanes



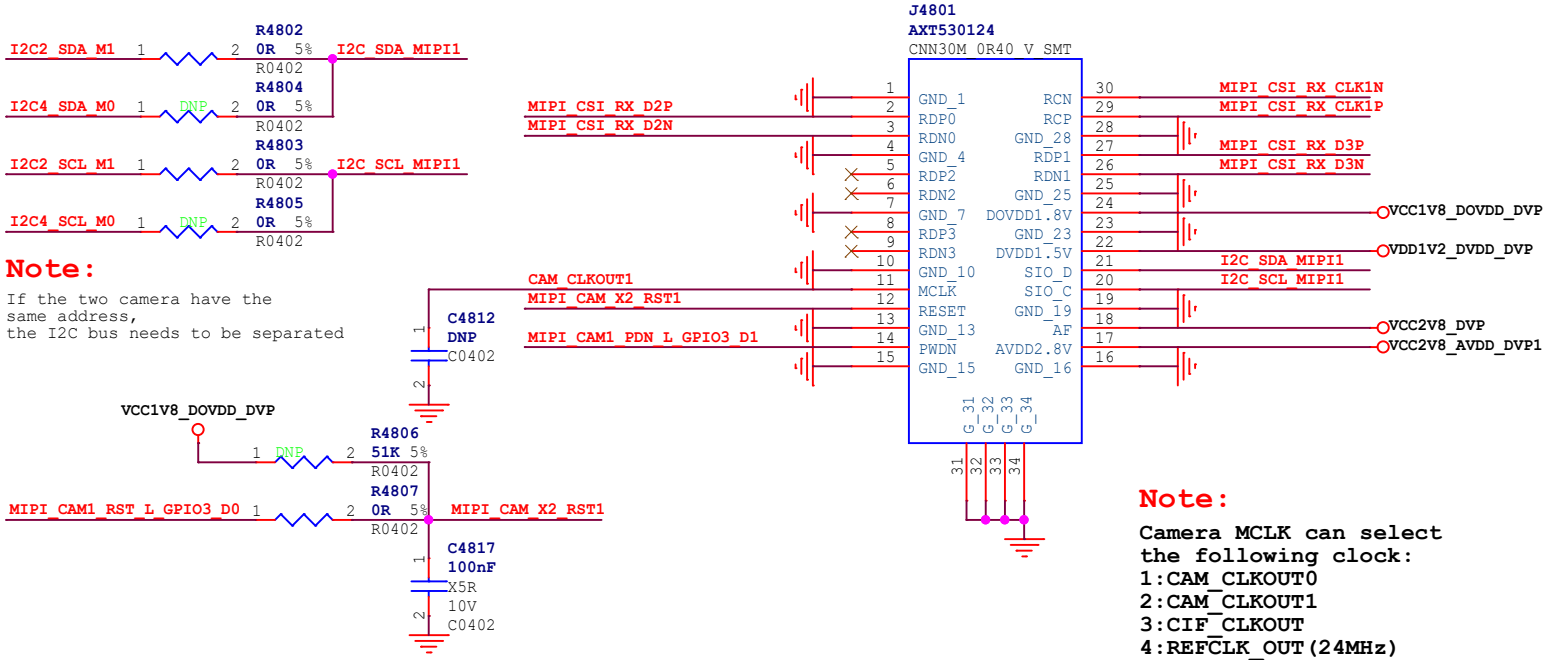
Note:

Camera MCLK can select the following clock:
1:CAM_CLKOUT0
2:CAM_CLKOUT1
3:CIF_CLKOUT
4:REFCLK_OUT(24MHz)

Attention to the voltage matching

Close to FPC Connector

Camera1:MIPI_CSI_RX 2Lanes



Note:

If the two camera have the same address, the I2C bus needs to be separated

Note:

Camera MCLK can select the following clock:
1:CAM_CLKOUT0
2:CAM_CLKOUT1
3:CIF_CLKOUT
4:REFCLK_OUT(24MHz)

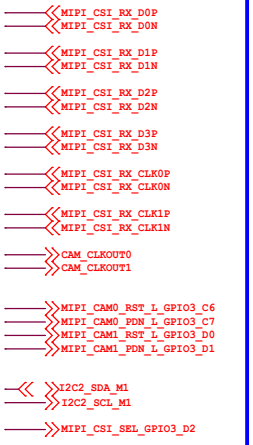
Attention to the voltage matching

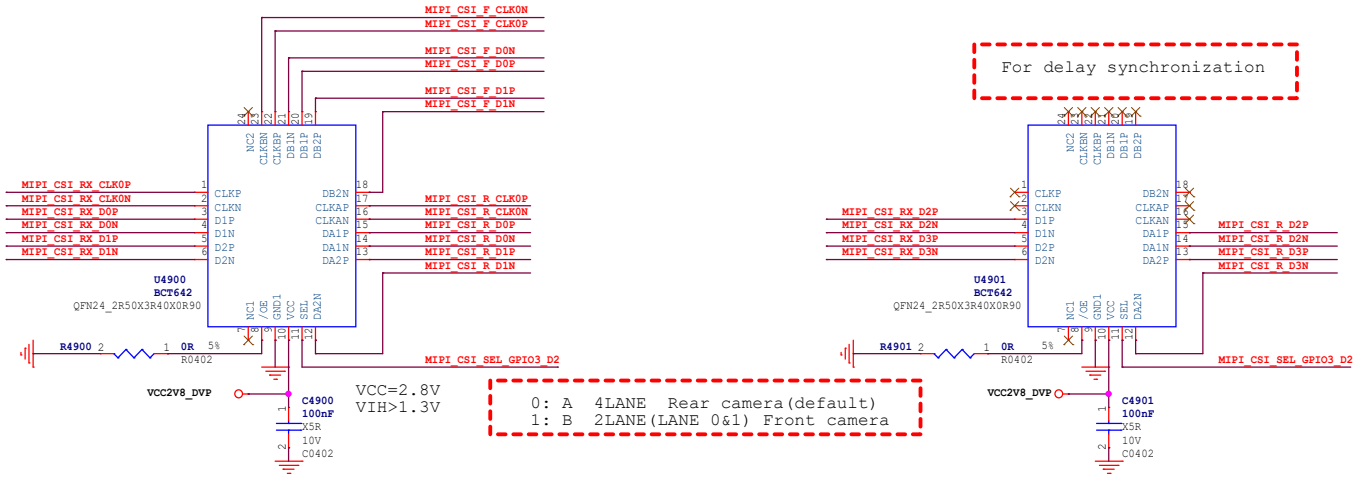
Close to FPC Connector

Rockchip Confidential

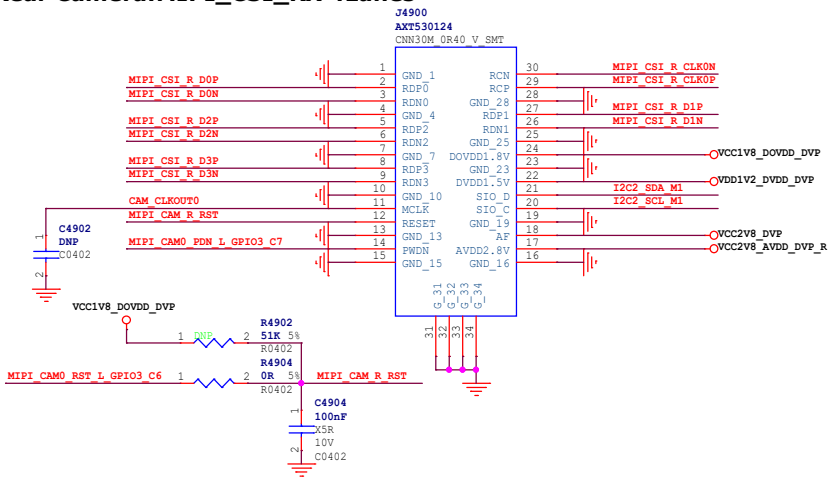
Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	48.VI-Camera_MIPI_CSI_2x 2Lanes				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	38 of 61

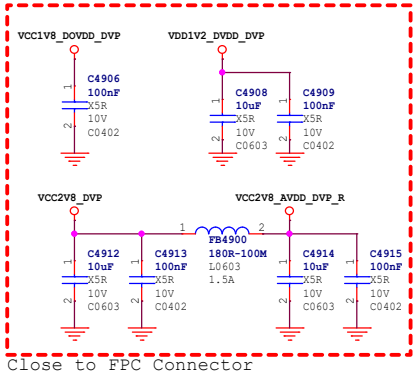
MIPI Camera:MIPI_CSI_RX 1 x 4Lanes or 1x 2Lanes



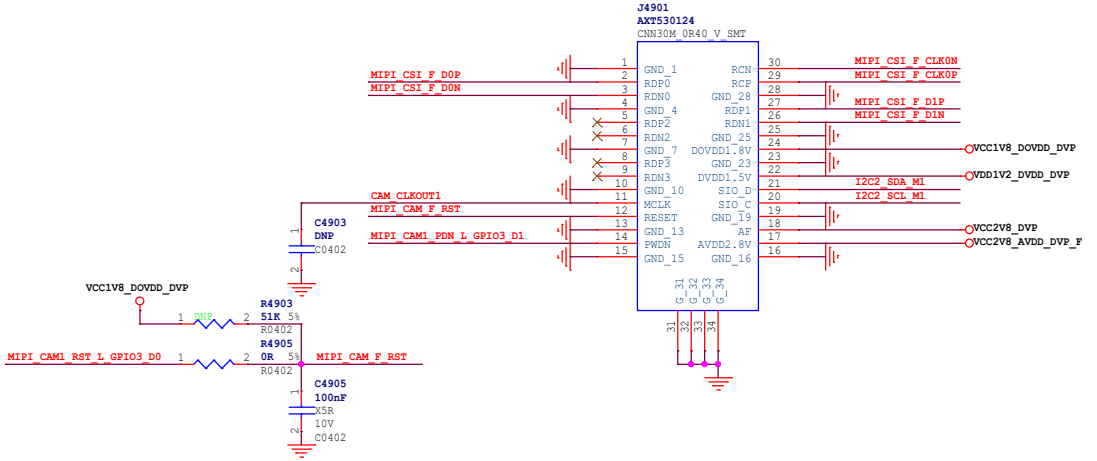


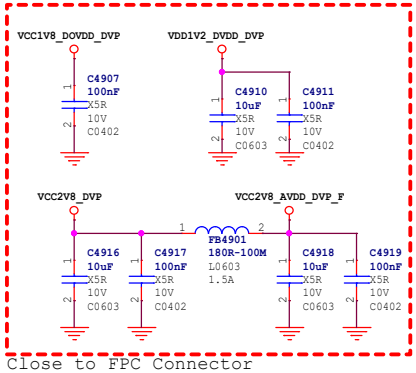
Rear Camera:MIPI_CSI_RX 4Lanes





Front Camera:MIPI_CSI_RX 2Lanes





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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	49.VI-Camera_MIPI_CSI_4or2Lanes				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	39 of 61

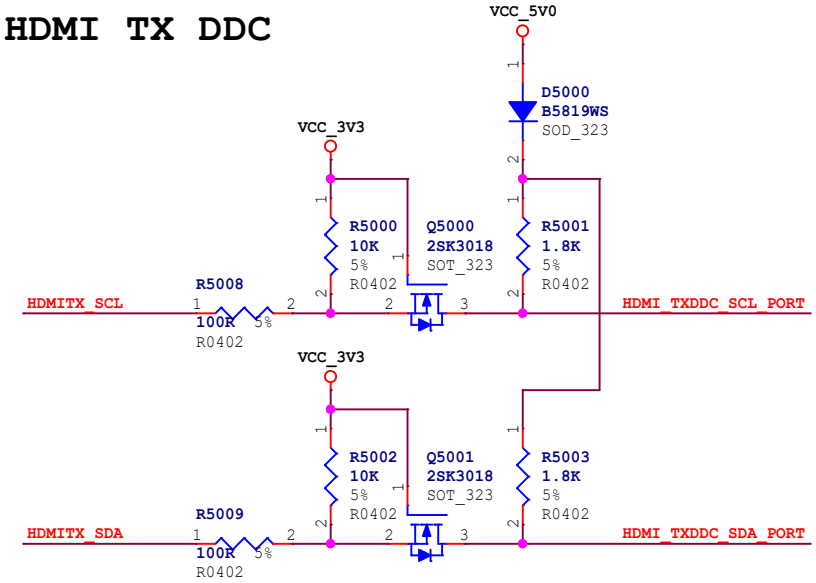
HDMI2.0 TX

HDMI_TX2P_PORT
HDMI_TX2N_PORT
HDMI_TX1P_PORT
HDMI_TX1N_PORT
HDMI_TX0P_PORT
HDMI_TX0N_PORT
HDMI_TXCLKP_PORT
HDMI_TXCLKN_PORT

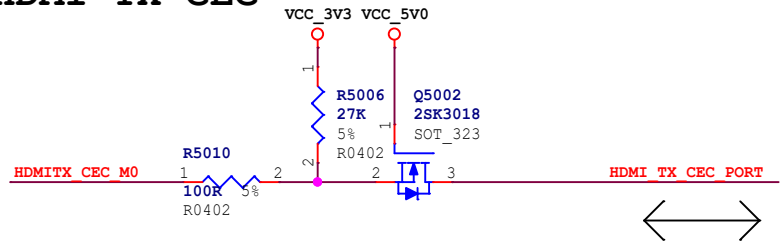
HDMITX_SCL
HDMITX_SDA

HDMITX_CEC_M0
HDMI_TX_HPDIN

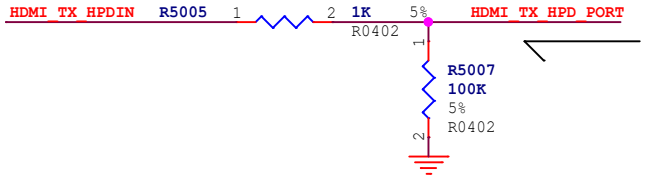
HDMI TX DDC



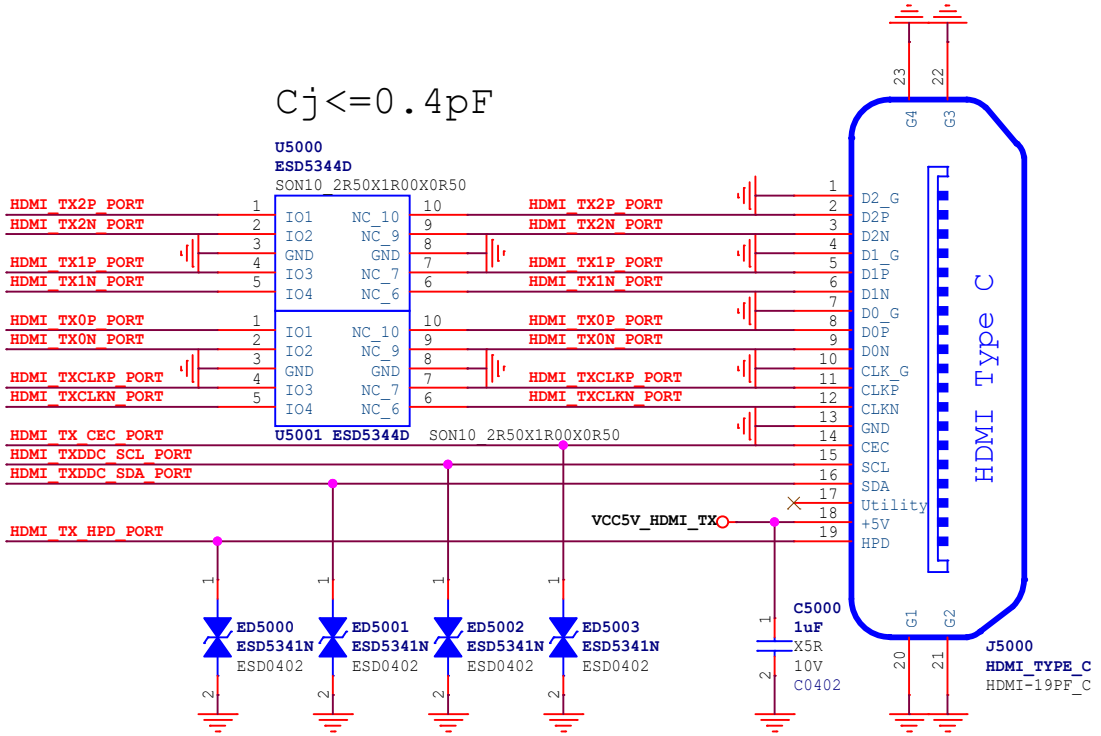
HDMI TX CEC



HDMI TX HPD

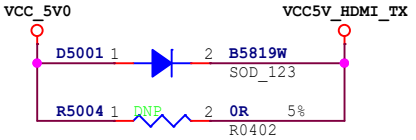


Cj<=0.4pF



Note:

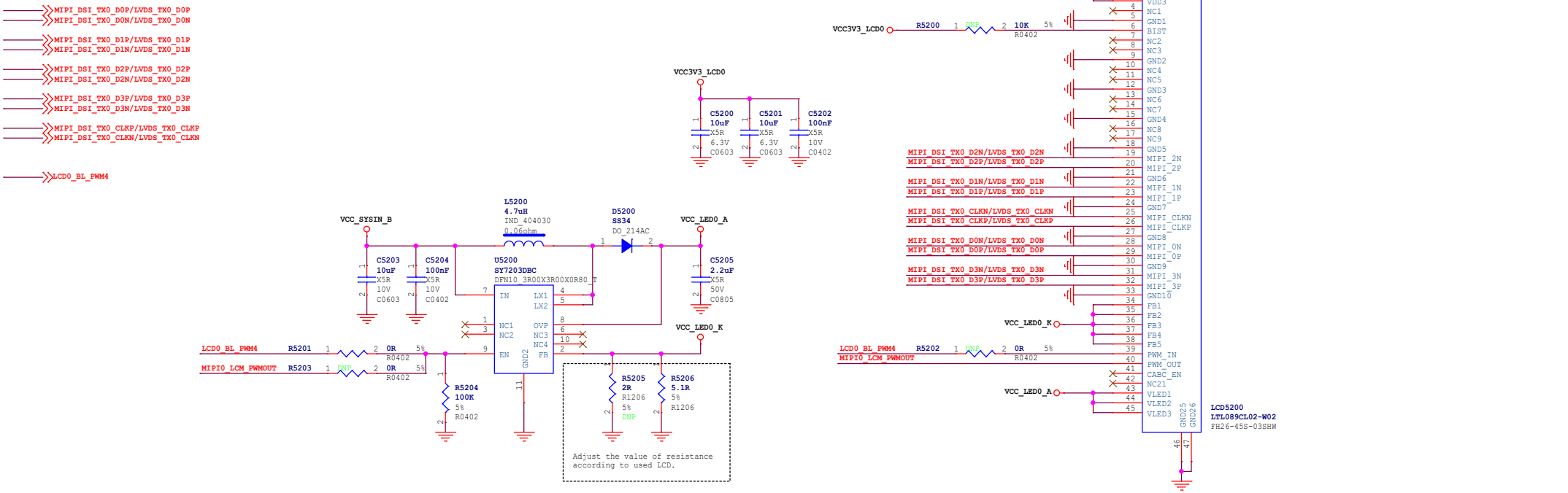
This reference circuit use TYPE C by default
If the project needs TYPE A, you must replace the library with TYPE A.



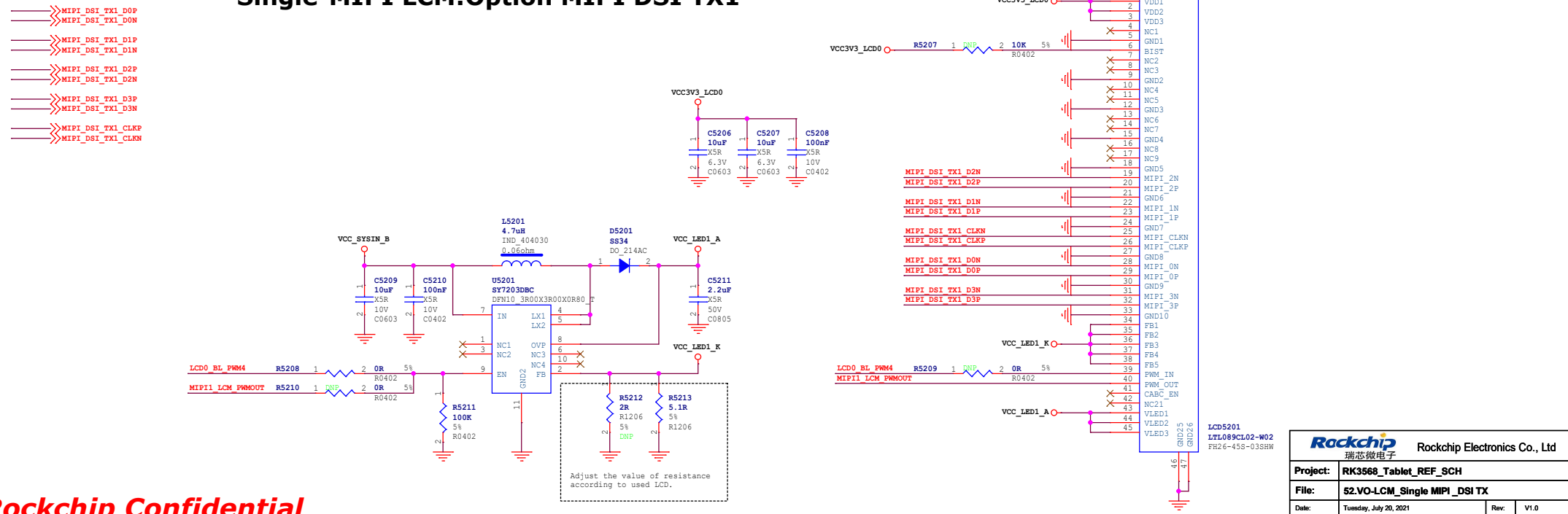
Rockchip Confidential

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	50.VO-HDMI2.0 TX				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	40 of 61

Single-MIPI LCM:Default MIPI DSI TX0



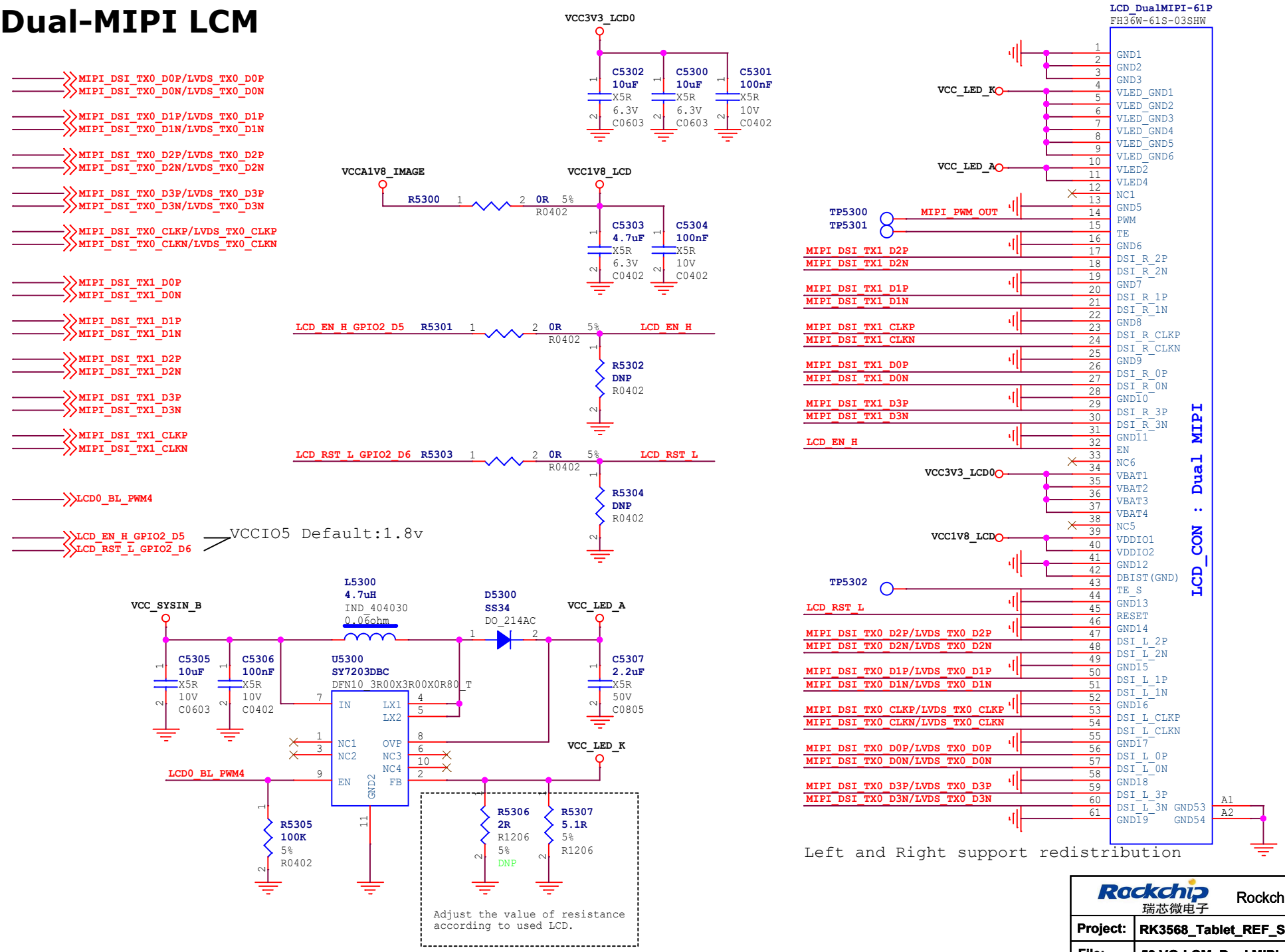
Single-MIPI LCM:Option MIPI DSI TX1



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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	52.VO-LCM_Single MIPI _DSI TX				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	41 of 61

Dual-MIPI LCM



Left and Right support redistribution

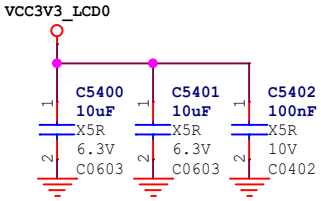
Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	53.VO-LCM_Dual MIPI_DSI TX				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	42 of 61

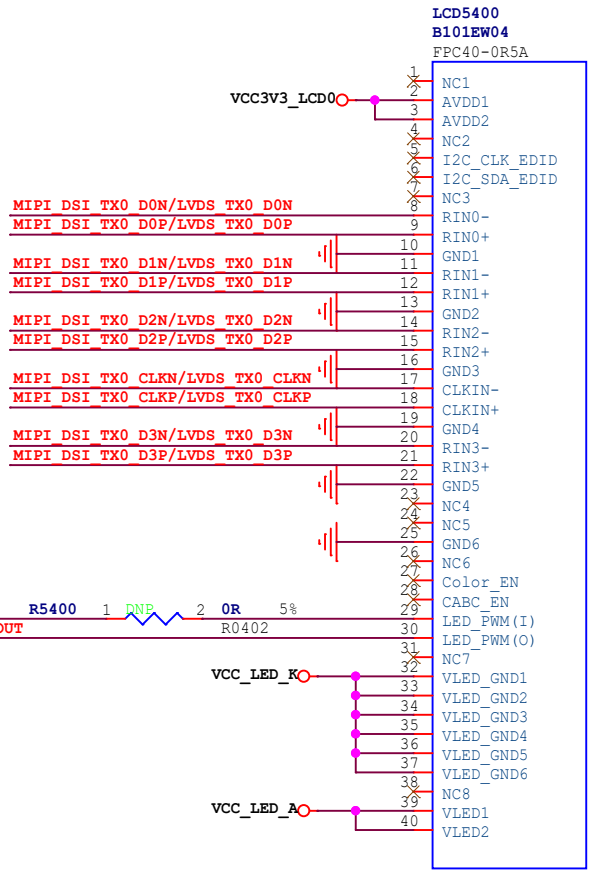
Rockchip Confidential

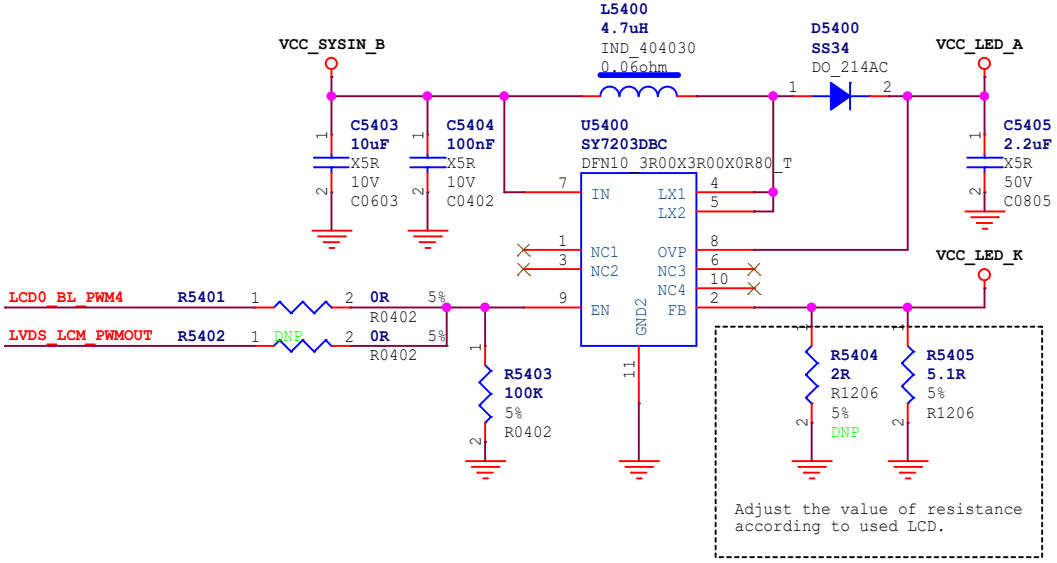
Single-LVDS LCM

MIPI_DSI_TX0_D0P/LVDS_TX0_D0P
MIPI_DSI_TX0_D0N/LVDS_TX0_D0N
MIPI_DSI_TX0_D1P/LVDS_TX0_D1P
MIPI_DSI_TX0_D1N/LVDS_TX0_D1N
MIPI_DSI_TX0_D2P/LVDS_TX0_D2P
MIPI_DSI_TX0_D2N/LVDS_TX0_D2N
MIPI_DSI_TX0_D3P/LVDS_TX0_D3P
MIPI_DSI_TX0_D3N/LVDS_TX0_D3N
MIPI_DSI_TX0_CLKP/LVDS_TX0_CLKP
MIPI_DSI_TX0_CLKN/LVDS_TX0_CLKN

LCD0_BL_PWM4



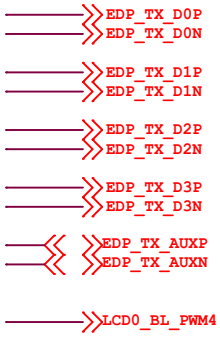


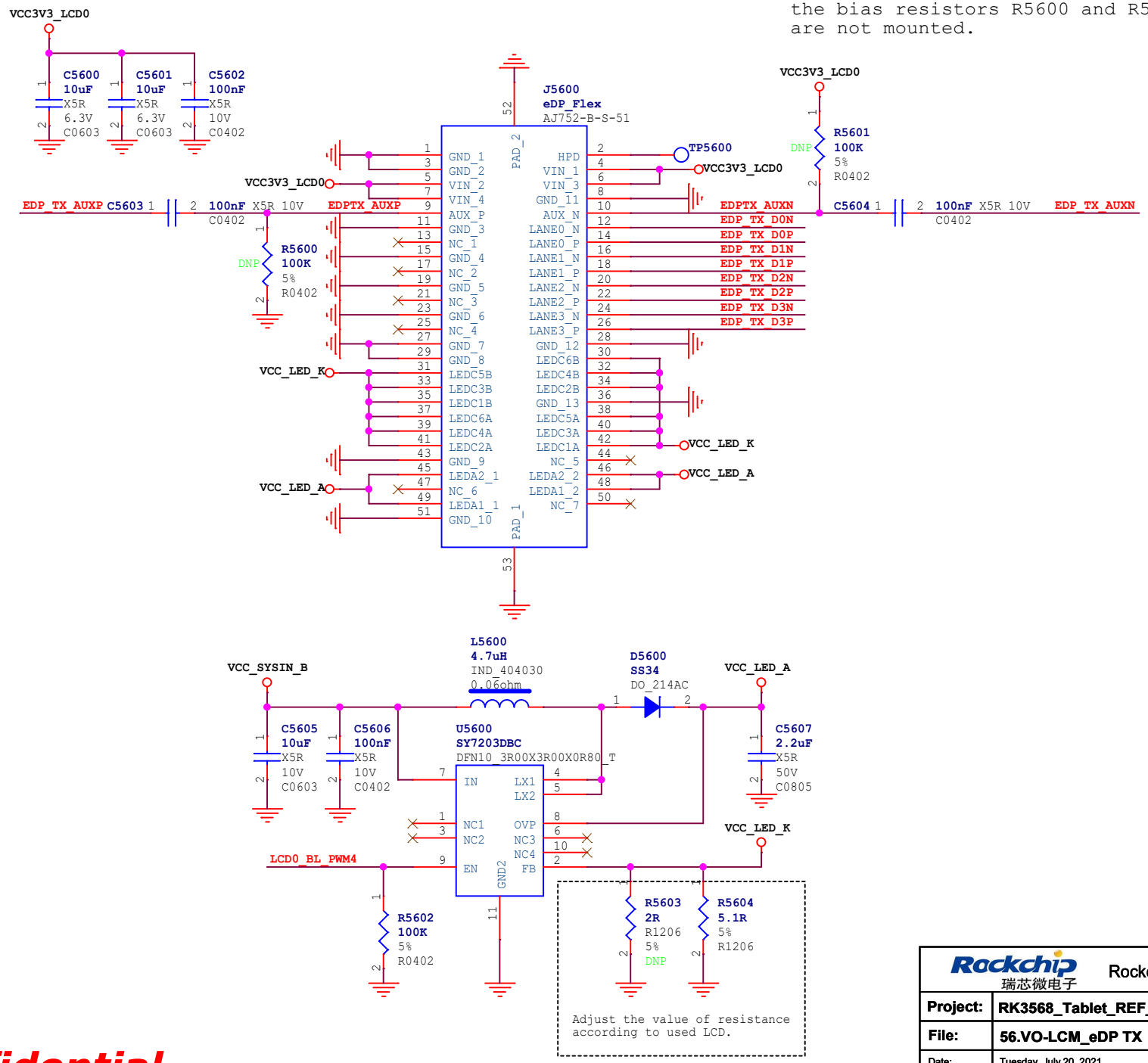


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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	54.VO-LCM_Single LVDS TX				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	43 of 61

Single-eDP LCM





Note:

For EDP displays with edp1.2a or above, the bias resistors R5600 and R5601 of aux are not mounted.

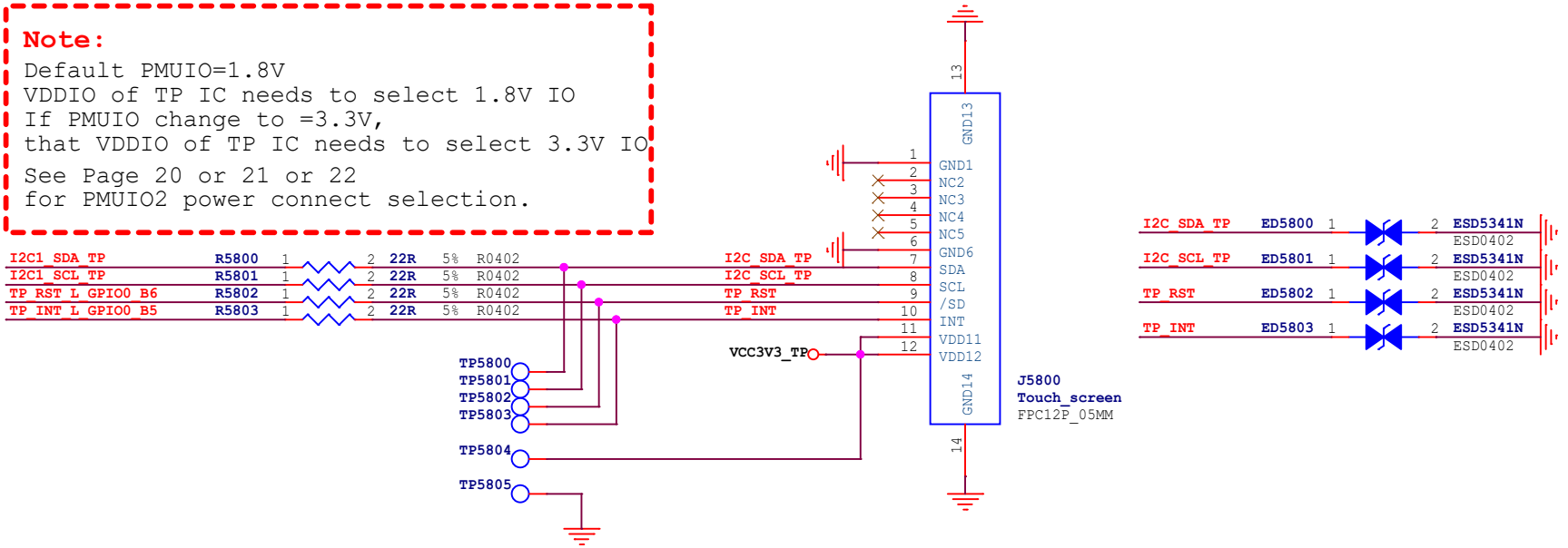
Rockchip Confidential

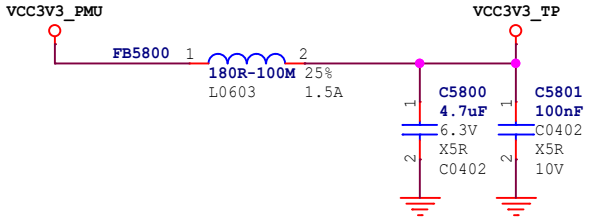
Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	56.VO-LCM_eDP TX				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	44 of 61

Touch Panel connector



Note:
Default PMUIO=1.8V
VDDIO of TP IC needs to select 1.8V IO
If PMUIO change to =3.3V,
that VDDIO of TP IC needs to select 3.3V IO
See Page 20 or 21 or 22
for PMUIO2 power connect selection.



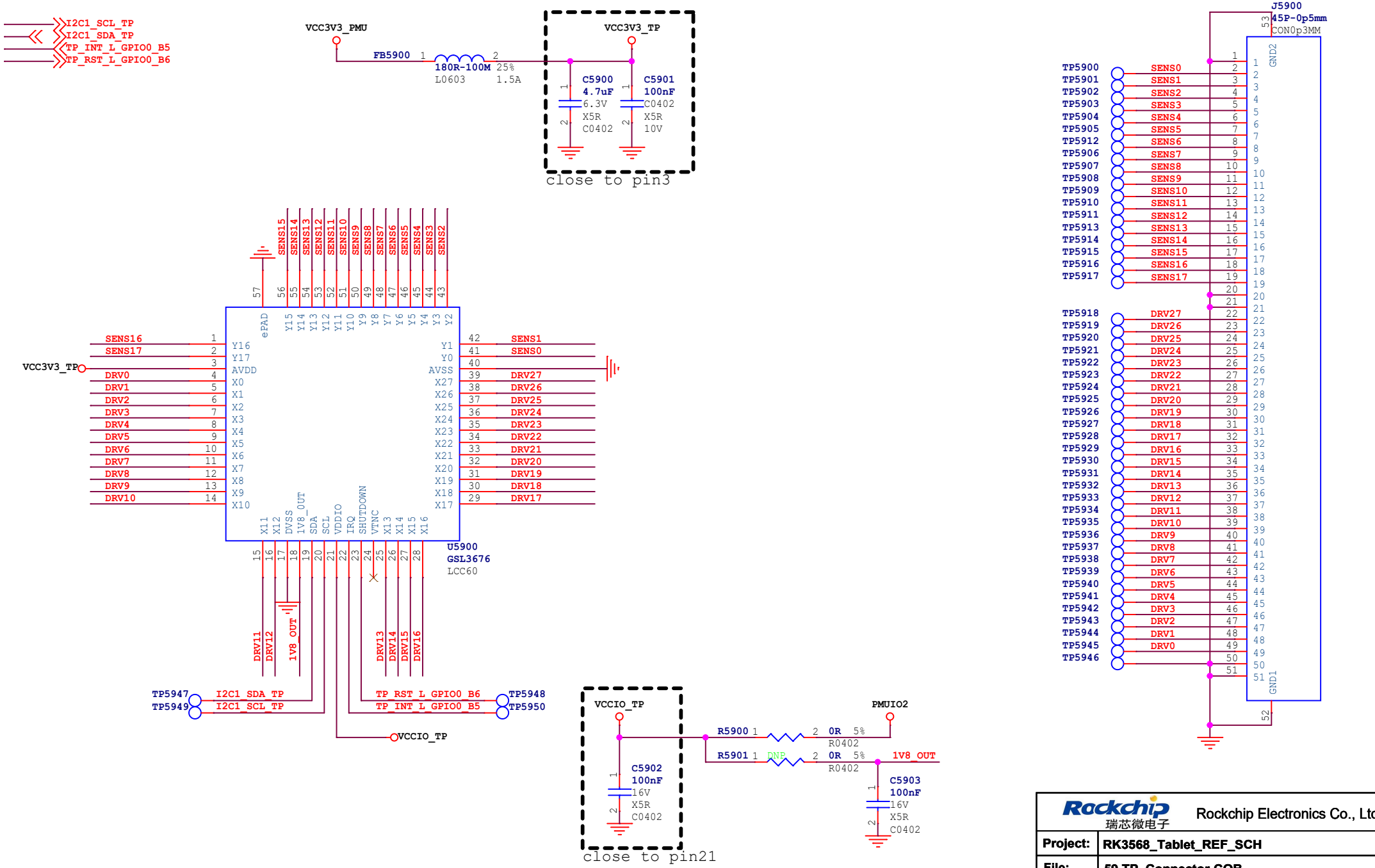


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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	58.TP Connector_COF				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	45 of 61

TP COB Connector

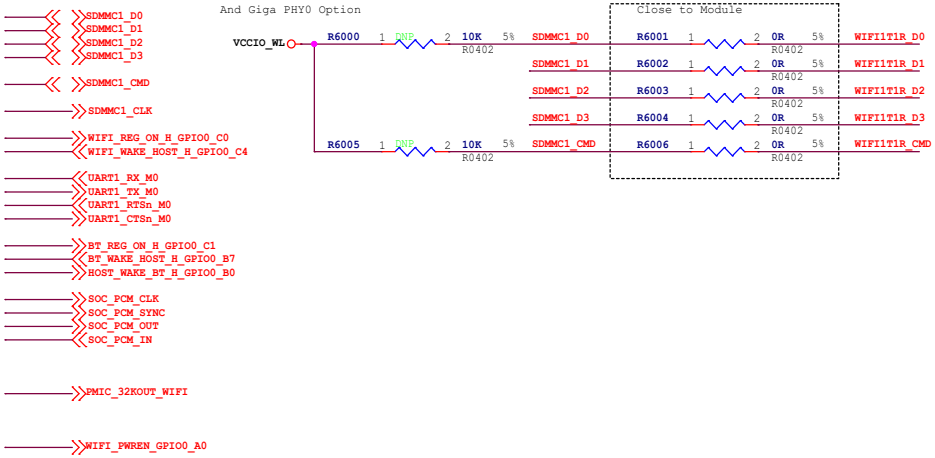
Note:26x14 channel,
suit for Touch Panel size≤12''

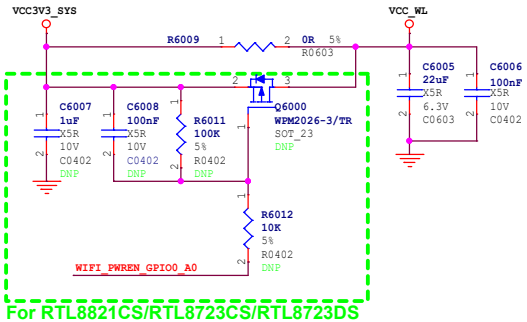


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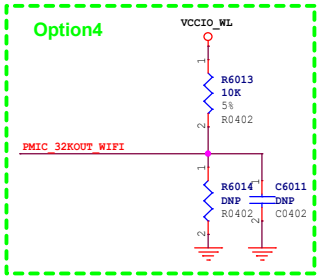
Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	59.TP_Connector-COB				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	46 of 61

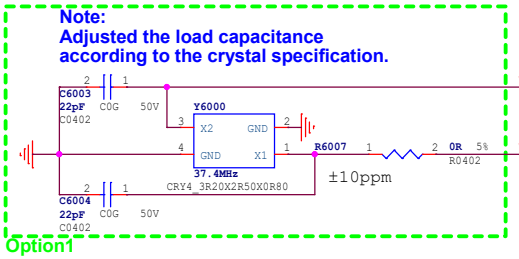
SDIO WIFI/BT MODULE





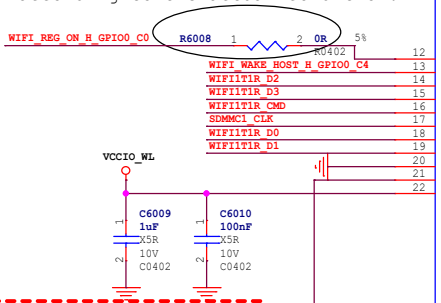
Note:
The maximum peak current is 600mA
Close to WIFI module



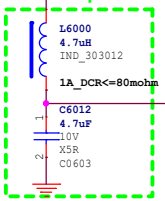


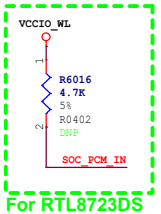
50 Ohm RF trace

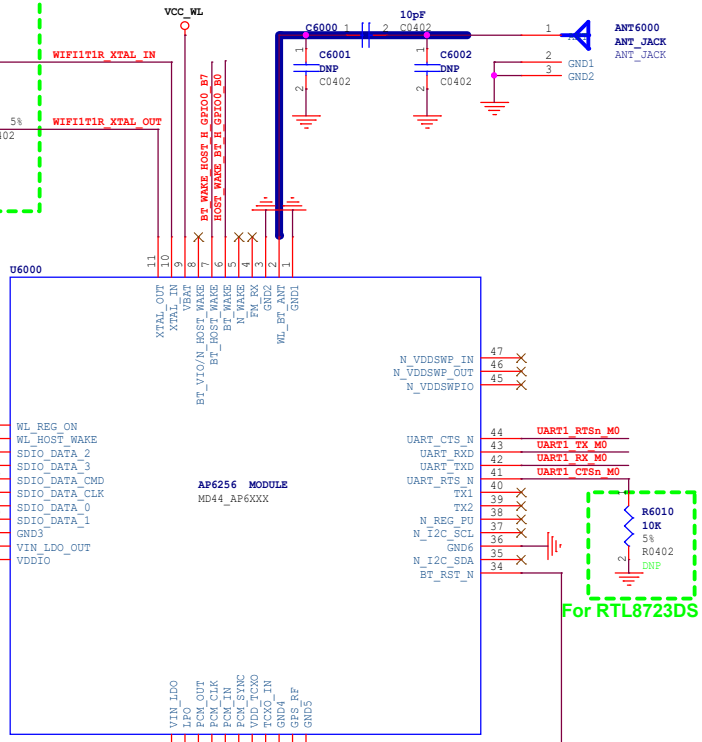
Using RTL8189ETV/FTV modules,
please notice
WIFI REG ON is on pin12 or pin34,
choose
according to the actual condition.



Note:
See Page 20 or 21 or 22
for VCCIO_WL power
connect selection.







Using RTL8189ETV/FTV modules, please notice
WIFI REG ON is on pin12 or pin34, choose
according to the actual condition.

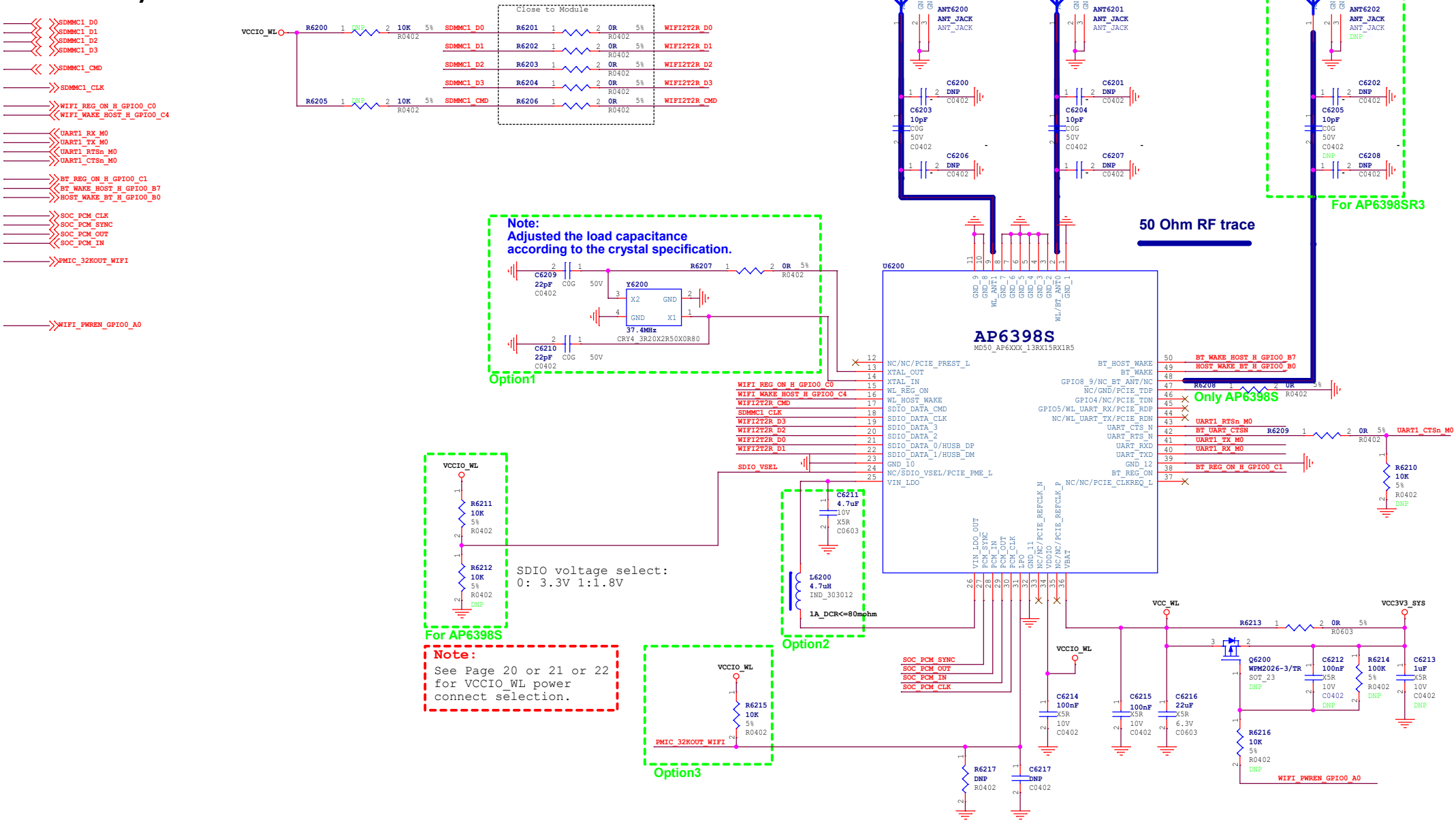
OPTION	WIFI				BT	Crystals	VDDIO	Option1	Option2	Option3	Option4
	a	b/g/n	ac	5GHz							
AW-CM256SM	Yes	Yes	Yes	Yes	4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	Yes
AP6236/AP6212	No	Yes	No	No	4.2/4.0	26MHz	1.71-3.6V	Yes	Yes	No	Yes
AP6256/AP6255	Yes	Yes	Yes	Yes	5.0/4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	Yes
RTL8189ETV Module F89FTSM12-W3	No	Yes	No	No	No	Module Integrated	1.8-3.3V	No	No	No	No
RTL8723BS Module F23BDSM23-W2	No	Yes	No	No	4.0	Module Integrated	1.62-3.6V	No	No	No	No
RTL8723DS Module 6223A-SRD	No	Yes	No	No	4.2	Module Integrated	1.62-3.6V	No	No	No	No
QCA9377 Module 8223A-SR	Yes	Yes	Yes	Yes	4.2	Module Integrated	1.7-3.45V	No	No	No	Yes
RTL8821CS Module 6221A-SRC	Yes	Yes	Yes	Yes	4.2	Module Integrated	1.7-3.45V	No	No	No	No

Note:
Yes: option circuit be mounted
No: option circuit not be mounted

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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd
Project:	RK3568_Tablet_REF_SCH
File:	60.WIFI/BT-SDMMC1_1T1R + UART
Date:	Tuesday, July 20, 2021
Rev:	V1.0
Designed by:	Zhangdz
Reviewed by:	Default
Sheet:	47 of 61

SDIO WIFI/BT MODULE-2T2R



OPTION	WIFI				BT	Crystals	VDDIO	Option1	Option2	Option3
	a	b/g/n	ac	5GHz						
AP6398S	Yes	Yes	Yes	Yes	5.0	37.4MHz	1.71-3.6V	Yes	Yes	Yes
AP6356S	Yes	Yes	Yes	Yes	4.1	37.4MHz	1.71-3.6V	Yes	Yes	Yes
RTL8822BS Module	Yes	Yes	Yes	Yes	4.1	Module Integrated	1.71-3.6V	No	No	No

Note:
Yes: option circuit be mounted
No: option circuit not be mounted

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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	62.WIFI/BT-SDMMC1_2T2R + UART				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	48 of 61

PCIe WIFI6/BT MODULE-2T2R

WIFI_REG_ON_H_GPIO0_C0
WIFI_WAKE_HOST_H_GPIO0_C4
UART1_RX_M0
UART1_TX_M0
UART1_RTSn_M0
UART1_CTSn_M0
BT_REG_ON_H_GPIO0_C1
BT_WAKE_HOST_H_GPIO0_B7
HOST_WAKE_BT_H_GPIO0_B0
SOC_PCM_CLK
SOC_PCM_SYNC
SOC_PCM_OUT
SOC_PCM_IN
PMIC_32KOUT_WIFI

PCIE20_TXP
PCIE20_TXN
PCIE20_RXP
PCIE20_RXN
PCIE20_REFCLKP
PCIE20_REFCLKN
PCIE20_CLKREQn_M1
PCIE20_WAKEn_M1
PCIE20_PERSTn_M1

VCCIO5 IO Domain Default:1.8v

Note:
Adjusted the load capacitance according to the crystal specification.

ANT6400 ANT_JACK
ANT6401 ANT_JACK
ANT6402 ANT_JACK
C6400 DNP C0402
C6401 DNP C0402
C6402 DNP C0402
C6403 10pF C0G 50V C0402
C6404 DNP C0402
C6405 DNP C0402
C6406 DNP C0402
R6400 0R 5% R0402
R6401 0R 5% R0402

50 Ohm RF trace
For AP6275PR3

C6407 22pF C0G 50V C0402
C6409 22pF C0G 50V C0402
Y6400 37.4MHz CRY4_3R20X2R50X0R80
R6402 0R 5% R0402

U6400
PCIE_PERST_L
XTAL_IN
XTAL_OUT
WL_REG_ON
WL_HOST_WAKE/WL_GPIO_0
NC1
NC2
BT_PCM_OUT
BT_PCM_IN
BT_PCM_SYNC
BT_PCM_CLK
GND10
PCIE_PME_L
CBUCK_0P9
BT_HOST_WAKE
BT_WAKE
NC4
PCIE_TX_P
PCIE_TX_N
PCIE_RX_P
PCIE_RX_N
BT_UART_CTS_N
BT_UART_RTS_N
BT_UART_RXD
BT_UART_TXD
GND14
BT_REG_ON
PCIE_CLKREQ_L
CSR_VLX
GND11
ASR_VLX
ABUCK_1P12
GND12
LPO
GND13
PCIE_RCLK_N
VDDIO
PCIE_RCLK_P
VBAT
AP6275P/AP6275PR3
MD50_AP6XXX_13RX15RX1R5

PCIE20_PERSTn_1V8
WIFI_REG_ON_H_GPIO0_C0
WIFI_WAKE_HOST_H_GPIO0_C4
SOC_PCM_IN
SOC_PCM_OUT
SOC_PCM_SYNC
SOC_PCM_CLK
PCIE20_WAKEn_1V8

BT_WAKE_HOST_H_GPIO0_B7
HOST_WAKE_BT_H_GPIO0_B0
PCIE_TDP C6408 100nF C0201 X5R 10V PCIE20_RXP
PCIE_TDN C6410 100nF C0201 X5R 10V PCIE20_RXN
PCIE_RDP C6411 100nF C0201 X5R 10V PCIE20_TXP
PCIE_RDN C6412 100nF C0201 X5R 10V PCIE20_TXN
UART1_RTSn_M0
UART1_CTSn_M0
UART1_TX_M0
UART1_RX_M0
BT_REG_ON_H_GPIO0_C1
PCIE20_CLKREQn_1V8

C6413 4.7uF 10V X5R C0603

PCIE20_CLKREQn_M1 R6403 22R 5% R0402 PCIE20_CLKREQn_1V8
PCIE20_WAKEn_M1 R6405 22R 5% R0402 PCIE20_WAKEn_1V8
PCIE20_PERSTn_M1 R6406 22R 5% R0402 PCIE20_PERSTn_1V8

VCC_WL
C6414 100nF X5R 10V C0402
C6415 22uF X5R 10V C0805
R6404 0R 5% R0603
VCC3V3_SYS
Max:1.2A

L6400 2.2uH IND_303015 1A_DCR<=80mohm
L6401 2.2uH IND_303015 1A_DCR<=80mohm

VCCIO_WL
C6416 100nF X5R 10V C0402
C6417 4.7uF 10V X5R C0603
C6418 4.7uF 10V X5R C0603

Note:
See Page 20 or 21 or 22 for VCCIO_WL power connect selection.

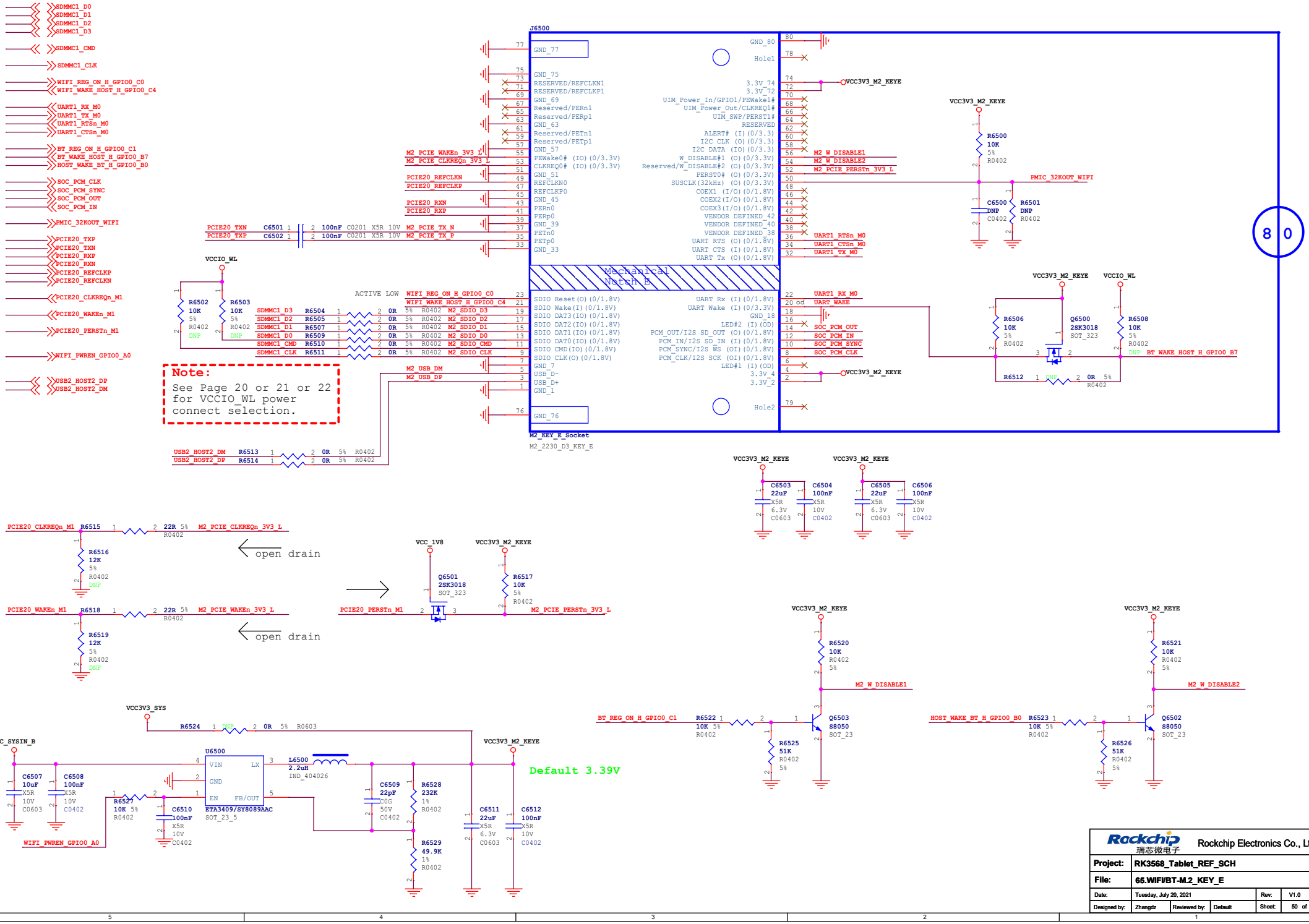
VCCIO_WL
R6407 10K 5% R0402
PMIC_32KOUT_WIFI
R6408 DNP R0402
C6421 DNP C0402

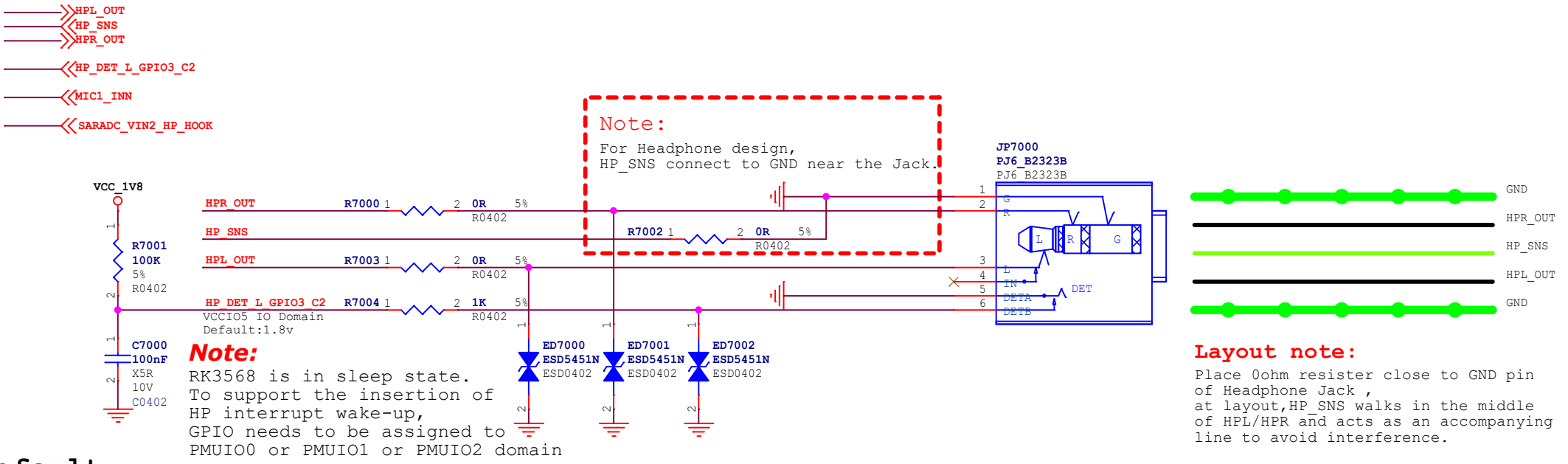
PCIE_REFCLKP
PCIE_REFCLKN
C6419 100pF C0201 X5R 25V PCIE20_REFCLKP
C6420 100pF C0201 X5R 25V PCIE20_REFCLKN

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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd			
Project:	RK3568_Tablet_REF_SCH			
File:	64.WIFI6/BT-PCIe_2T2R + UART			
Date:	Tuesday, July 20, 2021		Rev:	V1.0
Designed by:	Zhangdz	Reviewed by: Default	Sheet:	49 of 61

WIFI/BT-M.2_KEY_E



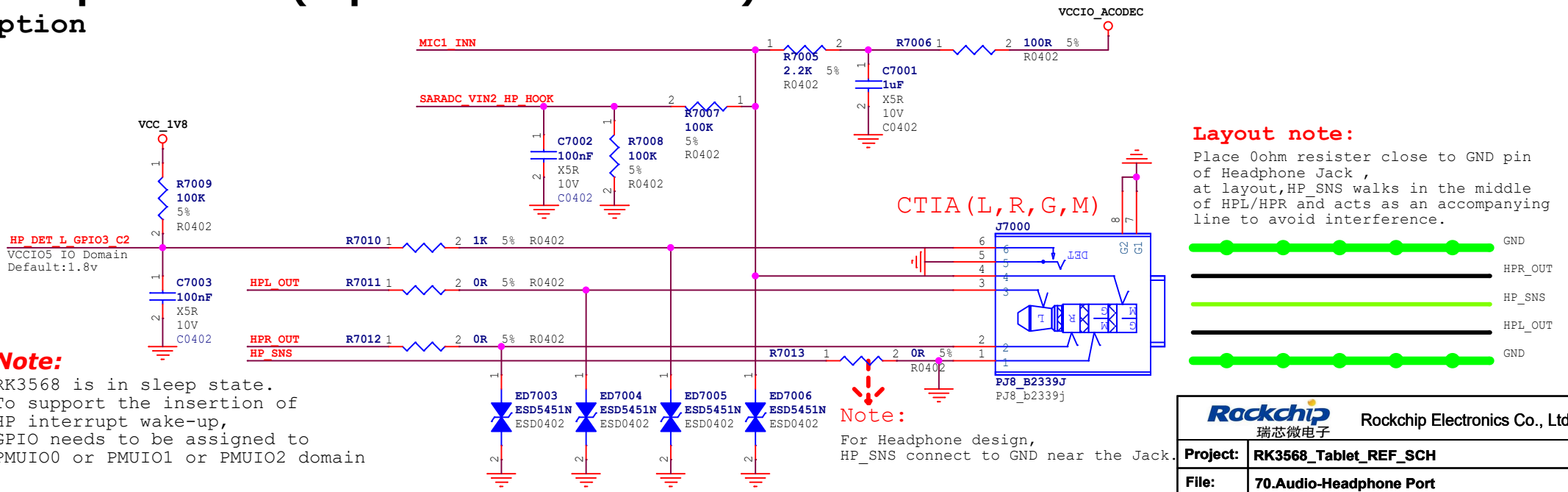


Default

Headphone Jack(3-pole with DET)

Headphone Jack(4-pole with DET & MIC)

Option



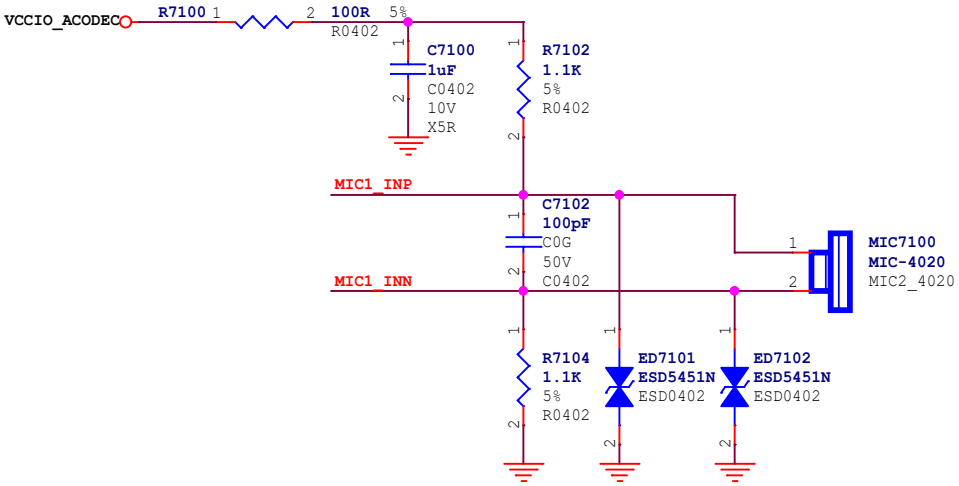
Rockchip Confidential

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	70.Audio-Headphone Port				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	51 of 61

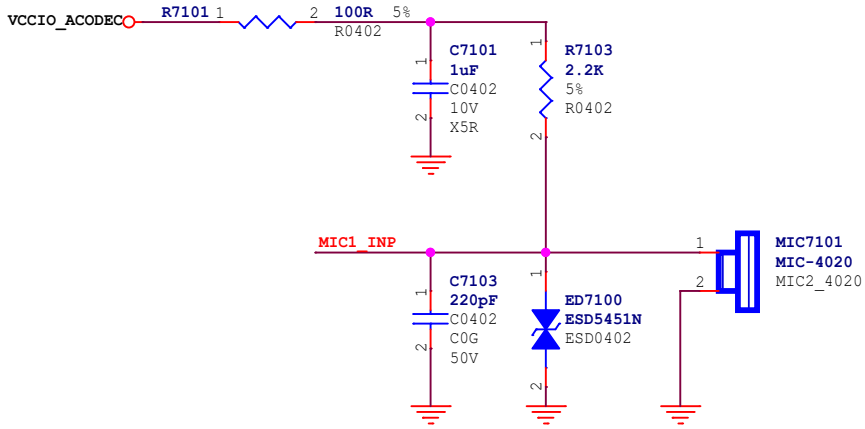
SPKN_OUT
SPKP_OUT

MIC1_INP
MIC1_INN

Default
MIC for 3-pole Headphone Jack

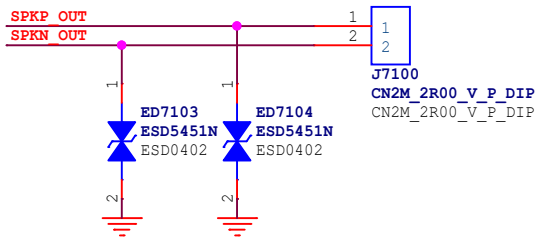


Option
MIC for 4-pole Headphone Jack



SPK

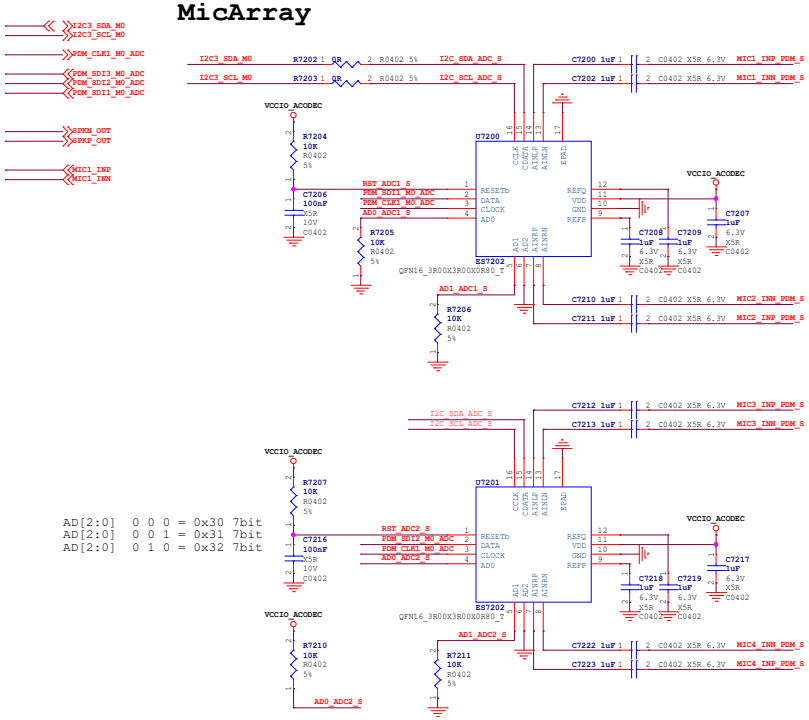
Note:8ohm/1.3W
Speaker Output



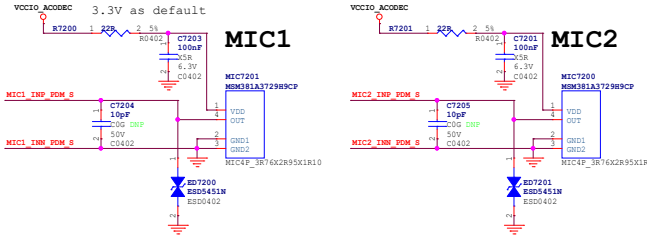
Rockchip Confidential

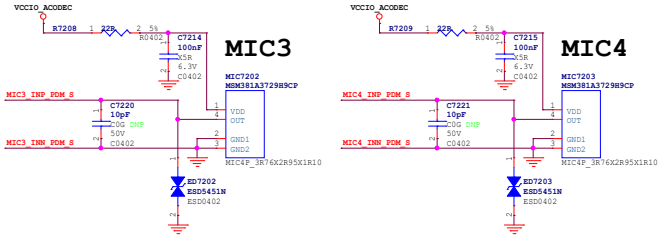
Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	71.Audio-SingleMic+RK809_SPK				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	52 of 61

MicArray

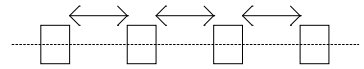


AD[2:0] 0 0 0 = 0x30 7bit
AD[2:0] 0 0 1 = 0x31 7bit
AD[2:0] 0 1 0 = 0x32 7bit



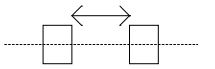


Equal spacing arrangement; according to mic algorithm



For 4MIC Array Applications

Equal spacing arrangement; according to mic algorithm

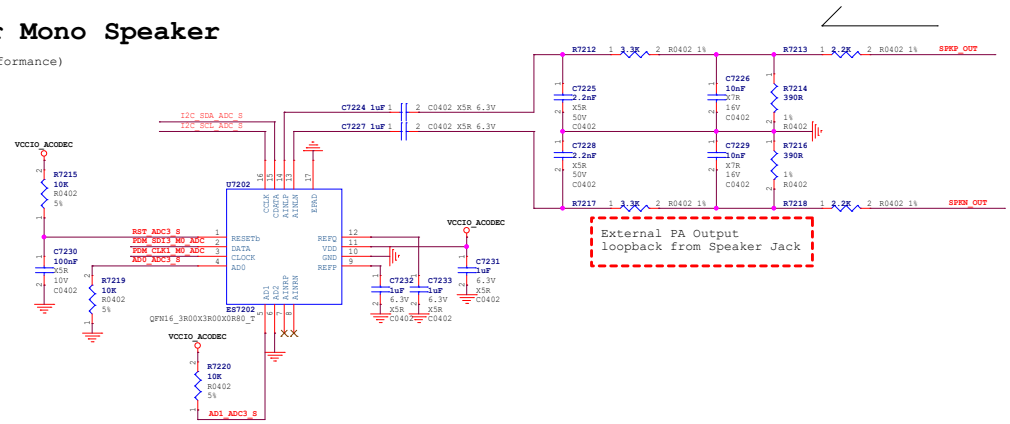


For 2MIC Array Applications

Option1

Loopback for Mono Speaker

(External ADC, better performance)

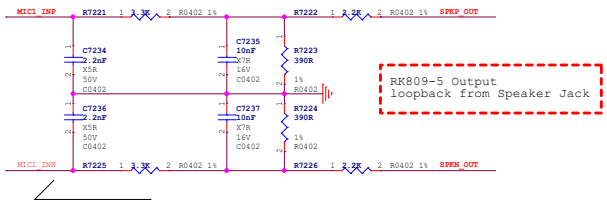


Option2

Loopback for Mono Speaker

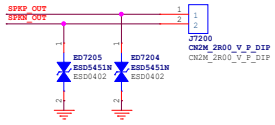
(Available while No MIC is connected to RK809-5)

(cost-effective)



RK809-5 SPK

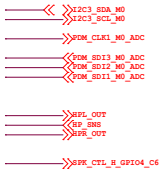
Note:8ohm/1.3W
Speaker Output

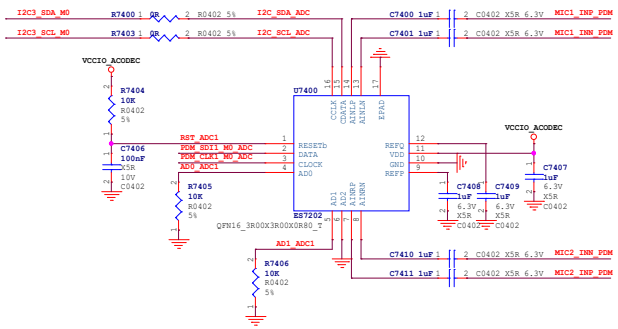


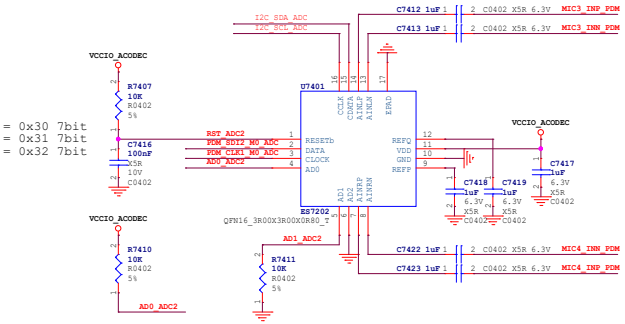
Rockchip Confidential

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd		
Project:	RK3568_Tablet_REF_SCH		
File:	72.Audio-MicArray+RK809_SPK		
Date:	Tuesday, July 20, 2021	Rev:	V1.0
Designed by: Zhangtz	Reviewed by: Default	Sheet:	53 of 61

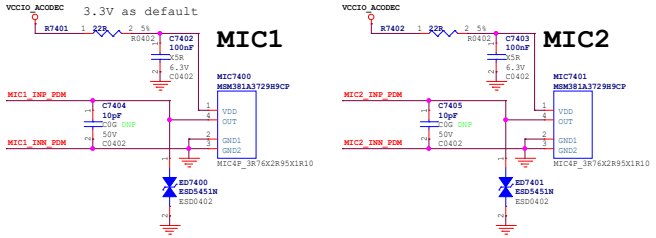
MicArray



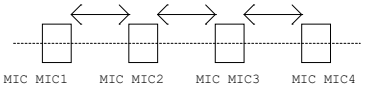




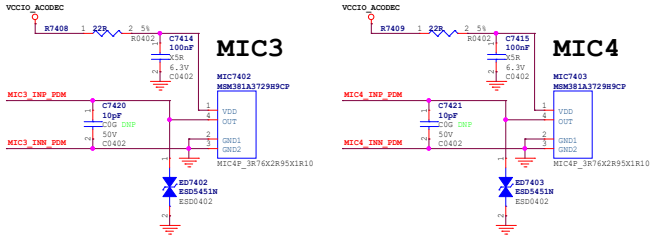
AD[2:0] 0 0 0 = 0x30 7bit
AD[2:0] 0 0 1 = 0x31 7bit
AD[2:0] 0 1 0 = 0x32 7bit



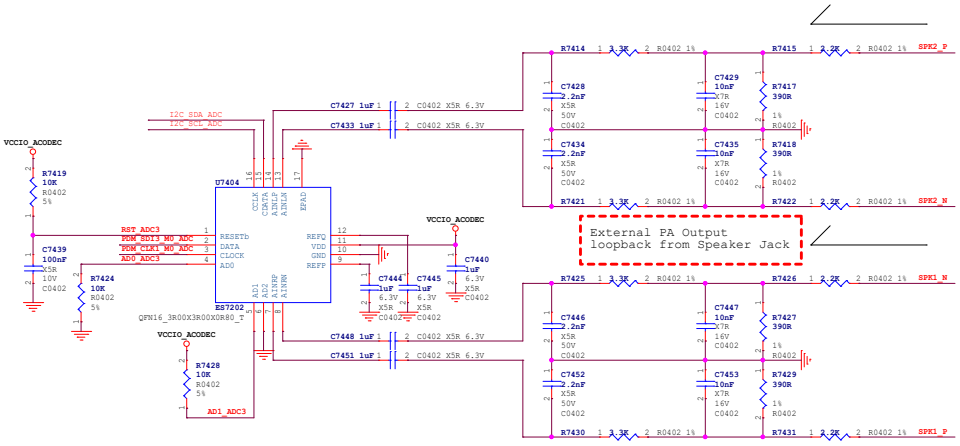
Equal spacing arrangement; according to mic algorithm



For 4MIC Array Applications

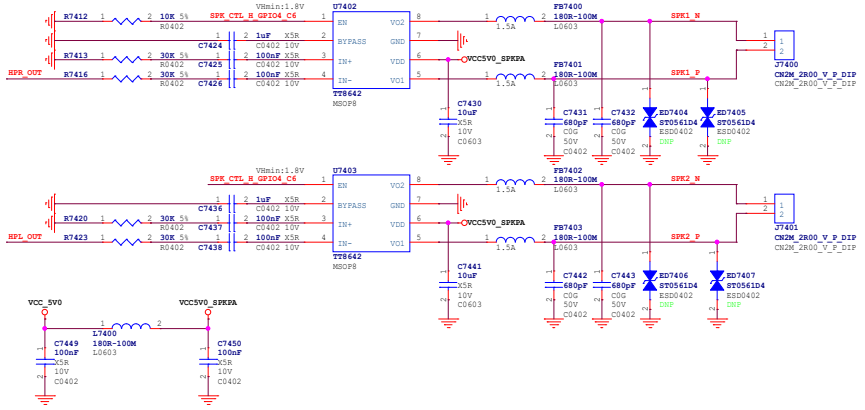


Loopback for Dual Speakers

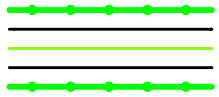


Speaker Output

Note:4ohm/3W



Layout note:



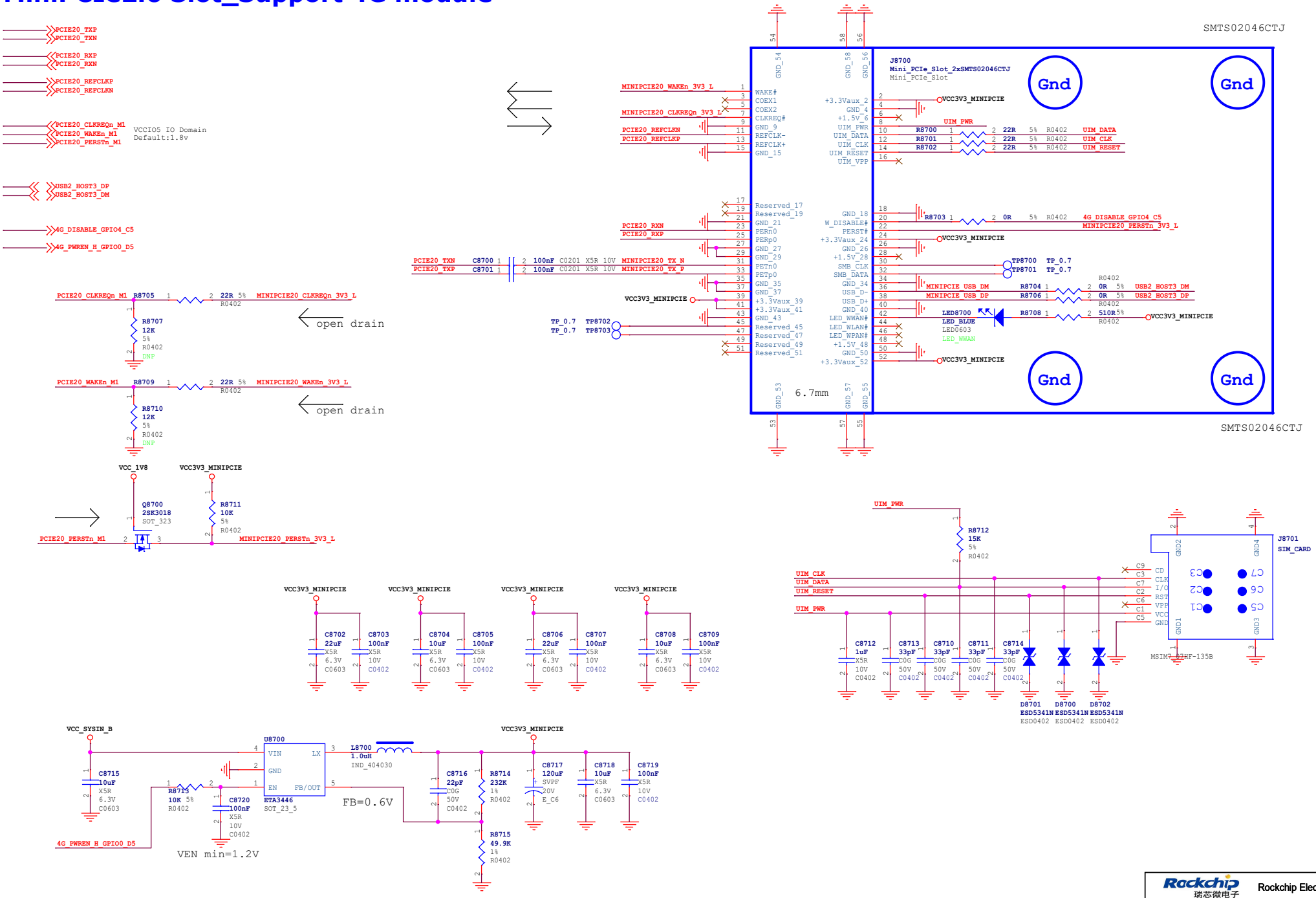


For NO Headphone design,
HP_SNS connect to GND near the PMIC.

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd		
Project:	RK3568_Tablet_REF_SCH		
File:	74.Audio-MicArray+EXT Dual_SPK		
Date:	Tuesday, July 20, 2021	Rev:	V1.0
Designed by: Zhangdz	Reviewed by: Default	Sheet:	54 of 61

Rockchip Confidential

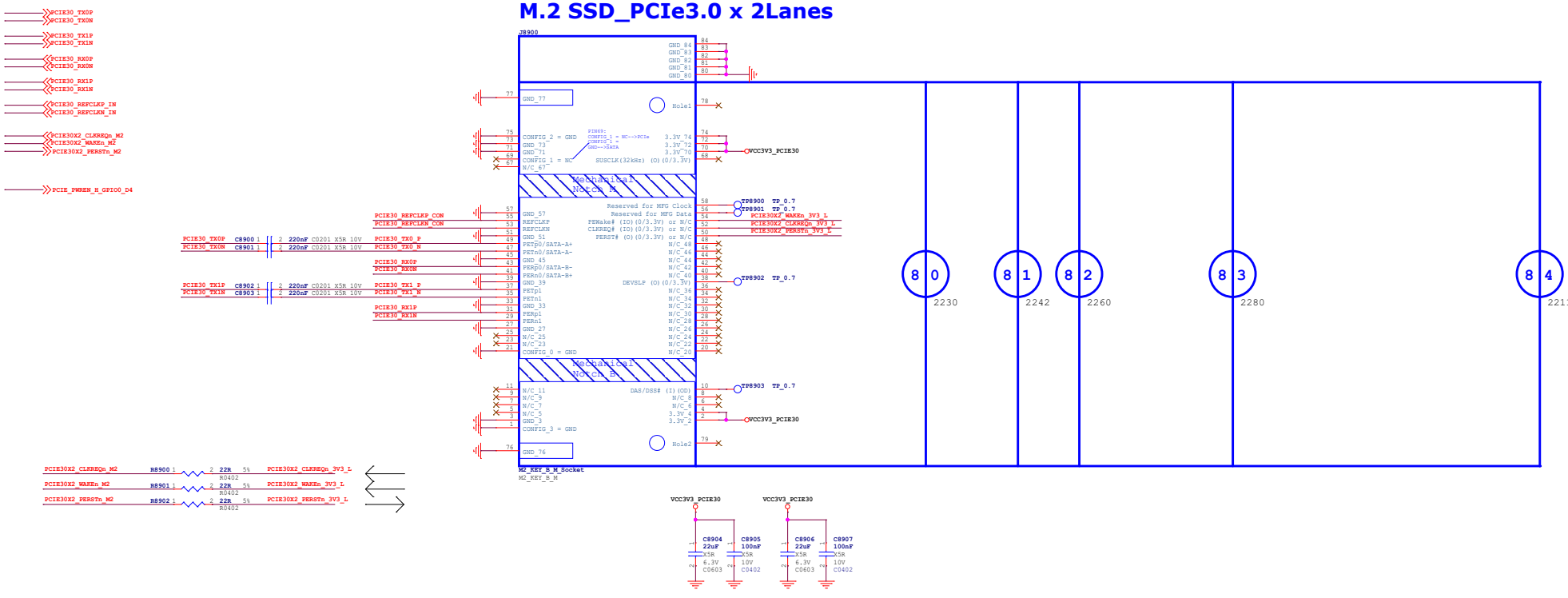
MiniPCIe2.0 Slot_Support 4G module

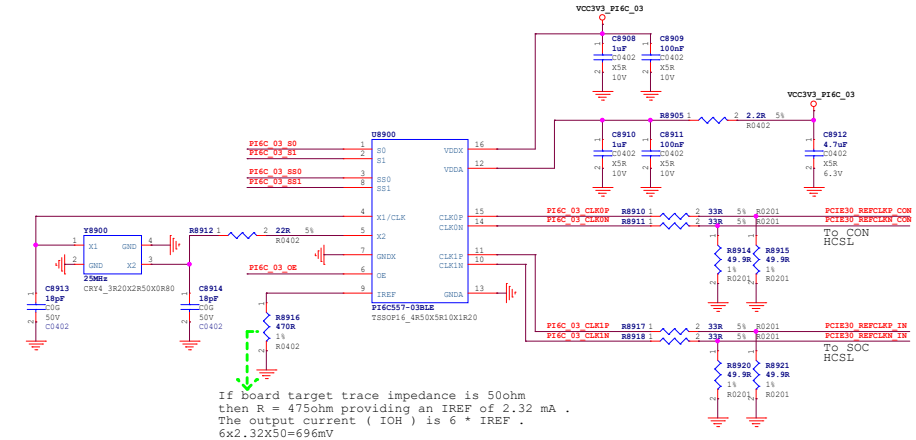


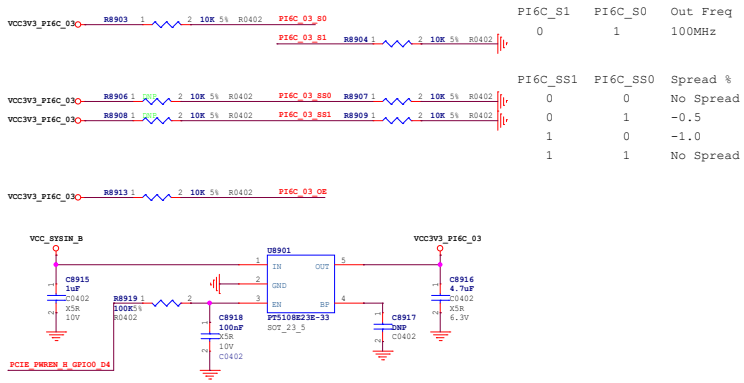
Rockchip Confidential

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	87.MiniPCIe2.0 Slot_With 4G Fun				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	55 of 61

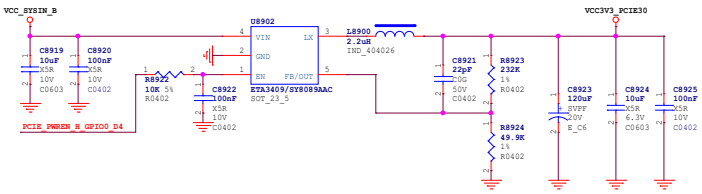
M.2 SSD_PCIe3.0 x 2Lanes







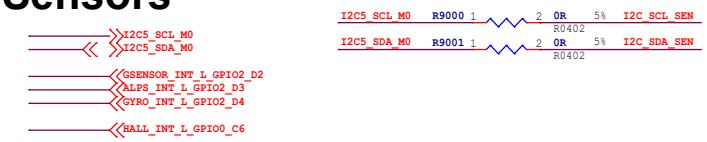
Default 3.39V



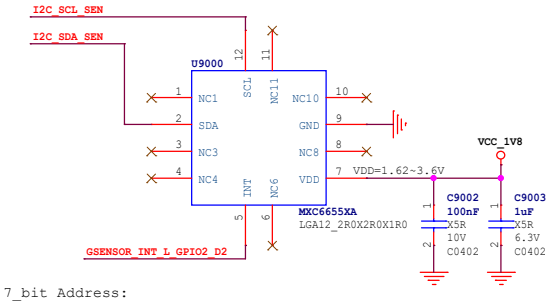
Rockchip Confidential

Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd
Project:	RK3568_Tablet_REF_SCH
File:	89.PCIE-M.2 SSD_KEY_B_M
Date:	Tuesday, July 20, 2021
Rev:	V1.0
Designed by:	Zhangtz
Reviewed by:	Default
Sheet:	56 of 61

Sensors

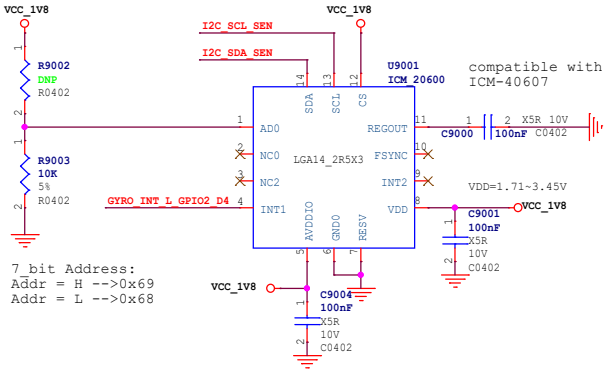


Gyroscope+Accelerometer



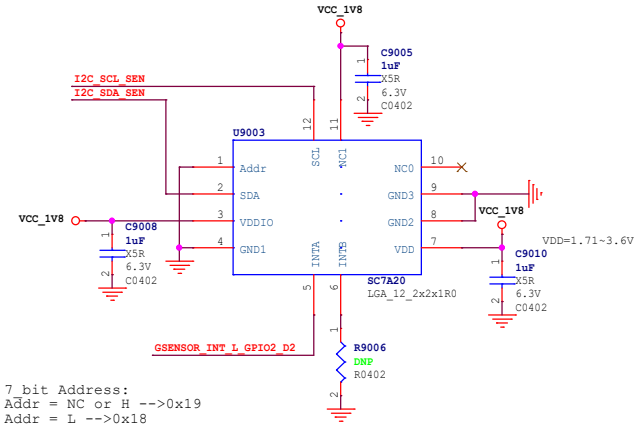
7_bit Address:
Fixed 0x15

Gyroscope+Accelerometer



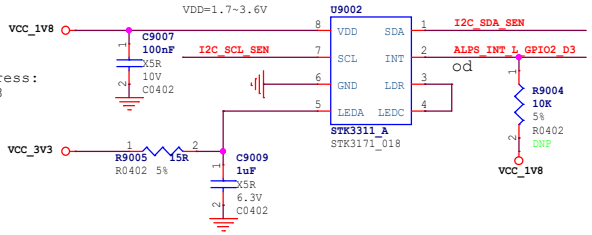
7_bit Address:
Addr = H -->0x69
Addr = L -->0x68

Accelerometer



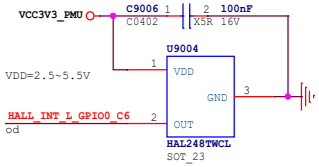
7_bit Address:
Addr = NC or H -->0x19
Addr = L -->0x18

Ambient Light & Proximity Sensor

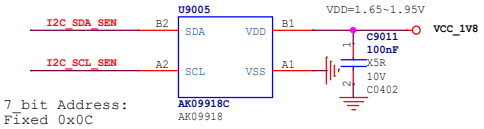


7_bit Address:
Fixed 0x48

Hall-Sensor



M-Sensor



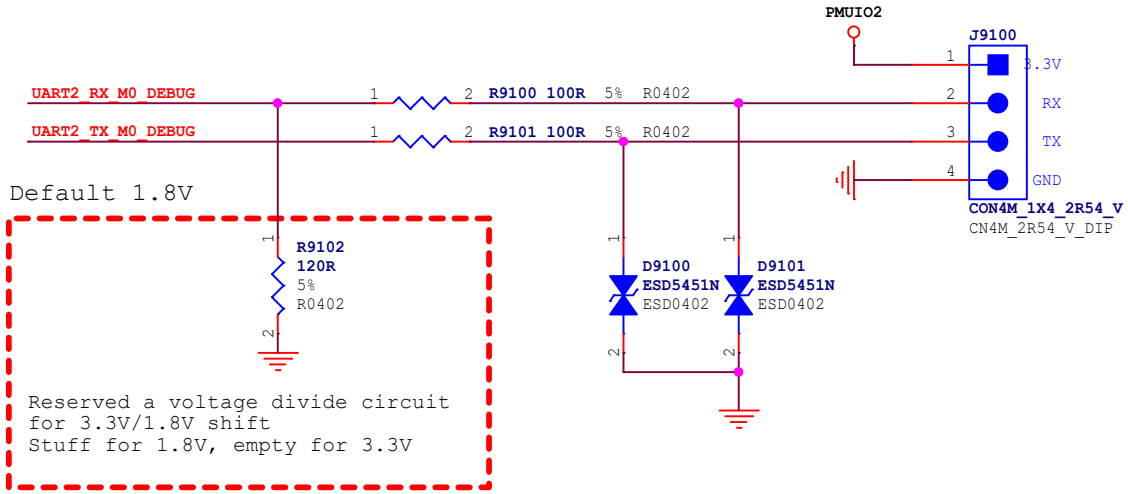
7_bit Address:
Fixed 0x0C

Rockchip Confidential

Rockchip 瑞芯微电子 Rockchip Electronics Co., Ltd					
Project:	RK3568_Tablet_REF_SCH				
File:	90.Sensors				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	57 of 61

UART2_RX_M0_DEBUG
UART2_TX_M0_DEBUG

Debug UART2



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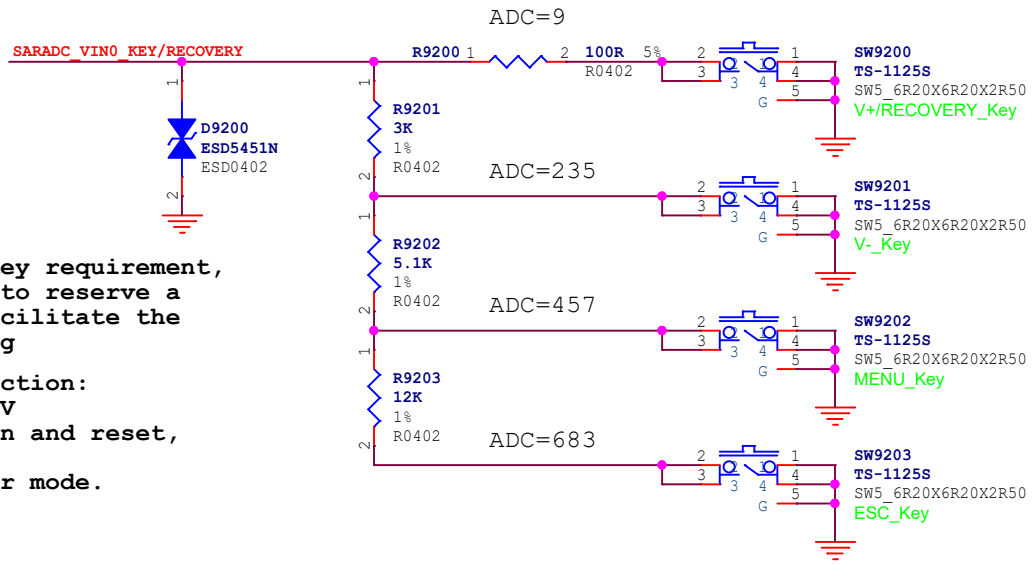
Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	91.Debug UART				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	58 of 61

Key Array

SARADC_VIN0_KEY/RECOVERY

RESETn

PMIC_PWRON

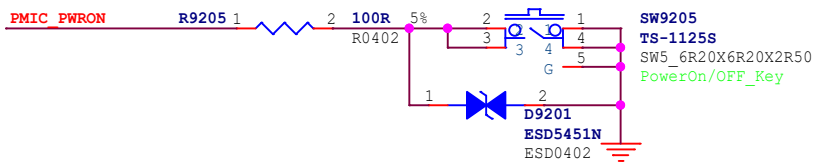


Note:

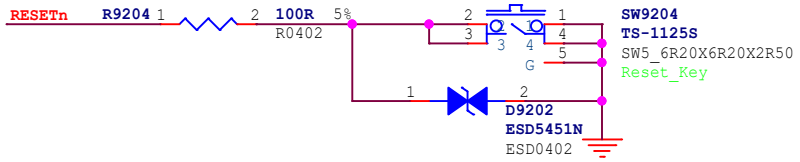
If there is no Key requirement, It is suggested to reserve a SW9200 Key to facilitate the development debug

RECOVERY_Key function:
If SARADC_VIN0=0V at after power on and reset, then system will enter into loader mode.

PowerOn/OFF_Key



Reset_Key



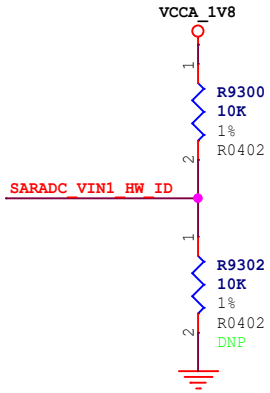
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Rockchip 瑞芯微电子	Rockchip Electronics Co., Ltd				
Project:	RK3568_Tablet_REF_SCH				
File:	92.KEY Array				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	59 of 61

SARADC_VIN1_HW_ID

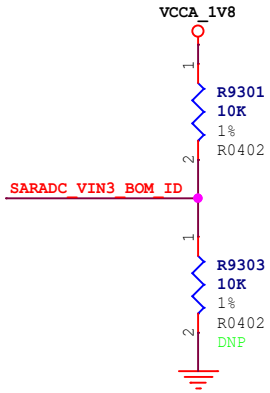
SARADC_VIN3_BOM_ID

HW_ID



SARADC_VIN1	Up Resistance	Down Resistance
HW_ID0	10K	DNP
HW_ID1	10K	110K
HW_ID2	20K	100K
HW_ID3	33K	100K
HW_ID4	18K	36K
HW_ID5	36K	51K
HW_ID6	51K	51K
HW_ID7	51K	36K
HW_ID8	36K	18K
HW_ID9	100K	33K
HW_ID10	100K	20K
HW_ID11	110K	10K
HW_ID12	DNP	10K

BOM_ID

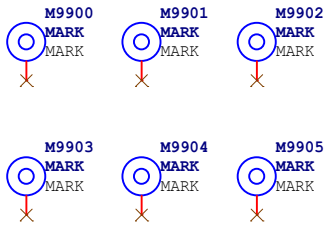


SARADC_VIN3	Up Resistance	Down Resistance
BOM_ID0	10K	DNP
BOM_ID1	10K	110K
BOM_ID2	20K	100K
BOM_ID3	33K	100K
BOM_ID4	18K	36K
BOM_ID5	36K	51K
BOM_ID6	51K	51K
BOM_ID7	51K	36K
BOM_ID8	36K	18K
BOM_ID9	100K	33K
BOM_ID10	100K	20K
BOM_ID11	110K	10K
BOM_ID12	DNP	10K

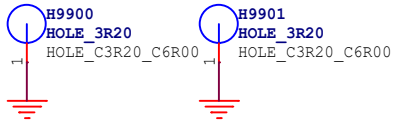
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Project:	RK3568_Tablet_REF_SCH				
File:	93.HW_ID/BOM_ID				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	60 of 61

PCB Mark Point

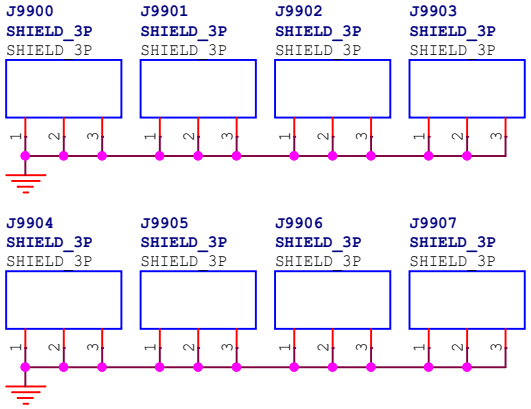


Mechanical Hole



Heatsink

Shield



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Project:	RK3568_Tablet_REF_SCH				
File:	99.Mark/Hole/Heatsink				
Date:	Tuesday, July 20, 2021			Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet:	61 of 61